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Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures

--Manuscript Draft--

Manuscript Number:	ICHMT-D-26-05613R1
Full Title:	Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures
Article Type:	Research Paper
Keywords:	Thermal-wave propagation; Thermoelastic double-porosity media; Corrugated interface; Love waves; Wave attenuation.
Abstract:	The present study examines coupled thermal-wave transport and guided-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. The model consists of a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space, where thermal interactions, pore coupling, and interface irregularities jointly influence energy transport and wave characteristics. Using generalized thermoelastic double-porosity theory, the coupled displacement, temperature, and porosity fields are formulated under anti-plane shear motion, and an analytical dispersion relation for Love-wave propagation is derived by imposing appropriate boundary and interface continuity conditions. Numerical investigations quantify the influences of corrugation amplitude, corrugation wave number, porosity parameters, frequency, and propagation modes on phase velocity, attenuation, stress penetration, temperature, and heat-flux distributions. The results demonstrate that interface corrugation strongly modifies Love-wave dispersion, particularly in the low- and intermediate-frequency regimes, whereas its influence becomes weaker at higher frequencies. Increasing porosity reduces the effective thermoelastic stiffness and consequently lowers the phase velocity. Attenuation increases with frequency, while lower-order modes exhibit stronger attenuation than higher-order modes because of their greater interaction with the coupled thermo-porous region. Thermal analysis further shows that larger thermo-porous decay parameters produce stronger near-interface temperature and heat-flux localization but reduce the effective thermal penetration depth. A smooth-interface limiting case is recovered, providing a consistency check for the proposed formulation. These findings clarify the coupled roles of interface geometry, double porosity, and thermal transport in controlling wave dispersion, attenuation, and energy localization, with potential relevance to thermal-barrier materials, porous composites, subsurface characterization, and thermo-mechanical monitoring of layered structures.
Response to Reviewers:	Response to the Editor and Reviewers' Comments We sincerely thank the Editor and Reviewers for their valuable and constructive comments on our manuscript. We have carefully addressed all the comments and revised the manuscript accordingly. All changes are highlighted in yellow in the revised manuscript. Our point-by-point responses are provided below. Reviewer 1 Manuscript Number: ICHMT-D-26-05613 Title: Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures Recommendation: Major Revision The manuscript investigates thermoelastic Love-wave propagation in a double-porosity layered medium with a corrugated interface. The topic is relevant to the scope of the journal and the analytical formulation appears to be carefully developed. However, in its current form, the manuscript does not sufficiently demonstrate its scientific novelty or its contribution to heat-transfer research. Several aspects require substantial clarification and improvement before the work can be considered for publication. 1. The novelty of the proposed model should be stated more explicitly. The manuscript should clearly explain how the present work advances beyond the existing literature on thermoelastic Love waves, double-porosity media, and corrugated interfaces. Response: Thank you for this important comment. We agree that the novelty of the

proposed model needed to be stated more explicitly. Accordingly, the final part of the Introduction has been substantially strengthened to distinguish the present work from previous studies on thermoelastic Love waves, double-porosity media, and irregular/corrugated interfaces. The revised manuscript now clearly explains that the principal contribution is the unified treatment of thermoelastic coupling, two interacting porosity fields, periodic interface corrugation, and mode-dependent Love-wave propagation. We have also clarified that the model provides information on dispersion, attenuation, stress penetration, and coupled thermo-porosity field localization within the same analytical framework.

Added text in the introduction section

Although thermoelastic wave propagation, double-porosity effects, and interface irregularities have individually received considerable attention, their combined influence on coupled thermal-wave and Love-wave transport in a layered medium remains insufficiently understood. Most existing studies have focused either on thermoelastic porous media with regular interfaces, elastic or porous media with interface irregularities, or double-porosity systems without explicitly examining the combined role of thermal interaction and periodic corrugation. Consequently, the manner in which a corrugated interface modifies the interaction among thermal fields, two interconnected porosity systems, guided-wave dispersion, attenuation, and energy localization has not been adequately clarified.

The novelty of the present study therefore lies in developing a unified analytical framework for a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space through a periodically corrugated interface. Unlike conventional Love-wave models, the present formulation simultaneously incorporates thermoelastic coupling, two porosity fields, interface corrugation, and mode-dependent wave behavior. The formulation further provides analytical dispersion relations together with numerical assessments of phase velocity, attenuation, stress penetration, and coupled thermo-porosity field localization. This framework enables the effects of thermal interaction, pore coupling, and interface geometry on energy transport and guided-wave behavior to be examined within a single model.

2. The literature review is overly descriptive. It would benefit from a more critical discussion that clearly identifies the limitations of previous studies and better establishes the motivation for the present work.

Comment 2: The literature review is overly descriptive. It would benefit from a more critical discussion that clearly identifies the limitations of previous studies and better establishes the motivation for the present work.

Response: Thank you for this valuable comment. We agree that the literature review in the original manuscript was primarily descriptive. Accordingly, the Introduction has been substantially revised to provide a more critical and comparative assessment of the existing literature. Rather than only summarizing individual studies, the revised discussion now highlights the limitations of previous models, particularly their separate treatment of thermal effects, single/double porosity, material heterogeneity, and interface irregularity. The limitations of conventional single-porosity formulations and the comparatively limited consideration of corrugated interfaces within thermoelastic double-porosity systems have also been clarified. Based on this critical assessment, the research gap has been explicitly identified concerning the combined effects of thermoelastic double-porosity coupling, periodic interface corrugation, thermal interaction, and propagation modes on dispersion, attenuation, energy localization, and stress penetration. These revisions provide a clearer motivation for the unified analytical framework developed in the present study.

Location in revised manuscript: Section 1, Introduction - the literature-review paragraphs and the research-gap discussion immediately preceding the paragraph describing the objectives and contributions of the present study.

3. Since the journal primarily focuses on heat and mass transfer, the manuscript should better emphasize its contribution to thermal transport. At present, the discussion is largely centered on wave dispersion and attenuation, while the heat-transfer aspects receive comparatively limited attention.

Response:

Thank you for this important observation. We agree that the thermal-transport aspects of the proposed formulation required greater emphasis. Accordingly, the revised manuscript has been expanded to provide a more explicit analysis of the thermal response of the thermoelastic double-porosity system. Two additional results have

been included to illustrate the variation of normalized temperature and normalized heat flux with dimensionless penetration depth for different values of the thermo-porous decay parameter. The temperature profiles demonstrate the progressive attenuation of the thermal disturbance away from the interface and show that a larger decay parameter produces stronger thermal localization and a reduced penetration depth. The corresponding heat-flux distributions further show that larger decay parameters generate stronger heat-flux gradients near the interface, followed by more rapid attenuation within the medium. These results provide a direct thermal interpretation of the analytical solution and complement the existing analyses of dispersion, attenuation, stress penetration, and coupled thermo-porosity fields. The associated discussion has also been expanded to clarify the relationship between thermal transport, interface localization, and thermoelastic energy dissipation.

Added text in the manuscript:

Two new figures and their corresponding physical interpretations have been added in the Numerical Results and Discussion section: Fig. 18 and Fig. 19:

Fig. 18. Variation of normalized temperature Θ^* with dimensionless penetration depth $\zeta=z/h$ for different values of the thermo-porous decay parameter S .

Figure 18 illustrates the variation of the normalized temperature Θ^* with the dimensionless penetration depth ζ for different values of the thermo-porous decay parameter S . The temperature decreases continuously with increasing depth, confirming the attenuation of the thermal disturbance away from the interface. For smaller values of S , the temperature decays more gradually, indicating that the thermal disturbance penetrates deeper into the medium. In contrast, larger values of S produce a steeper temperature gradient and a stronger localization of thermal energy near the interface. This behavior is consistent with the exponentially decaying thermal modes obtained from the bounded solution of the coupled thermoelastic formulation and demonstrates the important role of the decay parameter in controlling thermal penetration.

Fig. 19. Variation of normalized heat flux q^* with dimensionless penetration depth $\zeta=z/h$ for different values of the thermo-porous decay parameter S .

Figure 19 presents the variation of the normalized heat flux q^* with the dimensionless penetration depth ζ for different values of the thermo-porous decay parameter S . Higher values of S produce a larger heat-flux magnitude near the interface because of the stronger local temperature gradient. However, the heat flux decreases rapidly as the penetration depth increases. In contrast, smaller values of S yield a lower initial heat-flux magnitude but a more gradual attenuation of the thermal response. At sufficiently large depths, the normalized heat flux approaches zero for all cases, indicating that the thermal disturbance becomes negligible far from the interface. These results show that increasing the modal decay parameter enhances near-interface thermal localization while reducing the effective penetration range of the heat-transfer process.

8.1 Thermal-Transport Characteristics of the Coupled System

In the present formulation, the thermal field is intrinsically coupled with the mechanical and double-porosity fields through the governing thermoelastic equations.

Consequently, variations in temperature interact with elastic deformation and the two porosity fields, resulting in a coupled redistribution of thermal and mechanical energy within the layered medium. The thermal response is therefore characterized in terms of normalized temperature and heat flux with dimensionless penetration depth, which provides direct insight into the localization and penetration of thermal disturbances within the medium.

The thermal-field analysis complements the wave-propagation results by establishing the connection between thermal transport and the attenuation of coupled disturbances away from the interface. It also provides a physical basis for interpreting how thermo-porous interactions influence energy localization within the layered structure. The corresponding temperature and heat-flux characteristics are presented in Figs. 18 and 19, followed by the dispersion, attenuation, and stress-response analyses.

4. The numerical results are mostly descriptive. The authors should provide stronger physical explanations for the observed trends rather than only describing the behaviour

of the plotted curves.

Response:

Thank you for this valuable comment. The Numerical Results and Discussion section has been carefully revised to strengthen the physical interpretation of the observed numerical trends. In particular, the discussions accompanying the figures have been expanded to explain the underlying physical mechanisms rather than merely describing the variation of the plotted curves. The revised discussion now addresses the roles of interface corrugation, wave-energy confinement, mode-dependent penetration, thermo-porosity coupling, effective stiffness, interfacial scattering, and energy dissipation in determining the phase velocity and attenuation characteristics. Particular attention has also been given to explaining the different responses of lower- and higher-order propagation modes and the frequency-dependent sensitivity of the waves to corrugation and porosity parameters. These additions provide a clearer physical interpretation of the numerical results and substantially improve the discussion of the coupled thermoelastic double-porosity wave-propagation behavior.

Location in revised manuscript: Section 8, Numerical Results and Discussion-physical explanations accompanying Figs. 2-17 have been revised.

5. The manuscript would be considerably strengthened by validating the proposed formulation through comparison with an analytical solution, an appropriate limiting case, or previously published results.

Response:

Thank you for this important and constructive suggestion. In the revised manuscript, we have strengthened the validation of the proposed formulation by explicitly considering an appropriate limiting case. Specifically, the corrugation amplitude parameter is set to $\epsilon=0$, for which the periodically corrugated interface reduces to a smooth interface and the corrugation-dependent contributions vanish. Consequently, the proposed formulation recovers the corresponding smooth-interface thermoelastic double-porosity Love-wave model. The numerical behavior of this limiting case is also demonstrated through the smooth-interface results and its comparison with the corrugated-interface configuration. The smooth and stable dispersion behavior obtained for $\epsilon=0$, together with the expected asymptotic convergence at higher frequencies, provides a consistency check for the developed dispersion formulation. This limiting-case verification has now been explicitly incorporated into the revised manuscript and discussed in connection with Fig. 17.

Location in revised manuscript: Section 8, Numerical Results and Discussion, under a new paragraph/subsection entitled "Validation of the formulation through a limiting case," and an additional validation statement has been included in the discussion of Fig. 17.

Text added to the manuscript

Validation of the formulation through a limiting case

To verify the consistency of the proposed formulation, a limiting-case analysis is performed by considering $\epsilon=0$. Under this condition, the periodically corrugated interface reduces to a smooth interface, and all corrugation-dependent contributions in the dispersion relation vanish. The resulting formulation therefore represents Love-wave propagation in a thermoelastic double-porosity layered medium with a smooth interface. Mathematically, this limiting condition can be represented as $\epsilon \rightarrow 0$ "corrugated interface" $\rightarrow$ "smooth interface".

The numerical solution corresponding to $\epsilon=0$ exhibits a smooth dispersion response and approaches the expected limiting behavior with increasing dimensionless frequency. The successful recovery of the smooth-interface configuration provides a useful limiting-case consistency check for the proposed corrugated-interface formulation.

We have added this text in the Fig. 17 discussion

The smooth-interface result also provides a limiting-case verification of the present formulation, since setting $\epsilon=0$ eliminates the corrugation contribution and recovers the corresponding smooth-interface propagation behavior.

6. The practical engineering significance of the proposed model should be discussed in greater detail. The authors should better explain how the present findings may contribute to realistic heat-transfer or thermoelastic engineering applications.

Response:

Thank you for this valuable suggestion. The revised manuscript has been expanded to

provide a clearer discussion of the practical engineering significance of the proposed thermoelastic double-porosity Love-wave model. In particular, the numerical findings have been related to engineering and geophysical systems involving layered porous materials, irregular interfaces, coupled thermal transport, and guided-wave propagation. The analysis demonstrates how interface corrugation, double-porosity parameters, thermal interaction, and propagation modes can influence phase velocity, attenuation, stress localization, thermal penetration, and heat-flux behavior. These observations are relevant to the characterization and design of layered porous structures in which mechanical-wave propagation and heat transfer occur simultaneously. The revised discussion further highlights the potential usefulness of the model for subsurface characterization, non-destructive evaluation of layered porous materials, thermal-barrier and porous composite systems, and guided-wave-based monitoring of thermo-mechanically coupled structures. The corresponding practical implications have now been incorporated into the Numerical Results and Discussion section and further emphasized in the Conclusions.

Location in revised manuscript: Section 8, Numerical Results and Discussion, after the discussion of the final figure. A shorter application statement incorporated into the Conclusion section.

Text added to the manuscript

Engineering significance and practical implications. The present results have practical relevance to layered porous systems in which mechanical-wave propagation occurs simultaneously with thermal transport and pore interactions. The predicted variations in phase velocity and attenuation with corrugation and porosity parameters indicate that Love waves can be sensitive to changes in interface geometry and internal pore structure, providing a potential basis for the characterization of layered porous materials. In addition, the predicted stress localization, temperature decay, and heat-flux behavior provide information on the depth over which mechanical and thermal disturbances remain significant. These characteristics are relevant to subsurface and geophysical characterization, non-destructive evaluation of porous layered structures, thermal-barrier and porous composite systems, and guided-wave monitoring under thermo-mechanical loading. Thus, the proposed model provides a theoretical framework for assessing how interface irregularity, double porosity, and thermal coupling jointly affect wave transmission and heat-transfer characteristics in realistic layered media.

Added in Conclusion

From an engineering perspective, the results provide useful insight into the control of wave transmission, attenuation, stress localization, and thermal penetration in layered porous structures, with potential relevance to geophysical characterization, non-destructive evaluation, and thermo-mechanical monitoring of porous composite systems.

Overall, the manuscript has potential, but substantial revision is required to improve its scientific contribution, physical interpretation, and overall relevance to the journal.

Reviewer 2

In this paper, the authors investigated the “Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures”. I have gone through the entire manuscript. All the equations are systematically presented. The theoretical results are well supplemented by the graphical presentations. The novelty and applications of the present problem are clearly defined. Overall, the manuscript is a good piece of scientific work. It is fit for publication in this journal. However, the authors must incorporate the following points in their manuscript:

1. Abstract should be précised; however, major result can be described in the abstract.
Response: Thank you for this helpful suggestion. The Abstract has been carefully revised and refined to provide a more precise and focused presentation of the study. The revised Abstract now briefly describes the physical model, analytical formulation, principal controlling parameters, and major numerical findings. In particular, it highlights the effects of interface corrugation, porosity, frequency, and propagation modes on phase velocity, attenuation, thermal-wave transport, and energy dissipation. The practical significance of the obtained results for porous thermoelastic layered systems has also been concisely stated. These revisions improve the clarity of the Abstract

while ensuring that the principal outcomes of the study are explicitly communicated.

Updated abstract

“The present study examines coupled thermal-wave transport and guided-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. The model consists of a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space, where thermal interactions, pore coupling, and interface irregularities jointly influence energy transport and wave characteristics. Using generalized thermoelastic double-porosity theory, the coupled displacement, temperature, and porosity fields are formulated under anti-plane shear motion, and an analytical dispersion relation for Love-wave propagation is derived by imposing appropriate boundary and interface continuity conditions. Numerical investigations quantify the influences of corrugation amplitude, corrugation wave number, porosity parameters, frequency, and propagation modes on phase velocity, attenuation, stress penetration, temperature, and heat-flux distributions. The results demonstrate that interface corrugation strongly modifies Love-wave dispersion, particularly in the low- and intermediate-frequency regimes, whereas its influence becomes weaker at higher frequencies. Increasing porosity reduces the effective thermoelastic stiffness and consequently lowers the phase velocity. Attenuation increases with frequency, while lower-order modes exhibit stronger attenuation than higher-order modes because of their greater interaction with the coupled thermo-porous region. Thermal analysis further shows that larger thermo-porous decay parameters produce stronger near-interface temperature and heat-flux localization but reduce the effective thermal penetration depth. A smooth-interface limiting case ($\epsilon=0$) is recovered, providing a consistency check for the proposed formulation. These findings clarify the coupled roles of interface geometry, double porosity, and thermal transport in controlling wave dispersion, attenuation, and energy localization, with potential relevance to thermal-barrier materials, porous composites, subsurface characterization, and thermo-mechanical monitoring of layered structures.”

2. What are the advantages of employed technique?

Response to Reviewer

Thank you for this important comment. The advantages of the employed analytical technique have now been clarified in the revised manuscript. The harmonic-wave approach combined with the matrix formulation transforms the coupled thermoelastic double-porosity governing equations into a compact algebraic/eigenvalue framework, allowing the mechanical, thermal, and two porosity fields to be treated simultaneously. The technique enables an explicit dispersion relation to be obtained while retaining the effects of thermal coupling, double porosity, and interface corrugation. Moreover, the first-order treatment of the corrugated boundary provides a computationally efficient way of incorporating small interface irregularities without requiring a full numerical discretization of the entire domain. The resulting formulation also facilitates systematic parametric studies of phase velocity, attenuation, thermal penetration, stress localization, and different propagation modes. These advantages have now been explicitly stated in the revised manuscript.

We have added at the end of Section 6, Dispersion Relation Formulation, The employed analytical technique offers several advantages for the present coupled problem. The harmonic-wave representation reduces the governing partial differential equations to a tractable system of ordinary differential equations, while the matrix formulation enables the mechanical, thermal, and two porosity fields to be treated within a unified framework. Furthermore, the first-order treatment of the corrugated interface incorporates the influence of small interface irregularities without requiring full-domain numerical discretization. The resulting dispersion formulation is computationally efficient and facilitates systematic evaluation of phase velocity, attenuation, thermal penetration, stress localization, and mode-dependent wave behavior over a wide range of governing parameters.

3. Author should enhance the physical description of the model.

Response: Thank you for this valuable suggestion. The physical description of the model has been strengthened in Section 2. The revised text now explains the physical roles of the two interacting porosity fields, the finite guiding layer and underlying half-space, the corrugation amplitude and wave number, and the anti-plane shear configuration. In addition, the physical influence of interface corrugation on wave-energy redistribution, dispersion, attenuation, and penetration has been clarified.

These additions provide a clearer physical interpretation of the proposed thermoelastic double-porosity layered configuration while maintaining consistency with the mathematical formulation and Fig. 1.

4. The authors should add some more lines in the numerical results and discussion section to strengthen these sections.

Response: Thank you for this valuable suggestion. The Numerical Results and Discussion section has been substantially expanded in the revised manuscript. Additional discussions have been incorporated throughout the section to provide deeper physical interpretation of the numerical results rather than merely describing the trends of the plotted curves. In particular, the revised discussion explains the effects of interface corrugation, propagation mode, porosity parameters, wave confinement, interfacial scattering, thermo-porosity coupling, stress penetration, and energy dissipation on the phase velocity and attenuation characteristics. Furthermore, additional results concerning stress-amplitude penetration, spatial thermo-porosity field intensity, the combined effects of wave number and corrugation amplitude, smooth-versus corrugated-interface behavior, temperature distribution, and heat-flux variation have been included to provide a more comprehensive interpretation of the coupled thermoelastic response. A limiting-case verification for the smooth-interface condition ($\epsilon=0$) has also been incorporated. These additions considerably strengthen the numerical analysis and provide clearer physical insight into the coupled thermal-wave and Love-wave propagation mechanisms.

We have thoroughly revised the manuscript, particularly Section 8, "Numerical Results and Discussion," by expanding the discussions accompanying Figs. 2-19.

5. It is beneficial for the readers to add more explanations about the novel contribution of this manuscript from theoretical/experimental viewpoints.

Response to Reviewer

Thank you for this constructive suggestion. The novel contributions of the present study have been clarified further from both theoretical and practical/experimental viewpoints. From the theoretical perspective, the present formulation integrates generalized thermoelasticity, double-porosity effects, and periodic interface corrugation within a unified Love-wave propagation framework. The model simultaneously accounts for the coupled displacement, temperature, and two porosity fields and provides a generalized dispersion relation for examining phase velocity, attenuation, and thermal-wave behavior. In addition, the smooth-interface condition is recovered as a limiting case, providing a consistency check for the formulation.

From the experimental/practical viewpoint, the predicted sensitivity of phase velocity, attenuation, stress penetration, temperature, and heat flux to porosity and interface corrugation provides measurable quantities that could be used in future experimental characterization of layered porous materials. In particular, changes in guided-wave phase velocity and attenuation may serve as indicators of variations in pore structure or interface geometry, while temperature and heat-flux distributions can provide complementary information on thermo-mechanical coupling. Since the present work is theoretical and numerical, no experimental validation is claimed; rather, the results provide a theoretical basis and specific response quantities that may guide future experimental investigations.

We have added the following paragraph at the end of the Introduction

The novelty of the present study lies in the unified treatment of generalized thermoelasticity, double-porosity coupling, and periodic interface corrugation in Love-wave propagation through a layered medium. Unlike formulations that consider these effects separately, the present model simultaneously captures their influence on dispersion, attenuation, stress penetration, temperature, and heat-flux characteristics. Theoretically, the derived dispersion framework provides a systematic means of quantifying the coupled effects of interface geometry, thermal interaction, and two pore systems, while also recovering the smooth-interface configuration as a limiting case. From an experimental viewpoint, the predicted variations in phase velocity and attenuation provide potentially measurable signatures of changes in porosity and interface geometry, whereas temperature and heat-flux responses offer complementary indicators of thermo-mechanical coupling. Although experimental validation is beyond the scope of the present theoretical study, these predicted quantities provide a basis for future guided-wave and thermal measurements in layered porous materials.

	6. There are some misprints that can be corrected in the revised version for running a spell check and making the manuscript better. Response: Thank you for the comment. We have carefully proofread the entire manuscript and corrected the typographical, spelling, grammatical, and formatting errors throughout the revised version. 7. Please check all the references again and correct them as per journal format. Response: Thank you for the comment. We have carefully checked and revised all the references to ensure consistency and compliance with the journal's prescribed reference format. The in-text citations and reference list have also been cross-checked and corrected accordingly. Based on the comprehensive assessment of your paper and after minor revision, I am pleased to recommend it for publication
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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Response to the Editor and Reviewers' Comments

We sincerely thank the Editor and Reviewers for their valuable and constructive comments on our manuscript. We have carefully addressed all the comments and revised the manuscript accordingly. All changes are highlighted in yellow in the revised manuscript. Our point-by-point responses are provided below.

Reviewer 1

Manuscript Number: ICHMT-D-26-05613

Title: Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures

Recommendation: Major Revision

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1. The novelty of the proposed model should be stated more explicitly. The manuscript should clearly explain how the present work advances beyond the existing literature on thermoelastic Love waves, double-porosity media, and corrugated interfaces.

Response: Thank you for this important comment. We agree that the novelty of the proposed model needed to be stated more explicitly. Accordingly, the final part of the Introduction has been substantially strengthened to distinguish the present work from previous studies on thermoelastic Love waves, double-porosity media, and irregular/corrugated interfaces. The revised manuscript now clearly explains that the principal contribution is the unified treatment of thermoelastic coupling, two interacting porosity fields, periodic interface corrugation, and mode-dependent Love-wave propagation. We have also clarified that the model provides information on dispersion, attenuation, stress penetration, and coupled thermo-porosity field localization within the same analytical framework.

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3. Since the journal primarily focuses on heat and mass transfer, the manuscript should better emphasize its contribution to thermal transport. At present, the discussion is largely centered on wave dispersion and attenuation, while the heat-transfer aspects receive comparatively limited attention.

Response:

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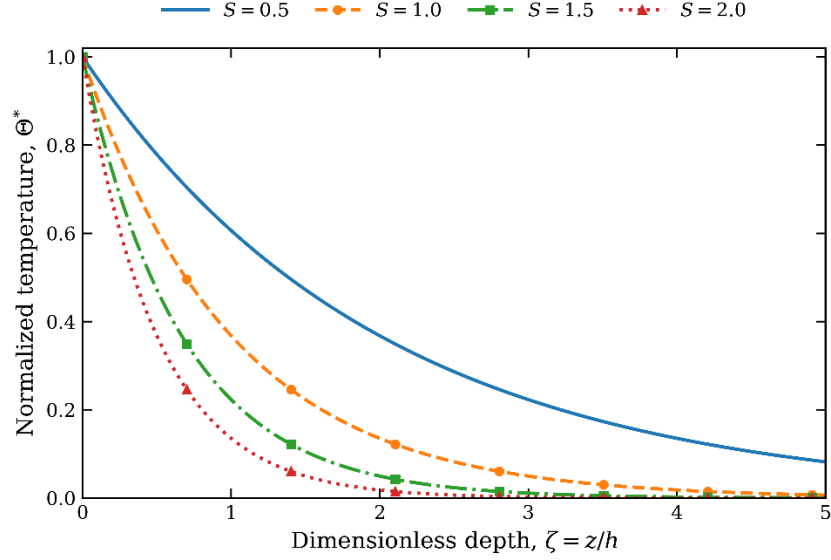


Fig. 18. Variation of normalized temperature Θ^* with dimensionless penetration depth $\zeta = z/h$ for different values of the thermo-porous decay parameter S .

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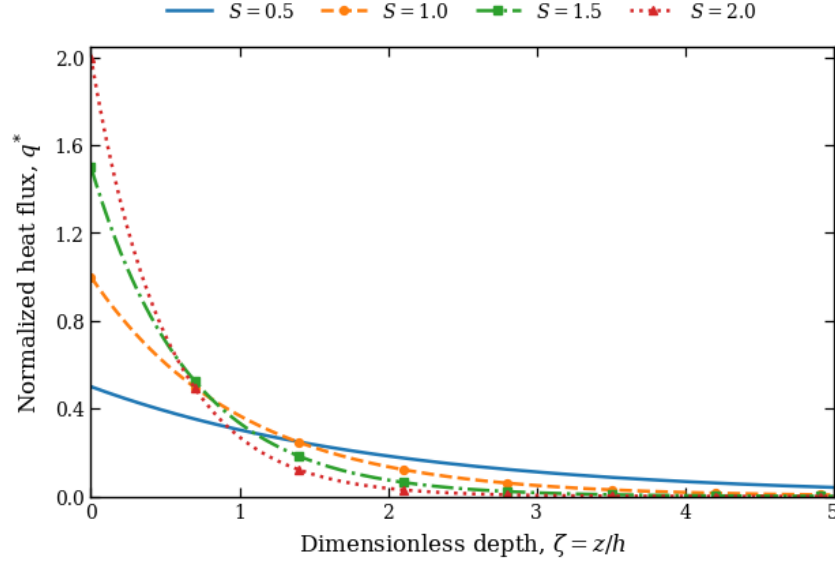


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In the present formulation, the thermal field is intrinsically coupled with the mechanical and double-porosity fields through the governing thermoelastic equations. Consequently, variations in temperature interact with elastic deformation and the two porosity fields, resulting in a coupled redistribution of thermal and mechanical energy within the layered medium. The thermal response is therefore characterized in terms of normalized temperature and heat flux with dimensionless penetration depth, which provides direct insight into the localization and penetration of thermal disturbances within the medium.

The thermal-field analysis complements the wave-propagation results by establishing the connection between thermal transport and the attenuation of coupled disturbances away from the interface. It also provides a physical basis for interpreting how thermo-porous interactions influence energy localization within the layered structure. The corresponding temperature and

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5. The manuscript would be considerably strengthened by validating the proposed formulation through comparison with an analytical solution, an appropriate limiting case, or previously published results.

Response:

Thank you for this important and constructive suggestion. In the revised manuscript, we have strengthened the validation of the proposed formulation by explicitly considering an appropriate limiting case. Specifically, the corrugation amplitude parameter is set to $\varepsilon = 0$, for which the periodically corrugated interface reduces to a smooth interface and the corrugation-dependent contributions vanish. Consequently, the proposed formulation recovers the corresponding smooth-interface thermoelastic double-porosity Love-wave model. The numerical behavior of this limiting case is also demonstrated through the smooth-interface results and its comparison with the corrugated-interface configuration. The smooth and stable dispersion behavior obtained for $\varepsilon = 0$, together with the expected asymptotic convergence at higher frequencies, provides a consistency check for the developed dispersion formulation. This limiting-case verification has now been explicitly incorporated into the revised manuscript and discussed in connection with Fig. 17.

Location in revised manuscript: Section 8, Numerical Results and Discussion, under a new paragraph/subsection entitled “Validation of the formulation through a limiting case,” and an additional validation statement has been included in the discussion of Fig. 17.

Text added to the manuscript

Validation of the formulation through a limiting case

To verify the consistency of the proposed formulation, a limiting-case analysis is performed by considering $\varepsilon = 0$. Under this condition, the periodically corrugated interface reduces to a smooth interface, and all corrugation-dependent contributions in the dispersion relation vanish. The resulting formulation therefore represents Love-wave propagation in a thermoelastic double-porosity layered medium with a smooth interface. Mathematically, this limiting condition can be represented as

$$\varepsilon \rightarrow 0 \Rightarrow \text{corrugated interface} \rightarrow \text{smooth interface.}$$

The numerical solution corresponding to $\varepsilon = 0$ exhibits a smooth dispersion response and approaches the expected limiting behavior with increasing dimensionless frequency. The successful recovery of the smooth-interface configuration provides a useful limiting-case consistency check for the proposed corrugated-interface formulation.

We have added this text in the Fig. 17 discussion

The smooth-interface result also provides a limiting-case verification of the present formulation, since setting $\varepsilon = 0$ eliminates the corrugation contribution and recovers the corresponding smooth-interface propagation behavior.

6. The practical engineering significance of the proposed model should be discussed in greater detail. The authors should better explain how the present findings may contribute to realistic heat-transfer or thermoelastic engineering applications.

Response:

Thank you for this valuable suggestion. The revised manuscript has been expanded to provide a clearer discussion of the practical engineering significance of the proposed thermoelastic double-porosity Love-wave model. In particular, the numerical findings have been related to engineering and geophysical systems involving layered porous materials, irregular interfaces, coupled thermal transport, and guided-wave propagation. The analysis demonstrates how interface corrugation, double-porosity parameters, thermal interaction, and propagation modes can influence phase velocity, attenuation, stress localization, thermal penetration, and heat-flux behavior. These observations are relevant to the characterization and design of layered porous structures in which mechanical-wave propagation and heat transfer occur simultaneously. The revised discussion

further highlights the potential usefulness of the model for subsurface characterization, non-destructive evaluation of layered porous materials, thermal-barrier and porous composite systems, and guided-wave-based monitoring of thermo-mechanically coupled structures. The corresponding practical implications have now been incorporated into the Numerical Results and Discussion section and further emphasized in the Conclusions.

Location in revised manuscript: Section 8, Numerical Results and Discussion, after the discussion of the final figure. A shorter application statement incorporated into the Conclusion section.

Text added to the manuscript

Engineering significance and practical implications. The present results have practical relevance to layered porous systems in which mechanical-wave propagation occurs simultaneously with thermal transport and pore interactions. The predicted variations in phase velocity and attenuation with corrugation and porosity parameters indicate that Love waves can be sensitive to changes in interface geometry and internal pore structure, providing a potential basis for the characterization of layered porous materials. In addition, the predicted stress localization, temperature decay, and heat-flux behavior provide information on the depth over which mechanical and thermal disturbances remain significant. These characteristics are relevant to subsurface and geophysical characterization, non-destructive evaluation of porous layered structures, thermal-barrier and porous composite systems, and guided-wave monitoring under thermo-mechanical loading. Thus, the proposed model provides a theoretical framework for assessing how interface irregularity, double porosity, and thermal coupling jointly affect wave transmission and heat-transfer characteristics in realistic layered media.

Added in Conclusion

From an engineering perspective, the results provide useful insight into the control of wave transmission, attenuation, stress localization, and thermal penetration in layered porous structures, with potential relevance to geophysical characterization, non-destructive evaluation, and thermo-mechanical monitoring of porous composite systems.

Overall, the manuscript has potential, but substantial revision is required to improve its scientific contribution, physical interpretation, and overall relevance to the journal.

Reviewer 2

In this paper, the authors investigated the “Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures”. I have gone through the entire manuscript. All the equations are systematically presented. The theoretical results are well supplemented by the graphical presentations. The novelty and applications of the present problem are clearly defined. Overall, the manuscript is a good piece of scientific work. It is fit for publication in this journal. However, the authors must incorporate the following points in their manuscript:

1. Abstract should be précised; however, major result can be described in the abstract.

Response: Thank you for this helpful suggestion. The Abstract has been carefully revised and refined to provide a more precise and focused presentation of the study. The revised Abstract now briefly describes the physical model, analytical formulation, principal controlling parameters, and major numerical findings. In particular, it highlights the effects of interface corrugation, porosity, frequency, and propagation modes on phase velocity, attenuation, thermal-wave transport, and energy dissipation. The practical significance of the obtained results for porous thermoelastic layered systems has also been concisely stated. These revisions improve the clarity of the Abstract while ensuring that the principal outcomes of the study are explicitly communicated.

Updated abstract

“The present study examines coupled thermal-wave transport and guided-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. The model consists of a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space, where thermal interactions, pore coupling, and interface irregularities jointly influence energy transport and wave characteristics. Using generalized thermoelastic double-porosity theory, the coupled displacement, temperature, and porosity fields are formulated under anti-plane shear motion, and an analytical dispersion relation for Love-wave propagation is derived by imposing appropriate boundary and interface continuity conditions. Numerical investigations quantify the influences of corrugation amplitude, corrugation wave number, porosity parameters, frequency, and propagation modes on phase velocity, attenuation, stress penetration, temperature, and heat-flux distributions. The results demonstrate that interface corrugation strongly modifies Love-wave dispersion, particularly in the low- and intermediate-frequency regimes, whereas its influence becomes weaker at higher frequencies. Increasing porosity reduces the effective thermoelastic stiffness and consequently lowers the phase velocity. Attenuation increases with frequency, while lower-order modes exhibit stronger attenuation than higher-order modes because of their greater interaction with the coupled thermo-porous region. Thermal analysis further shows that larger thermo-porous decay parameters produce stronger near-interface temperature and heat-flux localization but reduce the effective thermal penetration depth. A smooth-interface limiting case ($\varepsilon = 0$) is recovered, providing a consistency check for the proposed formulation. These findings clarify the coupled roles of interface geometry, double

porosity, and thermal transport in controlling wave dispersion, attenuation, and energy localization, with potential relevance to thermal-barrier materials, porous composites, subsurface characterization, and thermo-mechanical monitoring of layered structures.”

2. What are the advantages of employed technique?

Response to Reviewer

Thank you for this important comment. The advantages of the employed analytical technique have now been clarified in the revised manuscript. The harmonic-wave approach combined with the matrix formulation transforms the coupled thermoelastic double-porosity governing equations into a compact algebraic/eigenvalue framework, allowing the mechanical, thermal, and two porosity fields to be treated simultaneously. The technique enables an explicit dispersion relation to be obtained while retaining the effects of thermal coupling, double porosity, and interface corrugation. Moreover, the first-order treatment of the corrugated boundary provides a computationally efficient way of incorporating small interface irregularities without requiring a full numerical discretization of the entire domain. The resulting formulation also facilitates systematic parametric studies of phase velocity, attenuation, thermal penetration, stress localization, and different propagation modes. These advantages have now been explicitly stated in the revised manuscript.

We have added at the end of Section 6, Dispersion Relation Formulation,

The employed analytical technique offers several advantages for the present coupled problem. The harmonic-wave representation reduces the governing partial differential equations to a tractable system of ordinary differential equations, while the matrix formulation enables the mechanical, thermal, and two porosity fields to be treated within a unified framework. Furthermore, the first-order treatment of the corrugated interface incorporates the influence of small interface irregularities without requiring full-domain numerical discretization. The resulting dispersion formulation is computationally efficient and facilitates systematic evaluation of phase velocity, attenuation, thermal penetration, stress localization, and mode-dependent wave behavior over a wide range of governing parameters.

3. Author should enhance the physical description of the model.

Response: Thank you for this valuable suggestion. The physical description of the model has been strengthened in Section 2. The revised text now explains the physical roles of the two interacting porosity fields, the finite guiding layer and underlying half-space, the corrugation amplitude and wave number, and the anti-plane shear configuration. In addition, the physical influence of interface corrugation on wave-energy redistribution, dispersion, attenuation, and penetration has

been clarified. These additions provide a clearer physical interpretation of the proposed thermoelastic double-porosity layered configuration while maintaining consistency with the mathematical formulation and Fig. 1.

4. The authors should add some more lines in the numerical results and discussion section to strengthens these sections.

Response: Thank you for this valuable suggestion. The Numerical Results and Discussion section has been substantially expanded in the revised manuscript. Additional discussions have been incorporated throughout the section to provide deeper physical interpretation of the numerical results rather than merely describing the trends of the plotted curves. In particular, the revised discussion explains the effects of interface corrugation, propagation mode, porosity parameters, wave confinement, interfacial scattering, thermo-porosity coupling, stress penetration, and energy dissipation on the phase velocity and attenuation characteristics. Furthermore, additional results concerning stress-amplitude penetration, spatial thermo-porosity field intensity, the combined effects of wave number and corrugation amplitude, smooth- versus corrugated-interface behavior, temperature distribution, and heat-flux variation have been included to provide a more comprehensive interpretation of the coupled thermoelastic response. A limiting-case verification for the smooth-interface condition ($\varepsilon = 0$) has also been incorporated. These additions considerably strengthen the numerical analysis and provide clearer physical insight into the coupled thermal-wave and Love-wave propagation mechanisms.

We have thoroughly revised the manuscript, particularly Section 8, “Numerical Results and Discussion,” by expanding the discussions accompanying Figs. 2-19.

5. It is beneficial for the readers to add more explanations about the novel contribution of this manuscript from theoretical/experimental viewpoints.

Response to Reviewer

Thank you for this constructive suggestion. The novel contributions of the present study have been clarified further from both theoretical and practical/experimental viewpoints. From the **theoretical perspective**, the present formulation integrates generalized thermoelasticity, double-porosity effects, and periodic interface corrugation within a unified Love-wave propagation framework. The model simultaneously accounts for the coupled displacement, temperature, and two porosity fields and provides a generalized dispersion relation for examining phase velocity, attenuation, and thermal-wave behavior. In addition, the smooth-interface condition is recovered as a limiting case, providing a consistency check for the formulation.

From the experimental/practical viewpoint, the predicted sensitivity of phase velocity, attenuation, stress penetration, temperature, and heat flux to porosity and interface corrugation provides

measurable quantities that could be used in future experimental characterization of layered porous materials. In particular, changes in guided-wave phase velocity and attenuation may serve as indicators of variations in pore structure or interface geometry, while temperature and heat-flux distributions can provide complementary information on thermo-mechanical coupling. Since the present work is theoretical and numerical, no experimental validation is claimed; rather, the results provide a theoretical basis and specific response quantities that may guide future experimental investigations.

We have added the following paragraph at the end of the Introduction

The novelty of the present study lies in the unified treatment of generalized thermoelasticity, double-porosity coupling, and periodic interface corrugation in Love-wave propagation through a layered medium. Unlike formulations that consider these effects separately, the present model simultaneously captures their influence on dispersion, attenuation, stress penetration, temperature, and heat-flux characteristics. Theoretically, the derived dispersion framework provides a systematic means of quantifying the coupled effects of interface geometry, thermal interaction, and two pore systems, while also recovering the smooth-interface configuration as a limiting case. From an experimental viewpoint, the predicted variations in phase velocity and attenuation provide potentially measurable signatures of changes in porosity and interface geometry, whereas temperature and heat-flux responses offer complementary indicators of thermo-mechanical coupling. Although experimental validation is beyond the scope of the present theoretical study, these predicted quantities provide a basis for future guided-wave and thermal measurements in layered porous materials.

6. There are some misprints that can be corrected in the revised version for running a spell check and making the manuscript better.

Response: Thank you for the comment. We have carefully proofread the entire manuscript and corrected the typographical, spelling, grammatical, and formatting errors throughout the revised version.

7. Please check all the references again and correct them as per journal format.

Response: Thank you for the comment. We have carefully checked and revised all the references to ensure consistency and compliance with the journal's prescribed reference format. The in-text citations and reference list have also been cross-checked and corrected accordingly.

Based on the comprehensive assessment of your paper and after minor revision, I am pleased to recommend it for publication

Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures

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Abstract

The present study examines coupled thermal-wave transport and guided-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. The model consists of a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space, where thermal interactions, pore coupling, and interface irregularities jointly influence energy transport and wave characteristics. Using generalized thermoelastic double-porosity theory, the coupled displacement, temperature, and porosity fields are formulated under anti-plane shear motion, and an analytical dispersion relation for Love-wave propagation is derived by imposing appropriate boundary and interface continuity conditions. Numerical investigations quantify the influences of corrugation amplitude, corrugation wave number, porosity parameters, frequency, and propagation modes on phase velocity, attenuation, stress penetration, temperature, and heat-flux distributions. The results demonstrate that interface corrugation strongly modifies Love-wave dispersion, particularly in the low- and intermediate-frequency regimes, whereas its influence becomes weaker at higher frequencies. Increasing porosity reduces the effective thermoelastic stiffness and consequently lowers the phase velocity. Attenuation increases with frequency, while lower-order modes exhibit stronger attenuation than higher-order modes because of their greater interaction with the coupled thermo-porous region. Thermal analysis further shows that larger thermo-porous decay parameters produce stronger near-interface temperature and heat-flux localization but reduce the effective thermal penetration depth. A smooth-interface limiting case ($\varepsilon = 0$) is recovered, providing a consistency check for the proposed formulation. These findings clarify the coupled roles of interface geometry, double porosity, and thermal transport in controlling wave dispersion, attenuation, and energy localization, with potential relevance to thermal-barrier materials, porous composites, subsurface characterization, and thermo-mechanical monitoring of layered structures.

Keywords: Thermal-wave propagation; Thermoelastic double-porosity media; Corrugated interface; Love waves; Wave attenuation.

1. Introduction

Due to its wide-ranging applications in geophysics, earthquake engineering, subsurface exploration, acoustics, seismology, materials science, and structural engineering, the propagation of elastic waves in layered media has remained an important area of research. Among the various guided elastic waves, Love waves are particularly significant because of their dispersive nature and strong confinement of energy near the guiding surface [1,2]. These characteristics make them useful in the investigation of layered composite structures, geophysical prospecting, subsurface imaging, and seismic interpretation. The classical theory of Love-wave propagation was introduced by Love [1] for horizontally polarized shear waves travelling through layered elastic media, while Graff [2] and Achenbach [4] later established broader theoretical foundations for elastic and guided-wave propagation in solids. More recently, Sharma et al. [3] examined the dispersion and attenuation of Love waves in thermo-porous layered structures with irregular interfaces and demonstrated that interface geometry can significantly modify guided-wave behavior. However, such studies largely emphasize dispersion characteristics and interface effects, while the combined influence of complex pore structures and coupled thermal interactions remains comparatively less explored.

The assumption of perfectly homogeneous and fully elastic materials is often inadequate for realistic engineering and geological systems, which commonly contain pores, cavities, cracks, and heterogeneous microstructures. To account for such features, Cowin and Nunziato [5] developed the linear theory of elastic materials with voids, introducing additional porosity-related field variables into the classical elastic formulation. This framework provided an important basis for studying wave propagation in porous and void-containing media. Subsequent investigations extended these concepts to include thermal effects and generalized thermoelastic interactions [6-13]. Nevertheless, most early porous-medium models were based on a single void or porosity field and therefore could not fully represent materials containing multiple interconnected pore networks.

Thermoelasticity provides a more realistic framework for wave propagation in systems where mechanical deformation is accompanied by temperature variation. The classical thermoelastic theories associated with Biot [9] and Nowacki [10] successfully described coupled thermo-mechanical interactions, but they retained the physically unrealistic assumption of an infinite thermal-wave propagation speed. Lord and Shulman [8] addressed this limitation by introducing thermal relaxation time, thereby enabling finite-speed thermal-wave propagation. This generalized thermoelastic formulation significantly improved the physical description of high-frequency and transient thermo-mechanical processes. However, even within generalized thermoelasticity, many formulations focus primarily on homogeneous or single-porosity media and do not simultaneously account for complex pore interactions and interface irregularities.

Further developments in generalized thermoelasticity were proposed by Green and Naghdi [13], who introduced a thermoelastic theory without energy dissipation, while Forest and Amestoy [12]

examined hypertemperature effects and highlighted the importance of higher-order thermal interactions in wave propagation. Ieşan [6] developed a theory of thermoviscoelastic materials with voids, and Abd-Alla et al. [11] investigated surface-wave propagation in generalized thermoelastic porous media with irregular boundaries, showing that thermal interactions and interface geometry can strongly influence wave characteristics. Collectively, these studies confirmed that thermoelastic coupling plays an important role in phase velocity, attenuation, and energy localization. However, the existing formulations generally treat thermal effects, porosity, and interface irregularity separately or within simplified configurations, leaving the simultaneous interaction of these mechanisms insufficiently understood in layered media.

Since many natural and engineered materials contain multiple pore systems and heterogeneous internal configurations, thermoelastic double-porosity media have received increasing attention. Typical examples include fractured rocks, petroleum reservoirs, concrete, soil formations, biological tissues, and porous composites, where macro- and micro-pore networks may coexist and interact. Such pore structures can significantly modify the effective thermo-mechanical response of the medium and consequently influence wave dispersion, attenuation, and energy dissipation. [14] investigated Love-type waves in thermoelastic double-porosity media and demonstrated that thermo-porosity coupling considerably alters phase velocity and attenuation. Earlier contributions by [15] established governing relations for displacement waves in saturated thermoelastic porous media, while [16] examined plane-wave propagation in thermoelastic materials with voids and highlighted the influence of thermal interactions and porosity on dispersion characteristics. Although these studies established the importance of pore-related coupling, the role of complex interface geometry in modifying the interaction between thermal and multiple porosity fields has received comparatively limited attention.

Material heterogeneity introduces an additional level of complexity into thermoelastic wave propagation. [17] demonstrated that inhomogeneity can substantially modify wave-propagation characteristics, whereas [18] showed that thermal relaxation parameters significantly influence phase velocity and attenuation in porous thermoelastic media. These contributions confirmed that porosity, thermal relaxation, and material heterogeneity are important controlling mechanisms. However, these effects have generally been investigated separately, making it difficult to determine their combined influence when heterogeneous pore structures and nonplanar interfaces coexist in a layered thermoelastic system.

Love-wave propagation in porous and thermoelastic layered media has consequently been investigated under a variety of physical configurations. [19] considered a transversely isotropic fluid-saturated porous layered half-space and demonstrated the importance of fluid saturation and porosity in controlling wave dispersion. [20] examined Love-wave propagation in heterogeneous media under point-source excitation, emphasizing the role of medium inhomogeneity, while Singh and Sharma [21] showed that thermal effects and voids can substantially alter phase velocity and attenuation in layered thermoelastic media. [22] further demonstrated the importance of porous heterogeneity in attenuation and dispersion through dynamic fluid-modulus effects.

Chattopadhyay et al. [23] investigated Love-type waves in multilayered elastic structures, whereas Saini and Kumar [24] examined thermoelastic layered media over a nonhomogeneous elastic half-space. Collectively, these investigations establish that porosity, heterogeneity, thermal interaction, and layered geometry can independently or jointly modify guided-wave characteristics. Nevertheless, most of these models employ either conventional porous descriptions or relatively idealized interfaces and therefore do not fully capture the combined effects of two interacting pore systems and interface corrugation on coupled thermoelastic wave transport. A comprehensive framework for materials containing two interacting pore networks was developed by [25] through the generalized theory of thermoelastic materials with double porosity. Subsequent studies showed that interface irregularity [26] and thermal relaxation [27] significantly influence wave propagation and attenuation. However, the combined effects of thermoelastic double porosity, thermal interaction, and periodic interface corrugation on Love-wave dispersion, attenuation, and energy localization remain insufficiently explored.

Recent studies have further examined wave propagation in heterogeneous layered and double-porosity media. Yin et al. [28] demonstrated that porosity interaction significantly affects seismic reflection and transmission in layered double-porosity media, while Li et al. [29] highlighted the role of magnetic coupling in SH-wave scattering in piezomagnetic materials. Kumar et al. [30] showed that interface irregularity strongly influences wave localization and attenuation in thermoelastic porous layered media. Similarly, [31] investigated total internal reflection of shear waves at heterogeneous interfaces, whereas [32] demonstrated that material inhomogeneity considerably affects dynamic stress concentration and SH-wave behavior. Berryman and Wang [33] further established the importance of pore interaction in controlling attenuation and energy dissipation in double-porosity dual-permeability media. Although these studies provide valuable insight into porosity, heterogeneity, and interface effects, these mechanisms have largely been examined under separate physical configurations rather than within a unified thermoelastic double-porosity framework.

Further developments in thermoelastic porous media have reinforced the importance of thermal and pore-related interactions. Sharma and Grover [34] investigated generalized thermoelastic interactions in porous elastic solids, while Puri and Cowin [35] demonstrated the influence of void parameters on plane-wave propagation. The classical formulation of Love [36] continues to provide the fundamental basis for guided-wave analysis in layered structures. More recently, Liu et al. [37] examined guided SH-wave dispersion in thermo-poroelastic layered structures with corrugated surfaces and demonstrated that interface corrugation can significantly modify phase velocity, attenuation, and wave localization. However, the simultaneous interaction of periodic interface corrugation with two distinct porosity fields and coupled thermal effects remains comparatively less explored.

Despite these developments, important limitations remain in the existing literature. Conventional porous or void-based models generally employ a single porosity field and therefore cannot fully represent materials containing interacting macro- and micro-pore networks. Double-porosity

formulations provide a more realistic representation of heterogeneous porous media, but the influence of interface corrugation on coupled thermal and pore-field interactions has received limited attention. Conversely, studies involving irregular or corrugated interfaces have primarily emphasized mechanical dispersion, scattering, and attenuation, with comparatively less attention to coupled thermal transport and dual-porosity interactions.

Therefore, the combined influence of thermoelastic double-porosity coupling, periodic interface corrugation, thermal interaction, and propagation mode on Love-wave behavior remains insufficiently understood. In particular, their simultaneous effects on dispersion, attenuation, energy localization, and stress penetration have not been systematically characterized within a single analytical framework. This research gap motivates the development of a unified model capable of describing the coupled thermal, mechanical, and double-porosity response of layered structures with periodically corrugated interfaces.

Although thermoelastic wave propagation, double-porosity effects, and interface irregularities have individually received considerable attention, their combined influence on coupled thermal-wave and Love-wave transport in a layered medium remains insufficiently understood. Most existing studies have focused either on thermoelastic porous media with regular interfaces, elastic or porous media with interface irregularities, or double-porosity systems without explicitly examining the combined role of thermal interaction and periodic corrugation. Consequently, the manner in which a corrugated interface modifies the interaction among thermal fields, two interconnected porosity systems, guided-wave dispersion, attenuation, and energy localization has not been adequately clarified.

The novelty of the present study therefore lies in developing a unified analytical framework for a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space through a periodically corrugated interface. Unlike conventional Love-wave models, the present formulation simultaneously incorporates thermoelastic coupling, two porosity fields, interface corrugation, and mode-dependent wave behavior. The formulation further provides analytical dispersion relations together with numerical assessments of phase velocity, attenuation, stress penetration, and coupled thermo-porosity field localization. This framework enables the effects of thermal interaction, pore coupling, and interface geometry on energy transport and guided-wave behavior to be examined within a single model.

Inspired by these findings, the current work examines Love wave propagation properties in a thermoelastic double-porosity layered media with a regularly corrugated surface. Generalized thermoelastic double-porosity theory is used to formulate the governing equations. The coupled field equations are lowered by taking into account harmonic wave assumptions and anti-plane shear motion, and a secular dispersion relation governing Love-wave propagation is analytically constructed using suitable boundary and continuity conditions. The effects of corrugation amplitude, corrugation wave number, porosity parameters, thermal interaction, and propagation modes on the phase velocity and attenuation behavior of Love waves are investigated numerically.

Additionally, wave localization, energy dissipation, and the combined impact of thermo-porosity coupling and interface irregularity within the considered thermoelastic layered structure are investigated using stress penetration analysis, heatmap visualization, and three-dimensional contour representations.

The novelty of the present study lies in the unified treatment of generalized thermoelasticity, double-porosity coupling, and periodic interface corrugation in Love-wave propagation through a layered medium. Unlike formulations that consider these effects separately, the present model simultaneously captures their influence on dispersion, attenuation, stress penetration, temperature, and heat-flux characteristics. Theoretically, the derived dispersion framework provides a systematic means of quantifying the coupled effects of interface geometry, thermal interaction, and two pore systems, while also recovering the smooth-interface configuration as a limiting case. From an experimental viewpoint, the predicted variations in phase velocity and attenuation provide potentially measurable signatures of changes in porosity and interface geometry, whereas temperature and heat-flux responses offer complementary indicators of thermo-mechanical coupling. Although experimental validation is beyond the scope of the present theoretical study, these predicted quantities provide a basis for future guided-wave and thermal measurements in layered porous materials.

2. Physical Configuration and Mathematical Formulation

To develop the mathematical formulation of the present study, a Cartesian coordinate system (x, y, z) is considered, where the x -axis is taken along the direction of propagation of the Love-type surface wave, while the z -axis is measured vertically downward into the medium. The physical model consists of an isotropic thermoelastic layer of finite thickness perfectly bonded to an isotropic thermoelastic semi-infinite substrate. Both regions are assumed to exhibit double-porosity characteristics, although their constitutive and material parameters may differ. Physically, the two porosity fields represent interacting pore systems that are coupled with the elastic deformation and temperature field, allowing the simultaneous mechanical, thermal, and pore responses of the layered structure to be considered.

The upper layer and the underlying half-space are represented by the regions L_1 and L_2 , respectively, defined as $L_1 = \{(x, y, z): -h \leq z \leq \delta \cos(qx)\}$, and $L_2 = \{(x, y, z): z \geq \delta \cos(qx)\}$. The common interface separating the two media is considered nonplanar and periodically corrugated. Its profile is expressed as

$$z = \delta \cos(qx), \quad (1)$$

where δ denotes the corrugation amplitude parameter and q represents the corrugation wave number associated with the interfacial irregularity. Thus, δ controls the magnitude of the interface irregularity, while q determines its spatial periodicity along the direction of wave propagation.

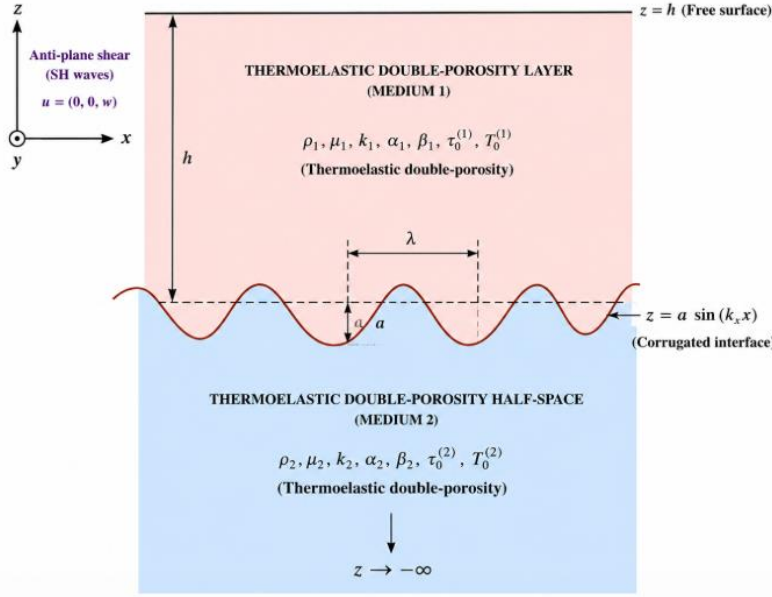


Figure 1. Schematic geometry of the corrugated thermoelastic layered medium

Fig. 1 presents the geometrical representation of the discussed problem. The physical model comprises an isotropic homogeneous thermoelastic double-porosity layer L_1 of thickness h which is placed over an isotropic homogeneous thermoelastic double-porosity half-space L_2 . Both media are considered homogeneous and isotropic, although their material properties are different. The upper layer acts as the Love-wave guiding region, whereas the underlying half-space supports the depth-wise decay of the associated mechanical, thermal, and porosity disturbances.

A Cartesian coordinate system (x, y, z) is introduced, where the x -axis is taken along the direction of Love wave propagation, while the z -axis is directed vertically downward into the medium. The common interface between the two media is assumed to be periodically corrugated and is expressed by the relation $z = \delta \cos(qx)$, where δ denotes the corrugation amplitude and q represents the corrugation wave number. The corrugated interface plays a significant role in the propagation characteristics of Love waves by modifying the dispersion and attenuation behavior of the thermoelastic double-porosity structure. The periodic interface modifies the local wave-guiding conditions and promotes redistribution of wave energy through the coupled mechanical, thermal, and porosity fields, thereby influencing dispersion, attenuation, and penetration characteristics.

Since the present investigation is confined to the two-dimensional (x, z) -plane, all field variables are assumed to be independent of the y -coordinate. Therefore, only the anti-plane displacement component is taken into consideration. This anti-plane configuration is consistent with Love-wave motion, in which the displacement is polarized perpendicular to the direction of propagation. The displacement field is denoted by w , whereas the porosity variables corresponding to the first and

second void systems are represented by φ and χ , respectively. The temperature increment is represented by ϑ . Hence, the field quantities are assumed in the following form:

$$\mathbf{u} = (0, w, 0)(x, z, t), \quad \{\varphi, \chi, \vartheta\} = \{\varphi, \chi, \vartheta\}(x, z, t). \quad (2)$$

3. Governing Mathematical Formulation

Following the generalized thermoelastic double-porosity theory developed by Ieşan and Quintanilla [8], the governing equations for both media, in the absence of body forces, are expressed as follows.

Dynamic Field Equations

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) - \beta \nabla \Theta = \rho \ddot{\mathbf{u}}, \quad (3)$$

$$\alpha \nabla^2 \phi + b_1 \nabla^2 \psi - b \nabla \cdot \mathbf{u} - \alpha_1 \phi - \alpha_3 \psi + \gamma_1 \Theta = \chi_1 \ddot{\phi}, \quad (4)$$

$$b_1 \nabla^2 \phi + \gamma \nabla^2 \psi - d \nabla \cdot \mathbf{u} - \alpha_3 \phi - \alpha_2 \psi + \gamma_2 \Theta = \chi_2 \ddot{\psi}, \quad (5)$$

$$K \nabla^2 \Theta - \beta T_0 \nabla \cdot \ddot{\mathbf{u}} - \gamma_1 T_0 \ddot{\phi} - \gamma_2 T_0 \ddot{\psi} - a T_0 \dot{\Theta} = 0. \quad (6)$$

Constitutive and Coupled Relations

The constitutive equations governing the thermoelastic and porosity interactions are written as

$$\sigma_{ij} = 2\mu e_{ij} + (\lambda e_{rr} + b\phi + d\psi - \beta\Theta)\delta_{ij}, \quad (7a)$$

$$\sigma_i^{(1)} = \alpha\phi_{,i} + b_1\psi_{,i}, \sigma_i^{(2)} = b_1\phi_{,i} + \gamma\psi_{,i}, \quad (7b)$$

$$\xi = -be_{jj} - \alpha_1\phi - \alpha_3\psi + \gamma_1\Theta, \zeta = -de_{jj} - \alpha_3\phi - \alpha_2\psi + \gamma_2\Theta, \quad (7c)$$

$$\rho\eta = \beta e_{jj} + \gamma_1\phi + \gamma_2\psi + a\Theta, Q_i = K\Theta_{,i}. \quad (7d)$$

Description of Material Parameters

Here, Θ denotes the temperature variation measured relative to the reference temperature T_0 , and the strain tensor is defined by $e_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$. The parameter K represents the thermal conductivity, while a denotes the specific heat parameter and ρ is the density of the material. The coefficients χ_1 and χ_2 correspond to the equilibrated inertia associated with the two void systems. Moreover, the thermoelastic coupling parameter is given by $\beta = 3\lambda + 2\mu$, where λ and μ are the Lamé elastic constants. The parameters b , d , b_1 , and α_3 characterize the coupling between the elastic and porous fields, whereas α_1 , α_2 , γ_1 , and γ_2 are constitutive coefficients associated with porosity and thermal interactions, respectively [8].

Substituting the assumed displacement representation given in equation (2) into the governing equations (3)-(6), the coupled system reduces to the following simplified form:

$$\mu \nabla^2 u = \rho \frac{\partial^2 u}{\partial t^2}, \quad (8)$$

$$\nabla^2(\alpha\phi + b_1\psi) - \alpha_1\phi - \alpha_3\psi + \gamma_1\Theta = \chi_1 \frac{\partial^2 \phi}{\partial t^2}, \quad (9)$$

$$\nabla^2(b_1\phi + \gamma\psi) - \alpha_3\phi - \alpha_2\psi + \gamma_2\Theta = \chi_2 \frac{\partial^2 \psi}{\partial t^2}, \quad (10)$$

$$K \nabla^2 \Theta - \gamma_1 T_0 \frac{\partial \phi}{\partial t} - \gamma_2 T_0 \frac{\partial \psi}{\partial t} - c_v \frac{\partial \Theta}{\partial t} = 0, \quad (11)$$

Where

$$c_v = aT_0.$$

These reduced equations describe the coupled propagation behavior of the displacement, porosity, and thermal fields within the thermoelastic double-porosity medium.

4. Harmonic Wave Solution

To analyze the propagation behavior of Love waves, the field variables are assumed to vary harmonically with space and time. Accordingly, the displacement, porosity, and temperature fields are expressed in the form

$$\{u, \phi, \psi, \Theta\}(x, z, t) = \{\bar{u}, \bar{\phi}, \bar{\psi}, \bar{\Theta}\}(z) e^{i(kx - \omega t)}, \quad (12)$$

where ω denotes the angular frequency and k represents the wave number.

Substituting the harmonic wave representation given in equation (12) into the reduced governing equations (8)-(11), the coupled partial differential equations transform into the following ordinary differential system:

$$\left(\frac{d^2}{dz^2} - k^2 + \frac{\omega^2}{c_t^2} \right) \bar{u} = 0, \quad (13)$$

$$\alpha \frac{d^2 \bar{\phi}}{dz^2} + b_1 \frac{d^2 \bar{\psi}}{dz^2} + (\omega^2 \chi_1 - \alpha_1 - \alpha k^2) \bar{\phi} + (-\alpha_3 - b_1 k^2) \bar{\psi} + \gamma_1 \bar{\Theta} = 0, \quad (14a)$$

$$b_1 \frac{d^2 \bar{\phi}}{dz^2} + \gamma \frac{d^2 \bar{\psi}}{dz^2} + (-\alpha_3 - b_1 k^2) \bar{\phi} + (\omega^2 \chi_2 - \alpha_2 - \gamma k^2) \bar{\psi} + \gamma_2 \bar{\Theta} = 0, \quad (14b)$$

$$K \left(\frac{d^2 \bar{\Theta}}{dz^2} - k^2 \bar{\Theta} \right) - \gamma_1 T_0 \dot{\bar{\phi}} - \gamma_2 T_0 \dot{\bar{\psi}} - c_v \dot{\bar{\Theta}} = 0. \quad (14c)$$

Introducing the shear-wave velocity $c_t = \sqrt{\frac{\mu}{\rho}}$, and using the relation $\omega = ck$, equation (13) can be rewritten as

$$\frac{d^2 \bar{u}^j}{dz^2} - k^2 \left(1 - \frac{c^2}{c_t^2} \right) \bar{u}^j = 0. \quad (15)$$

The above equations govern the propagation of coupled thermoelastic Love waves in the double-porosity layered structure.

The displacement field corresponding to the finite layer (L_1) and the underlying half-space (L_2) may be expressed in the following form:

$$\bar{u}^{(1)}(z) = G_1^{(1)} \cos(R_1 z) + G_2^{(1)} \sin(R_1 z), \quad (16)$$

$$\bar{u}^{(2)}(z) = G_1^{(2)} e^{-R_2 z}, \quad (17)$$

where $G_1^{(j)}$ and $G_2^{(1)}$ are arbitrary constants. The quantities R_1 and R_2 are given by $R_1 = k \sqrt{\frac{c^2}{c_{t1}^2} - 1}$, $R_2 = k \sqrt{1 - \frac{c^2}{c_{t2}^2}}$. To ensure bounded wave propagation in the half-space region, the exponentially increasing solution associated with equation (17) is omitted.

Further, equations (14a)-(14c) can be rewritten in a more compact form as

$$\alpha \bar{\phi}'' + b_1 \bar{\psi}'' - Q_{11} \bar{\phi} - Q_{12} \bar{\psi} + Q_{1\theta} \bar{\theta} = 0, \quad (18a)$$

$$b_1 \bar{\phi}'' + \gamma \bar{\psi}'' - Q_{21} \bar{\phi} - Q_{22} \bar{\psi} + Q_{2\theta} \bar{\theta} = 0, \quad (18b)$$

$$K \bar{\theta}'' + Q_{3\phi} \bar{\phi} + Q_{3\psi} \bar{\psi} - Q_{33} \bar{\theta} = 0, \quad (18c)$$

where

$$\begin{aligned} Q_{11} &= \alpha_1 + \alpha k^2 - \omega^2 \chi_1, Q_{12} = \alpha_3 + b_1 k^2, Q_{1\theta} = \gamma_1, \\ Q_{21} &= \alpha_3 + b_1 k^2, Q_{22} = \alpha_2 + \gamma k^2 - \omega^2 \chi_2, Q_{2\theta} = \gamma_2, \\ Q_{3\phi} &= i\omega \gamma_1 T_0, Q_{3\psi} = i\omega \gamma_2 T_0, Q_{33} = K k^2 - i\omega c_v. \end{aligned}$$

The above coupled system may alternatively be represented in matrix form as

$$L \begin{pmatrix} \bar{\phi} \\ \bar{\psi} \\ \bar{\theta} \end{pmatrix} = 0, \quad (19a)$$

where

$$L = \begin{pmatrix} \alpha \frac{d^2}{dz^2} - Q_{11} & b_1 \frac{d^2}{dz^2} - Q_{12} & Q_{1\theta} \\ b_1 \frac{d^2}{dz^2} - Q_{21} & \gamma \frac{d^2}{dz^2} - Q_{22} & Q_{2\theta} \\ Q_{3\phi} & Q_{3\psi} & K \frac{d^2}{dz^2} - Q_{33} \end{pmatrix}. \quad (19b)$$

The matrix approach gives a concise representation of the coupled thermoelastic double-porosity wave system and assists in deriving the dispersion relation for Love-wave propagation in the considered layered structure.

For the existence of a nontrivial solution in $\bar{\phi}$, the variables $\bar{\psi}$ and $\bar{\Theta}$ are eliminated from equations (18a)-(18b). Consequently, the determinant of the coefficient matrix L must vanish, i.e.,

$$\det L = 0.$$

This condition yields the following governing differential equation:

$$\sum_{n=0}^3 (-1)^n E_n \nabla^{2(3-n)} \bar{\phi} = 0, \quad (20)$$

where the coefficients E_n are defined as

$$\begin{aligned} E_0 &= K(\alpha\gamma - b_1^2), \\ E_1 &= -[K(\alpha Q_{22} + \gamma Q_{11} - 2b_1 Q_{21}) + Q_{33}(\alpha\gamma - b_1^2)] + [Q_{2\Theta}(b_1 Q_{3\phi} - \alpha Q_{3\psi}) + Q_{1\Theta}(b_1 Q_{3\psi} \\ &\quad - \gamma Q_{3\phi})], \\ E_2 &= K(Q_{11}Q_{22} - Q_{12}Q_{21}) + Q_{33}(\alpha Q_{22} + \gamma Q_{11} - 2b_1 Q_{21}) \\ &\quad - Q_{1\Theta}(Q_{3\phi}Q_{22} - Q_{3\psi}Q_{21}) + Q_{2\Theta}(Q_{3\psi}Q_{11} - Q_{3\phi}Q_{12}), \\ E_3 &= -Q_{33}(Q_{11}Q_{22} - Q_{12}Q_{21}) + Q_{1\Theta}(Q_{3\phi}Q_{22} - Q_{3\psi}Q_{21}) \\ &\quad - Q_{2\Theta}(Q_{3\psi}Q_{11} - Q_{3\phi}Q_{12}). \end{aligned}$$

Since the field variables are mutually coupled through equations (18e)-(18g), the variables $\bar{\psi}$ and $\bar{\Theta}$ can be expressed in terms of $\bar{\phi}$ as

$$\bar{\psi} = \sum_{j=1}^3 \xi_{2j} \bar{\phi}_j, \quad \bar{\Theta} = \sum_{j=1}^3 \xi_{3j} \bar{\phi}_j, \quad (21)$$

where

$$\begin{aligned} \xi_{2j} &= \frac{Q_{1\Theta}p_j - m_j Q_{2\Theta}}{n_j Q_{2\Theta} - Q_{1\Theta}q_j}, \\ \xi_{3j} &= \frac{m_j q_j - n_j p_j}{n_j Q_{2\Theta} - Q_{1\Theta}q_j}, \end{aligned}$$

with

$$\begin{aligned} m_j &= \alpha\mu_j - Q_{11}, \quad n_j = b_1\mu_j - Q_{12}, \\ p_j &= b_1\mu_j - Q_{21}, \quad q_j = \gamma\mu_j - Q_{22}, \\ r_j &= K\mu_j - Q_{33}. \end{aligned}$$

Equation (20) gives rise to three distinct wave modes, which may be written as $\bar{\phi} = \sum_{j=1}^3 \phi_j$, where each mode satisfies

$$\left(\frac{d^2}{dz^2} - k^2 + \frac{\omega^2}{c_j^2}\right)\phi_j = 0. \quad (22)$$

Hence, the complete solution corresponding to equations (20)-(21) can be expressed as

$$\{\phi, \psi, \Theta\} = \sum_{j=1}^3 \{1, \xi_{2j}, \xi_{3j}\} (U_j e^{S_j z} + V_j e^{-S_j z}) e^{i(kx - \omega t)}, \quad (23)$$

where U_j and V_j are arbitrary constants, $\omega = ck$, $S_j = k \sqrt{1 - \frac{c^2}{c_j^2}}$. The above expressions describe the coupled thermo-porous wave modes propagating through the considered layered structure.

5. Interface and Boundary Constraints

Since the governing equations are formulated for a bounded layered medium, suitable boundary conditions must be imposed to characterize the coupled stress, displacement, porosity, and thermal behavior at the boundaries of the structure.

1. The upper surface of the layer L_1 , located at $z = -H$, is assumed to be traction-free. Therefore, the shear stress component satisfies

$$\sigma_{23} = 0 \text{ at } z = -H. \quad (24a)$$

Further, the upper boundary is considered thermally insulated, which leads to

$$\frac{\partial \Theta}{\partial z} = 0 \text{ at } z = -H. \quad (24b)$$

2. The regions L_1 and L_2 are assumed to remain in perfect mechanical and thermal contact. Consequently, the following continuity conditions are imposed at the irregular interface

$$z = \varepsilon \sin(kx).$$

Continuity of displacement field

$$u^{(1)} = u^{(2)},$$

$$u^{(1)}(0) + \varepsilon \sin(kx) \frac{\partial u^{(1)}}{\partial z} = u^{(2)}(0) + \varepsilon \sin(kx) \frac{\partial u^{(2)}}{\partial z}. \quad (25a)$$

Continuity of stress field

$$\begin{aligned} \sigma_{23}^{(1)} &= \sigma_{23}^{(2)}, \\ \sigma_{32}^{(1)} + \varepsilon \sin(kx) \frac{\partial \sigma_{32}^{(1)}}{\partial z} &= \sigma_{32}^{(2)} + \varepsilon \sin(kx) \frac{\partial \sigma_{32}^{(2)}}{\partial z}. \end{aligned} \quad (25b)$$

Continuity of porosity fields

$$\phi^{(1)} = \phi^{(2)}, \psi^{(1)} = \psi^{(2)}. \quad (25c)$$

Continuity of thermal field

$$\theta^{(1)} = \theta^{(2)}. \quad (25d)$$

Since the interface geometry is irregular and described by $z = \varepsilon \sin(kx)$, $\varepsilon \ll 1$, the boundary conditions are expanded about the mean interface position $z = 0$ by applying the Taylor series approximation. Higher-order terms associated with the irregularity parameter are neglected due to the smallness of ε .

Thus, for any field quantity A , the approximation becomes $A(z = \varepsilon \sin(kx)) \approx A(0) + \varepsilon \sin(kx) \frac{\partial A}{\partial z}$. These boundary conditions govern the interaction of the coupled thermoelastic and porosity fields across the corrugated interface separating the two media.

6. Dispersion Relation Formulation

Substituting the solutions obtained from equations (16)-(17) and (23) into the boundary conditions given by equations (24a)-(25d), a system of linear algebraic equations involving the unknown constants E_j , F_j , and G_j , where $j = 1, 2, 3$, is obtained. The resulting system may be expressed in compact matrix form as

$$(E + \varepsilon E_0)X = 0, \quad (26a)$$

where

$$X = \left[G_1^{(1)} \ G_2^{(1)} \ G_1^{(2)} \ E_1^{(1)} \ E_2^{(1)} \ E_3^{(1)} \ F_1^{(1)} \ F_2^{(1)} \ F_3^{(1)} \ F_1^{(2)} \ F_2^{(2)} \ F_3^{(2)} \right]^T,$$

and E is a 12×12 coefficient matrix defined by

$$E = \begin{bmatrix} F_{3 \times 3} & 0 \\ 0 & H_{9 \times 9} \end{bmatrix}, E_0 = \begin{bmatrix} F_{0 \ 3 \times 3} & 0 \\ 0 & H_{0 \ 9 \times 9} \end{bmatrix}.$$

The matrix F is given by

$$F = \begin{bmatrix} R_1 e^{-R_1 H} & -R_1 e^{R_1 H} & 0 \\ \mu_1 R_1 & -\mu_1 R_1 & \mu_2 R_2 \\ 1 & 1 & -1 \end{bmatrix}, \quad (26b)$$

while

$$F_0 = \begin{bmatrix} R_1 & -R_1 & R_2 \\ \mu_1 R_1^2 & \mu_1 R_1^2 & \mu_2 R_2^2 \\ 0 & 0 & 0 \end{bmatrix}. \quad (26c)$$

The above matrix representation characterizes the coupled thermoelastic and porosity interactions responsible for Love-wave propagation in the layered structure with a corrugated interface.

The entries of the coupled-field block H are generated using the thermo-porous wave numbers $S_j^{(i)}$ together with the coupling parameters $H_j^{(i)}$ and $B_j^{(i)}$, and are written as

$$\begin{aligned} J_{1,j} &= H_j^{(1)} S_j^{(1)} e^{-S_j^{(1)} H}, J_{1,j+3} = -H_j^{(1)} S_j^{(1)} e^{S_j^{(1)} H}, \\ J_{2,j} &= B_j^{(1)} S_j^{(1)} e^{-S_j^{(1)} H}, J_{2,j+3} = -B_j^{(1)} S_j^{(1)} e^{S_j^{(1)} H}, \\ J_{3,j} &= H_j^{(1)} S_j^{(1)}, J_{3,j+3} = -H_j^{(1)} S_j^{(1)}, J_{3,j+6} = H_j^{(2)} S_j^{(2)}, \\ J_{4,j} &= B_j^{(1)} S_j^{(1)}, J_{4,j+3} = -B_j^{(1)} S_j^{(1)}, J_{4,j+6} = B_j^{(2)} S_j^{(2)}, \\ J_{5,j} &= J_{5,j+3} = -J_{5,j+6} = 1, \\ J_{6,j} &= J_{6,j+3} = \xi_{2j}^{(1)}, J_{6,j+6} = -\xi_{2j}^{(2)}, \\ J_{7,j} &= J_{7,j+3} = \xi_{3j}^{(1)}, J_{7,j+6} = -\xi_{3j}^{(2)}, \\ J_{8,j} &= -J_{8,j+3} = K^{(1)} S_j^{(1)} \xi_{3j}^{(1)}, \\ J_{8,j+6} &= K^{(2)} S_j^{(2)} \xi_{3j}^{(2)}, \\ J_{9,j} &= \xi_{3j}^{(1)} S_j^{(1)} e^{-S_j^{(1)} H}, J_{9,j+3} = -\xi_{3j}^{(1)} S_j^{(1)} e^{S_j^{(1)} H}, \end{aligned}$$

where

$$H_j^{(i)} = \alpha^{(i)} + b_1^{(i)} \xi_{2j}^{(i)},$$

and

$$B_j^{(i)} = b_1^{(i)} + \gamma^{(i)} \xi_{2j}^{(i)}.$$

Similarly, the coupled-field block H_0 takes the form

$$\begin{aligned} J_{1,j}^{(1)} &= S_j^{(1)}, J_{1,j+3}^{(1)} = -S_j^{(1)}, J_{1,j+6}^{(1)} = S_j^{(2)}, \\ J_{2,j}^{(1)} &= \xi_{2j}^{(1)} S_j^{(1)}, J_{2,j+3}^{(1)} = -\xi_{2j}^{(1)} S_j^{(1)}, J_{2,j+6}^{(1)} = \xi_{2j}^{(2)} S_j^{(2)}, \\ J_{3,j}^{(1)} &= \xi_{3j}^{(1)} S_j^{(1)}, J_{3,j+3}^{(1)} = -\xi_{3j}^{(1)} S_j^{(1)}, J_{3,j+6}^{(1)} = \xi_{3j}^{(2)} S_j^{(2)}, \\ J_{4,j}^{(1)} &= K^{(1)} S_j^{(1)2}, J_{4,j+6}^{(1)} = K^{(2)} S_j^{(2)2}. \end{aligned}$$

For a nontrivial solution of equation (26a) to exist, the determinant of the matrix E must vanish. Therefore,

$$|E^{(0)} + \varepsilon E^{(1)}| = 0. \quad (27)$$

Expanding the determinant in equation (27) and neglecting the terms containing second and higher powers of ε ($\varepsilon \ll 1$), one obtains

$$|E^{(0)}| + \varepsilon \frac{\partial |E|}{\partial \varepsilon} = 0, \quad (28a)$$

which further reduces to

$$|E| = |F| \cdot |J| = 0. \quad (28b)$$

By setting $|F| = 0$, and using the elements of matrix F from equation (26b), the governing relation corresponding to the classical mechanical wave front is obtained as $\mu_1 R_1 (e^{-R_1 H} - e^{R_1 H}) = \mu_2 R_2 (e^{-R_1 H} + e^{R_1 H})$. Substituting the expressions for R_1 and R_2 , the dispersion relation for the first Love-wave front becomes

$$\tan \left(kH \sqrt{\frac{c^2}{c_{t1}^2} - 1} \right) = \frac{\mu_2 \sqrt{1 - \frac{c^2}{c_{t2}^2}}}{\mu_1 \sqrt{\frac{c^2}{c_{t1}^2} - 1}}. \quad (29)$$

Equation (29) coincides with the classical dispersion relation for Love waves [36]. This wave mode remains unaffected by the porosity parameters, thermal field, and interface corrugation effects.

Furthermore, from equation (26a), the dispersion relation associated with the second wave front is obtained as

$$|J| + \varepsilon J_0 = 0, \quad (30)$$

where the perturbation term εJ_0 arises due to the first-order correction associated with the irregular boundary conditions and depends explicitly on the vertical wave numbers S_j and the thermoporous coupling coefficients.

After simplification, the explicit dispersion equation governing the second Love-wave front can be written as

$$\det[J(c, k, \omega)] + \varepsilon k \mathcal{F}(\omega, k, S_j, R, \xi_{2j}, \xi_{3j}) = 0. \quad (31a)$$

where the irregularity correction function $\mathcal{F}$ is written as $\mathcal{F} = \sum_{j=1}^3 [\mathcal{F}_j^{(\phi)} + \mathcal{F}_j^{(\psi)} + \mathcal{F}_j^{(\Theta)} + \mathcal{F}_j^{(\text{mech})}]$, and the individual components are defined as follows.

Void coupling (first kind)

$$\mathcal{F}_j^{(\phi)} = \frac{S_j^{(1)} - S_j^{(2)}}{S_j^{(1)} + S_j^{(2)}} (H_j^{(1)} S_j^{(1)} + H_j^{(2)} S_j^{(2)}), \quad (31b)$$

with

$$H_j^{(l)} = \alpha^{(l)} + b_1^{(l)} \xi_{2j}^{(l)}.$$

Void coupling (second kind)

$$\mathcal{F}_j^{(\psi)} = \frac{S_j^{(1)} - S_j^{(2)}}{S_j^{(1)} + S_j^{(2)}} (B_j^{(1)} S_j^{(1)} + B_j^{(2)} S_j^{(2)}) \cdot \frac{\xi_{2j}^{(1)} + \xi_{2j}^{(2)}}{2}, \quad (31c)$$

where

$$B_j^{(l)} = b_1^{(l)} + \gamma^{(l)} \xi_{2j}^{(l)}, l = 1, 2.$$

Thermal coupling

$$\mathcal{F}_j^{(\Theta)} = \frac{S_j^{(1)} - S_j^{(2)}}{S_j^{(1)} + S_j^{(2)}} (K_1 S_j^{(1)} \xi_{3j}^{(1)} + K_2 S_j^{(2)} \xi_{3j}^{(2)}) \frac{\xi_{3j}^{(1)} + \xi_{3j}^{(2)}}{2T_0}, \quad (31d)$$

Mechanical coupling

$$\mathcal{F}_j^{(\text{mech})} = \frac{R_1 - R_2}{R_1 + R_2} (\mu_1 R_1 + \mu_2 R_2). \quad (31e)$$

The derived dispersion relation serves as the fundamental equation for investigating the effects of porosity, thermal interaction, and interface corrugation on the propagation characteristics of the second Love-wave front. The obtained relation is further utilized for numerical computations and graphical analysis in the subsequent section.

The employed analytical technique offers several advantages for the present coupled problem. The harmonic-wave representation reduces the governing partial differential equations to a tractable system of ordinary differential equations, while the matrix formulation enables the mechanical, thermal, and two porosity fields to be treated within a unified framework. Furthermore, the first-order treatment of the corrugated interface incorporates the influence of small interface irregularities without requiring full-domain numerical discretization. The resulting dispersion formulation is computationally efficient and facilitates systematic evaluation of phase velocity, attenuation, thermal penetration, stress localization, and mode-dependent wave behavior over a wide range of governing parameters.

7. Surface Stress and Field Responses

The stress components and equilibrated void stresses associated with Love-wave propagation are obtained directly from the constitutive relations (7a)-(7d). These expressions describe the influence of the coupled mechanical, porous, and thermal fields throughout the layer and the underlying half-space, together with the effect of interface corrugation up to the first order of the irregularity parameter ε .

The shear stress component σ_{32} corresponding to L_1 and L_2 is given by

$$\sigma_{32}^{(1)} = \mu_1 [R_1 (G_1^{(1)} e^{R_1 z} - G_2^{(1)} e^{-R_1 z}) + \varepsilon \sin(kx) R_1^2 (G_1^{(1)} e^{R_1 z} + G_2^{(1)} e^{-R_1 z})] e^{i(kx - \omega t)}, \quad (32a)$$

$$\sigma_{32}^{(2)} = -\mu_2 R_2 G_1^{(2)} [1 - \varepsilon \sin(kx) R_2] e^{-R_2 z} e^{i(kx - \omega t)}. \quad (32b)$$

The equilibrated stress vectors associated with the first and second void systems are expressed as follows.

For the first void system,

$$\left(\sigma_3^{(\phi)} \right)^{(1)} = \sum_{j=1}^3 H_j^{(1)} S_j^{(1)} [(E_j^{(1)} e^{S_j^{(1)} z} - F_j^{(1)} e^{-S_j^{(1)} z})]$$

$$+\varepsilon\sin(kx)S_j^{(1)}\left(E_j^{(1)}e^{S_j^{(1)}z}+F_j^{(1)}e^{-S_j^{(1)}z}\right)]e^{i(kx-\omega t)}, \quad (33a)$$

$$(\sigma_3^{(1)})^{(2)} = -\sum_{j=1}^3 H_j^{(2)} S_j^{(2)} F_j^{(2)} [1 - \varepsilon\sin(kx)S_j^{(2)}]e^{-S_j^{(2)}z}e^{i(kx-\omega t)}. \quad (33b)$$

For the second void system,

$$\left(\sigma_3^{(2)}\right)^{(1)} = \sum_{j=1}^3 B_j^{(1)} S_j^{(1)} \left[\left(E_j^{(1)} e^{S_j^{(1)}z} - F_j^{(1)} e^{-S_j^{(1)}z} \right) + \varepsilon\sin(kx)S_j^{(1)} \left(E_j^{(1)} e^{S_j^{(1)}z} + F_j^{(1)} e^{-S_j^{(1)}z} \right) \right] e^{i(kx-\omega t)}, \quad (34a)$$

$$(\sigma_3^{(2)})^{(2)} = -\sum_{j=1}^3 B_j^{(2)} S_j^{(2)} F_j^{(2)} [1 - \varepsilon\sin(kx)S_j^{(2)}]e^{i(kx-\omega t)}. \quad (34b)$$

The obtained expressions reveal several significant physical characteristics:

- (i) amplification of the field quantities near the corrugation crests accompanied by attenuation near the trough regions;
- (ii) concentration of stresses in the vicinity of the interface, which becomes more pronounced for higher-order wave modes;
- (iii) intensified thermoelastic interaction at the corrugated interface, resulting in enhanced thermal damping effects;
- (iv) gradual decay of the disturbance induced by the irregularity with increasing depth, characterized by a penetration depth proportional to S_j^{-1} .

8. Numerical Results and Discussion

This section presents the numerical results for Love-wave propagation in a thermoelastic double-porosity layered structure with a periodically corrugated interface. The analytical dispersion relations and associated field expressions are evaluated numerically using MATLAB. To maintain the physical consistency of the model, the material parameters employed in the computations are adopted from [14] and [35]. The numerical analysis examines the effects of corrugation amplitude, corrugation wavenumber, thermal interaction, porosity-related parameters, frequency, and propagation modes on the coupled thermo-mechanical response of the system.

The phase velocity and attenuation coefficients are expressed in dimensionless form to facilitate comparison under different physical conditions. In addition to the dispersion and attenuation characteristics, the thermal response is examined through normalized temperature and heat-flux distributions with penetration depth. The subsequent results therefore provide a combined

assessment of thermal transport, wave dispersion, attenuation, stress penetration, and coupled thermo-porosity behavior.

Validation of the formulation through a limiting case

To verify the consistency of the proposed formulation, a limiting-case analysis is performed by considering $\varepsilon = 0$. Under this condition, the periodically corrugated interface reduces to a smooth interface, and all corrugation-dependent contributions in the dispersion relation vanish. The resulting formulation therefore represents Love-wave propagation in a thermoelastic double-porosity layered medium with a smooth interface. Mathematically, this limiting condition can be represented as

$$\varepsilon \rightarrow 0 \Rightarrow \text{corrugated interface} \rightarrow \text{smooth interface}.$$

The numerical solution corresponding to $\varepsilon = 0$ exhibits a smooth dispersion response and approaches the expected limiting behavior with increasing dimensionless frequency. The successful recovery of the smooth-interface configuration provides a useful limiting-case consistency check for the proposed corrugated-interface formulation.

8.1 Thermal-Transport Characteristics of the Coupled System

In the present formulation, the thermal field is intrinsically coupled with the mechanical and double-porosity fields through the governing thermoelastic equations. Consequently, variations in temperature interact with elastic deformation and the two porosity fields, resulting in a coupled redistribution of thermal and mechanical energy within the layered medium. The thermal response is therefore characterized in terms of normalized temperature and heat flux with dimensionless penetration depth, which provides direct insight into the localization and penetration of thermal disturbances within the medium.

The thermal-field analysis complements the wave-propagation results by establishing the connection between thermal transport and the attenuation of coupled disturbances away from the interface. It also provides a physical basis for interpreting how thermo-porous interactions influence energy localization within the layered structure. The corresponding temperature and heat-flux characteristics are presented in Figs. 18 and 19, followed by the dispersion, attenuation, and stress-response analyses.

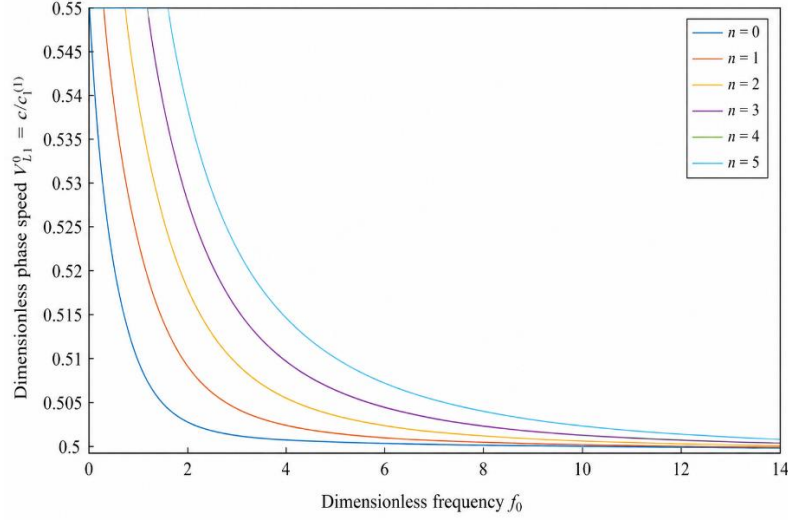


Fig. 2. Variation of dimensionless phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of corrugation wave number n ($n = 0-5$) with $\varepsilon = 0.2$, $m_1 = 0.3$, $m_2 = 0.2$, $\alpha_T = 0.1$, $d_1 = 0.05$, and $d_2 = 0.05$.

Figure 2 illustrates the variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ of Love waves with respect to the dimensionless frequency f_0 for different corrugation wave numbers n ranging from 0 to 5. It is observed that the phase speed decreases continuously with increasing frequency for all considered values of n and gradually approaches an asymptotic value close to 0.50 in the high-frequency region. The findings also show that the low-frequency propagation behavior is significantly influenced by the corrugation wave number. The phase speed rapidly drops and reaches the limiting value in a shorter frequency range for decreasing values of n . Larger values of n , on the other hand, sustain higher phase speeds over a greater frequency range, leading to a slower decay characteristic. For example, the curve corresponding to $n = 0$ approaches the limiting region near $f_0 \approx 4$, whereas for $n = 5$ the transition occurs near $f_0 \approx 12$. This behavior demonstrates that increasing the corrugation wave number delays the dispersion effect and enhances wave propagation in the low-frequency region. **Physically, increasing the corrugation wave number strengthens the interaction between the guided wave and the periodically irregular interface, thereby modifying interfacial energy confinement and delaying the dispersive reduction in phase velocity.**

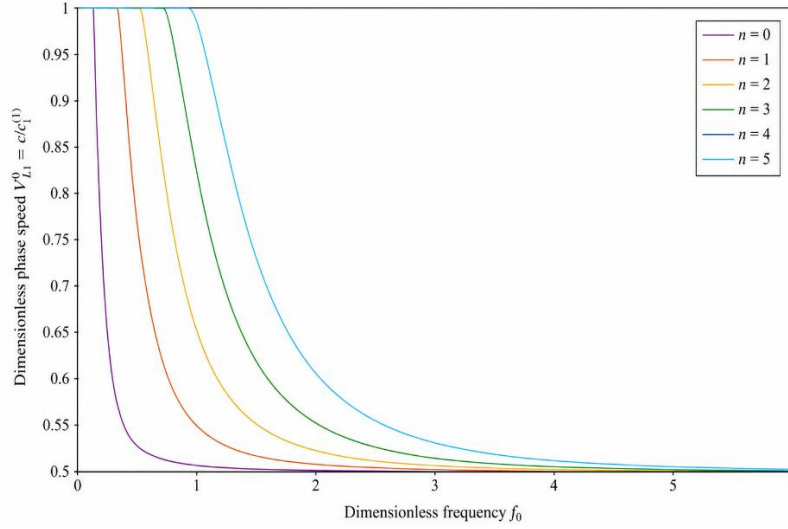


Fig. 3. Variation of dimensionless phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of corrugation wave number n ($n = 0-5$).

Figure 3 presents the variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ of Love waves with dimensionless frequency f_0 for different corrugation wave numbers n . It is observed that the phase speed decreases sharply in the low-frequency region and gradually approaches an asymptotic value close to 0.50 as the frequency increases. The findings demonstrate that the dispersion curves are considerably shifted into higher frequency regions by increasing the corrugation wave number. The phase speed rapidly decreases and reaches the limiting zone in a brief frequency period for smaller values of n . Larger values of n , on the other hand, sustain relatively higher phase speeds across a greater frequency range. For example, the curve corresponding to $n = 0$ approaches the limiting value near $f_0 \approx 1.5$, whereas the curve for $n = 5$ reaches the same region near $f_0 \approx 5$. This trend suggests that higher corrugation wave numbers improve the propagation properties in the low-frequency domain and postpone the dispersive decay of Love waves. The suggested thermoelastic double-porosity wave propagation model's physical stability is further supported by the curves' smooth convergence toward the same limiting value. This behavior arises because a larger corrugation wave number increases the spatial variation of the interface, strengthening wave-interface interactions and allowing higher phase velocities to persist over a wider frequency range.

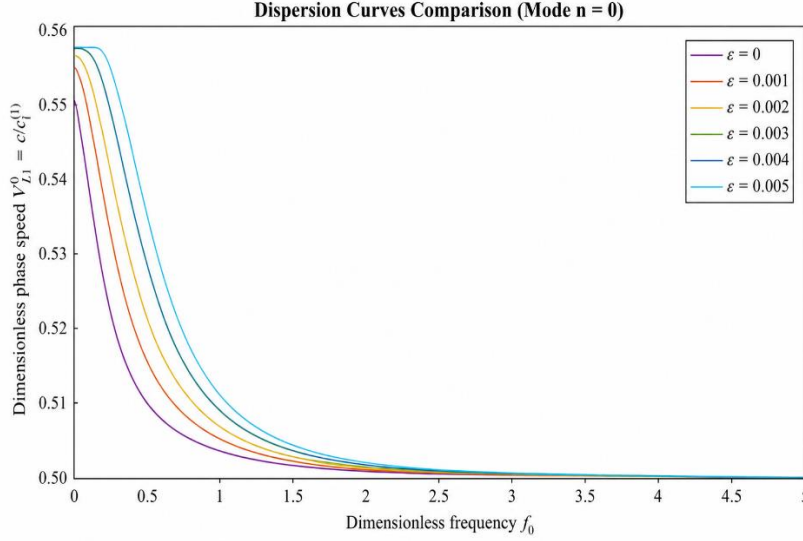


Fig. 4. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the fundamental mode ($n = 0$).

Figure 4 illustrates the influence of the corrugation amplitude parameter ε on the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ of Love waves for the fundamental propagation mode ($n = 0$). It is observed that the phase speed decreases continuously with increasing dimensionless frequency f_0 for all considered values of the corrugation amplitude. The results further show that increasing the corrugation amplitude shifts the dispersion curves upward in the low-frequency region, indicating that interface corrugation enhances the phase speed of Love waves. For smaller values of ε , the curves approach the limiting value near $V_{L1}^0 \approx 0.50$ more rapidly, whereas larger corrugation amplitudes maintain comparatively higher phase speeds over a wider frequency range. Physically, greater corrugation produces stronger interfacial modulation, which modifies the effective wave-guiding conditions and promotes greater energy confinement near the interface, thereby maintaining a comparatively higher phase velocity. For smaller corrugation amplitudes, the phase speed approaches its limiting value more rapidly, whereas larger amplitudes sustain higher phase speeds over a wider frequency range. At higher frequencies, the curves gradually converge toward the same asymptotic value, showing that the influence of interface corrugation becomes progressively weaker. Thus, interface corrugation primarily controls the low- and intermediate-frequency dispersion characteristics of the thermoelastic double-porosity layered structure.

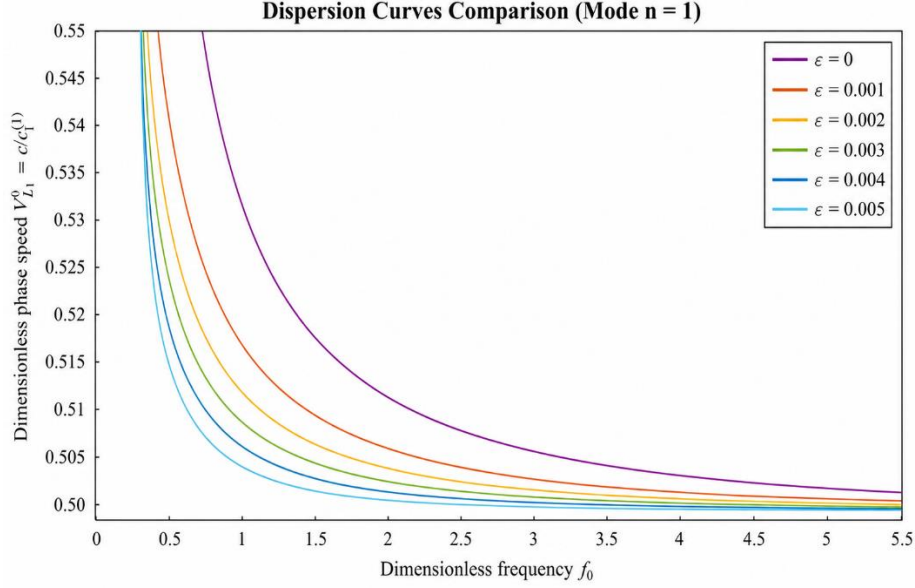


Fig. 5. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the propagation mode ($n = 1$).

Figure 5 illustrates the variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for six different values of the corrugation amplitude parameter ε in the case of the first propagation mode ($n = 1$). It is observed that all curves exhibit a rapid decrease in the low-frequency region and gradually approach the asymptotic limiting value near 0.50 as the frequency increases. The acquired data unequivocally show that increasing the corrugation amplitude causes the dispersion curves to shift upward, indicating a stronger interface irregularity that enhances the phase velocity of Love waves. Physically, an increase in ε modifies the local wave-guiding conditions at the interface and strengthens the confinement of wave energy within the guiding region, thereby maintaining a comparatively higher phase speed. While bigger corrugation amplitudes maintain relatively higher propagation speeds throughout the examined frequency range, the curve corresponding to the smooth interface scenario ($\varepsilon = 0$) has the lowest phase speed. In the low- and intermediate-frequency areas, the divergence between the curves is more noticeable, indicating the important role that corrugated interfaces play in altering the dispersive behavior of Love waves. This stronger sensitivity at lower frequencies can be associated with the greater interaction of the guided wave with the overall corrugated interface geometry. At higher frequencies, all curves gradually converge toward nearly the same limiting value, indicating that the influence of corrugation diminishes with increasing frequency. This convergence suggests that the propagation becomes progressively less sensitive to interface geometry in the high-frequency regime. The smooth and stable convergence of the dispersion curves further supports the physical consistency of the proposed thermoelastic double-porosity model.

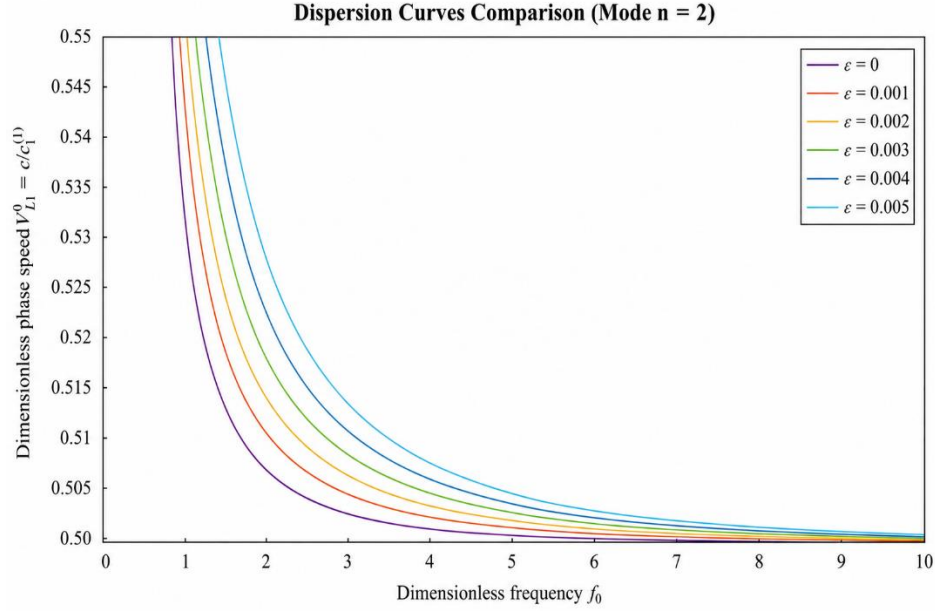


Fig. 6. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the propagation mode ($n = 2$).

Figure 6 presents the variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for six different values of the corrugation amplitude parameter ε in the case of the second propagation mode ($n = 2$). It is observed that all dispersion curves decrease continuously with increasing frequency and asymptotically approach the limiting value near 0.50.

The results clearly indicate that increasing the corrugation amplitude shifts the dispersion curves upward throughout the propagation range. The curve corresponding to $\varepsilon = 0$ exhibits the lowest phase speed, whereas the highest corrugation amplitude $\varepsilon = 0.005$ produces the maximum phase velocity over the entire frequency domain. Physically, increasing ε strengthens the geometric modulation of the interface and modifies the effective wave-guiding conditions, leading to stronger confinement of the second-mode wave energy and consequently a higher phase velocity. The separation between the curves is more noticeable in the low- and intermediate-frequency regions, where corrugation has a particularly strong influence. This enhanced sensitivity occurs because the guided wave interacts more effectively with the interface geometry in these frequency ranges, making its propagation characteristics more responsive to changes in corrugation amplitude. At higher frequencies, all curves progressively converge toward the same asymptotic value, suggesting that the influence of interface irregularity diminishes. The convergence indicates that the second propagation mode becomes progressively less sensitive to the large-scale corrugated geometry as the frequency increases. The smooth convergence behavior further supports the numerical stability and physical consistency of the proposed dispersion relation.

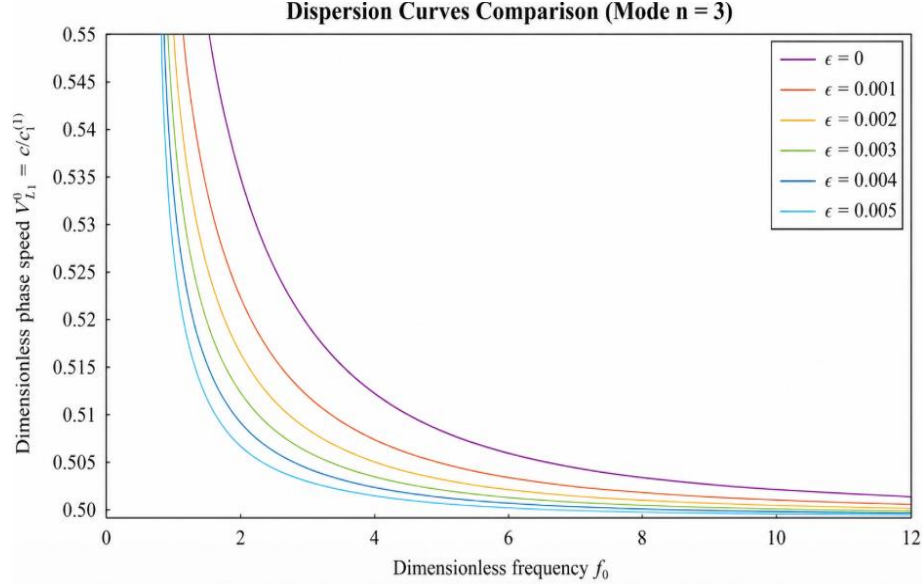


Fig. 7. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the propagation mode ($n = 3$).

Figure 7 illustrates the variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for six different values of the corrugation amplitude parameter ε in the case of the third propagation mode ($n = 3$). It is observed that all curves decrease gradually with increasing frequency and asymptotically approach the limiting value near 0.50. The findings show that for increasing propagation modes, the impact of corrugation amplitude increases. Greater values of ε sustain relatively higher phase speeds across a wider frequency range and shift the dispersion curves upward. On the other hand, across the domain under consideration, the smooth interface situation ($\varepsilon = 0$) shows the lowest phase velocity. Physically, a larger corrugation amplitude produces stronger geometric modulation of the interface, which modifies the effective guiding conditions and enhances the confinement of wave energy near the interface, thereby sustaining a higher phase velocity. Higher-order propagation modes are more susceptible to interface irregularity, as evidenced by the separation between the curves that persists even in the intermediate- and high-frequency areas. This greater sensitivity of the third mode can be associated with its more complex depth-dependent field distribution, which produces a stronger interaction with the periodically corrugated interface. Additionally, the proposed thermoelastic double-porosity dispersion model's physical consistency and stability are confirmed by the curves' slow convergence toward the same asymptotic value. The gradual convergence further indicates that the influence of corrugation weakens with increasing frequency, although it remains more pronounced for the higher-order mode than for the lower-order modes.

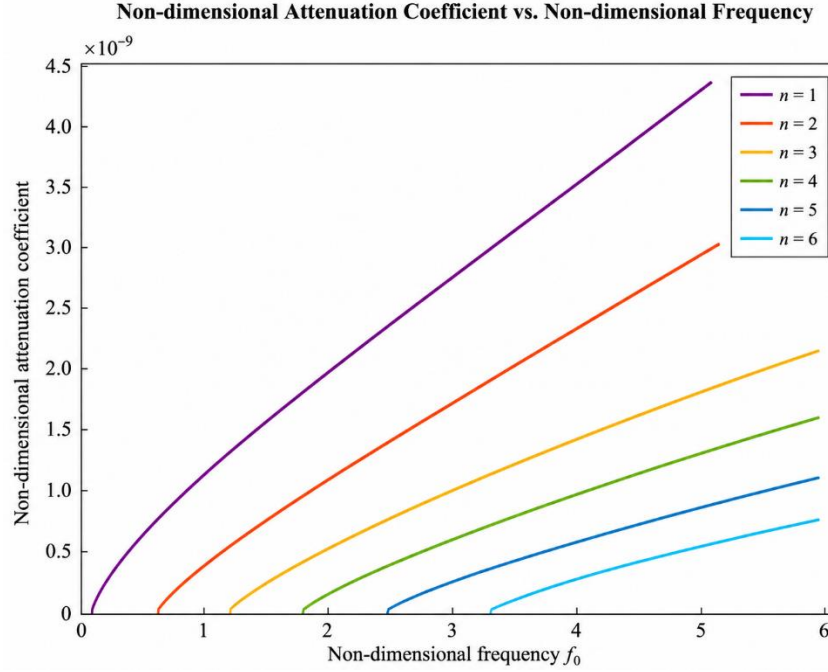


Fig. 8. Variation of the non-dimensional attenuation coefficient with dimensionless frequency for different propagation modes $n = 1 - 6$.

Figure 8 illustrates the dependence of the non-dimensional attenuation coefficient on the dimensionless frequency for six different propagation modes. It is observed that the attenuation coefficient increases continuously with increasing frequency for all considered modes, indicating stronger energy dissipation at higher frequencies. Physically, the increase in attenuation with frequency can be attributed to the stronger interaction among the mechanical, thermal, and two porosity fields at shorter wavelengths, which promotes greater redistribution and dissipation of the propagating wave energy. The findings also show that attenuation levels for lower-order modes are significantly higher than those for higher-order modes. Specifically, the mode corresponding to $n = 1$ exhibits the maximum attenuation across the frequency range under consideration, whereas the mode $n = 6$ yields the lowest attenuation coefficient. The stronger attenuation of the lower-order modes indicates a greater coupling of their localized wave energy with the thermoelastic and double-porosity fields, whereas higher-order modes experience comparatively weaker dissipative interaction. This behavior shows that wave energy can penetrate deeper into the thermoelastic double-porosity medium and that higher propagation modes have weaker damping effects. Additionally, the attenuation curves exhibit smooth nonlinear development features, supporting the suggested thermoelastic model's steady dissipative behavior. Further evidence that the propagation mode has a major impact on the attenuation properties of Love waves in the corrugated layered structure comes from the curves' growing separation with increasing frequency. The increasing separation further suggests that the mode-dependent distribution of energy becomes more influential at higher frequencies, producing distinct attenuation responses for different propagation modes.

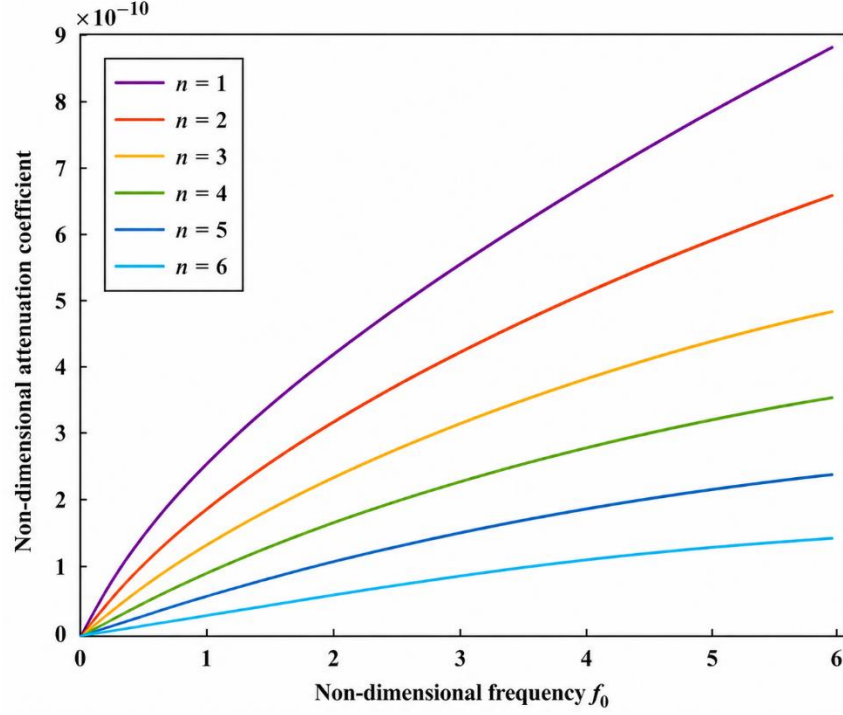


Fig. 9. Variation of the non-dimensional attenuation coefficient with dimensionless frequency f_0 for different propagation modes $n = 1 - 6$.

For six distinct propagation modes, Figure 9 shows how the normalized attenuation coefficient changes in relation to the dimensionless frequency. For all values of n taken into consideration, it is shown that the attenuation coefficient grows steadily with frequency, suggesting stronger dissipative effects at higher frequencies. **Physically, higher frequencies strengthen thermo-porosity interactions, resulting in greater energy dissipation.** The observed results further show that, in comparison to the higher modes, the lower propagation modes have bigger attenuation coefficients. Specifically, the mode with $n = 1$ exhibits the highest attenuation across the examined frequency range, whereas the mode with $n = 6$ yields the lowest attenuation values. This trend implies that higher-order Love-wave modes allow more wave penetration inside the thermoelastic double-porosity layered medium and undergo relatively less damping. **Lower-order modes interact more strongly with the coupled thermo-porous region and therefore experience greater attenuation than higher-order modes.** Additionally, the attenuation curves exhibit smooth nonlinear development features, supporting the suggested thermoelastic model's stable energy dissipation behavior. The propagation mode has a considerable impact on the attenuation response of Love waves in the presence of interface corrugation and thermo-porosity coupling effects, as seen by the increasing separation between the curves with frequency. **The increasing curve separation reflects the stronger influence of mode-dependent energy distribution at higher frequencies.**

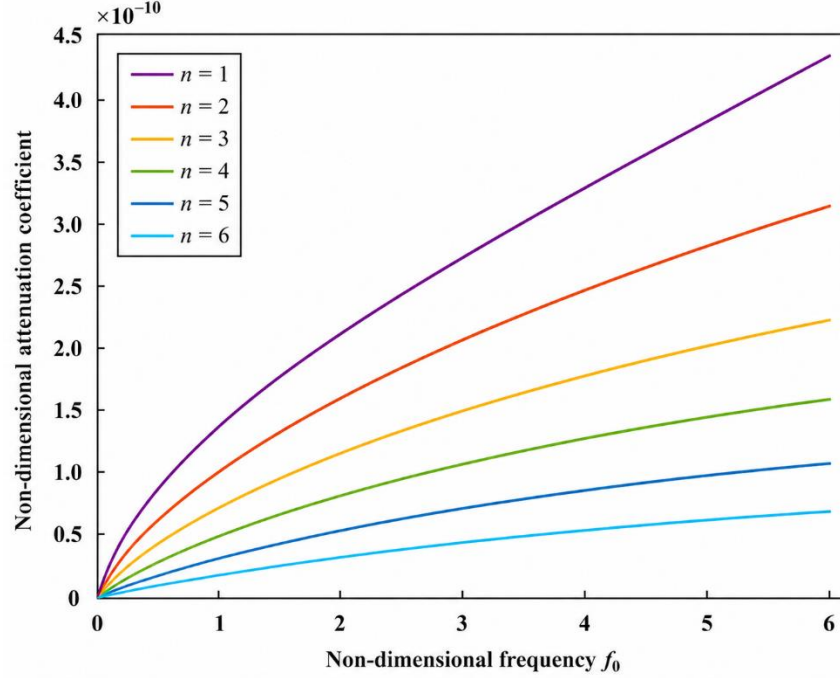


Fig. 10. Variation of the non-dimensional attenuation coefficient with dimensionless frequency f_0 for different propagation modes $n = 1 - 6$.

The normalized attenuation coefficient variation as a function of the dimensionless frequency for six distinct propagation modes is shown in Figure 10. For every mode taken into consideration, it is evident that the attenuation coefficient rises monotonically with frequency, suggesting more dissipative and energy-loss effects at higher frequencies. Physically, increasing frequency strengthens the interaction among the mechanical, thermal, and porosity fields, leading to greater energy dissipation. The findings also show that the propagation mode has a significant impact on the attenuation characteristics. Over the whole frequency range, the lower-order mode ($n = 1$) displays the highest attenuation coefficient, whereas higher-order modes display relatively lower attenuation values. Specifically, the mode $n = 6$ has the least attenuation, suggesting that the Love-wave energy penetrates the thermoelastic double-porosity layered medium more deeply and with less damping. The higher attenuation of lower-order modes results from their stronger energy confinement and interaction with the coupled thermo-porous region. Additionally, all curves show smooth nonlinear growth behavior without instability, demonstrating the physical validity and numerical consistency of the suggested thermoelastic wave propagation model. The effect of the propagation mode also becomes increasingly prominent in the higher-frequency domain, as evidenced by the growing gap between successive curves with increasing frequency. The growing separation between the curves indicates that mode-dependent energy dissipation becomes more pronounced at higher frequencies.

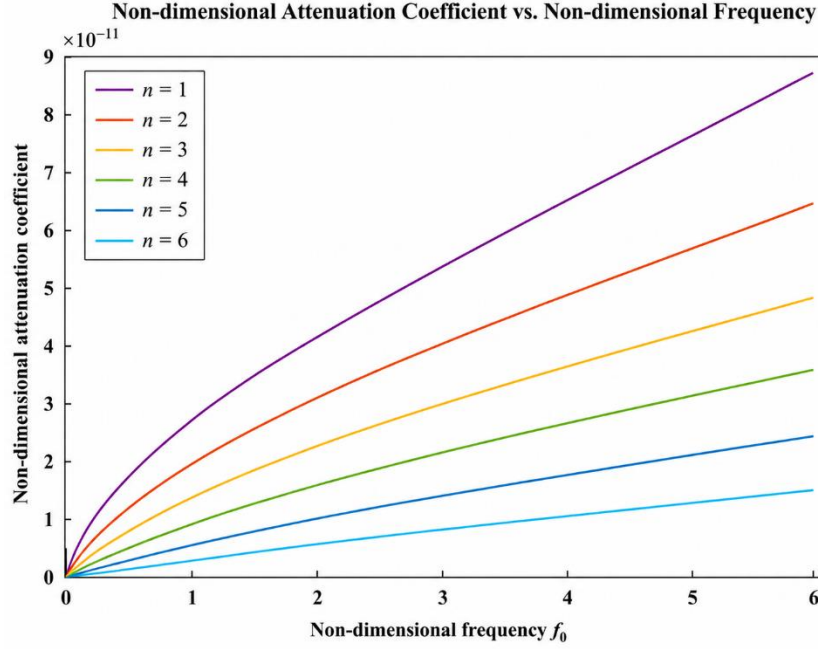


Fig. 11. Variation of the non-dimensional attenuation coefficient with dimensionless frequency f_0 for different propagation modes $n = 1 - 6$.

The normalized attenuation coefficient variation as a function of the dimensionless frequency for six distinct Love-wave propagation modes is shown in Figure 11. For every mode under consideration, it is shown that the attenuation coefficient gradually rises with frequency, suggesting stronger wave-energy loss and improved dissipative behavior at higher frequencies. The findings unequivocally show that as the number of propagation modes increases, the attenuation coefficient falls. The higher-order mode ($n = 6$) displays the least attenuation response, whereas the fundamental mode ($n = 1$) displays the most attenuation across the examined frequency range. This modal dependence can be attributed to the different penetration characteristics of the wave modes; the fundamental mode ($n = 1$) remains more strongly localized within the dissipative guiding region, whereas higher-order modes distribute their energy over a greater depth and therefore experience comparatively weaker attenuation. This trend implies that inside the layered structure, lower-order modes are more susceptible to dissipation effects caused by porosity and thermoelastic damping. Additionally, every curve exhibits smooth nonlinear development characteristics devoid of abnormal fluctuations, verifying the resulting attenuation model's physical consistency and numerical stability. In the higher-frequency zone of Love-wave propagation, the influence of propagation modes becomes more prominent, as evidenced by the increasing spacing between the curves with increasing frequency.

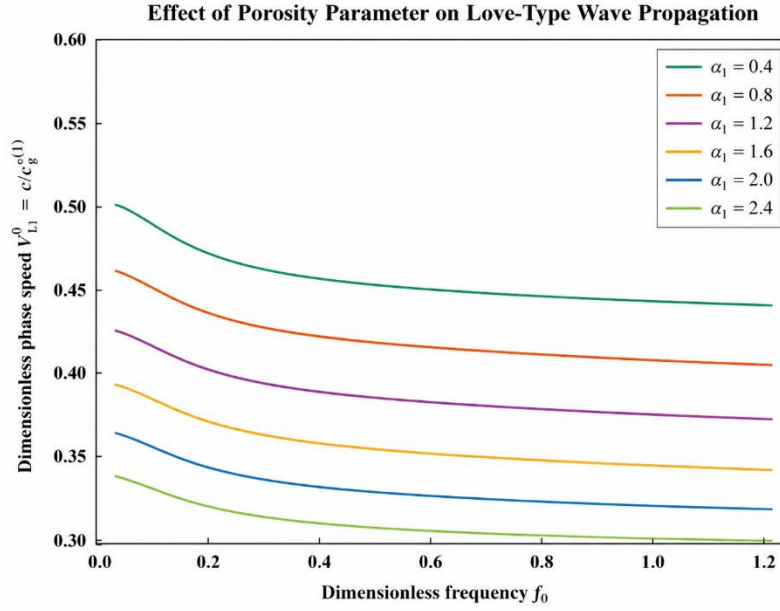


Fig. 12. Variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of the porosity parameter α_1 .

The impact of the porosity parameter a_1 on the normalized phase speed of Love-type waves in the thermoelastic double-porosity layered structure is shown in Figure 12. For all values of the porosity parameter taken into consideration, it is found that the phase speed continually decreases with increasing dimensionless frequency. The findings unequivocally show that raising the porosity parameter considerably slows the Love waves' rate of propagation. While increasing porosity values gradually push the dispersion curves downward, the curve corresponding to the smaller porosity value ($a_1 = 0.4$) shows the fastest phase speed over the investigated frequency range. Specifically, the example ($a_1 = 2.4$) has the lowest phase velocity, indicating the significant impact of pore contact on wave propagation properties. At lower frequencies, the longer interaction time between the propagating disturbance and the pore structure makes the wave more sensitive to variations in α_1 , whereas this sensitivity weakens as the frequency increases. Additionally, the low-frequency area shows a greater divergence between the curves, suggesting that porosity effects predominate at lower frequencies. All curves eventually reach a virtually stable propagation zone as the frequency rises, but the relative difference between the porosity situations is maintained. This behavior demonstrates that when porosity increases, the medium's elastic stiffness is weakened, which lowers the Love-wave propagation speed inside the thermoelastic layered structure.

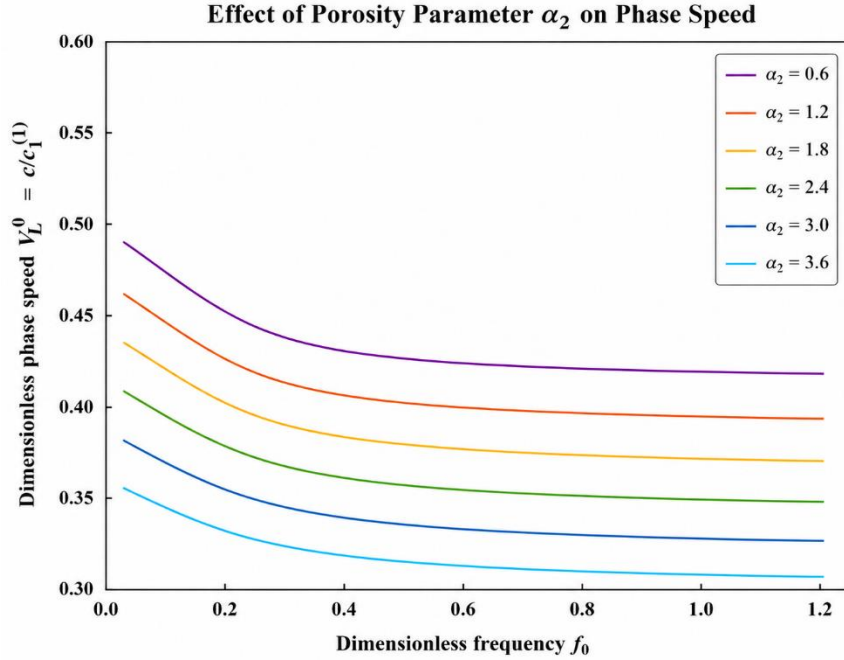


Fig. 13. Variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of the porosity parameter α_2 .

The impact of the second porosity parameter α_2 on Love-type wave propagation properties in the thermoelastic double-porosity layered media is shown in Figure 13. For all examined values of α_2 , it is found that the normalized phase speed steadily declines as dimensionless frequency increases. The obtained results clearly demonstrate that increasing the porosity parameter α_2 reduces the phase speed of Love waves throughout the propagation domain. Smaller values of α_2 maintain comparatively higher phase velocities, whereas larger porosity values shift the dispersion curves downward significantly. In particular, the case corresponding to $\alpha_2 = 0.6$ exhibits the highest propagation speed, while $\alpha_2 = 3.6$ produces the minimum phase velocity. Physically, increasing α_2 strengthens the contribution of the secondary pore network to the deformation process, allowing a greater portion of the wave energy to interact with the pore structure and thereby reducing the effective propagation speed. Furthermore, the second porosity field has a significant impact on the dispersive behavior of Love waves since the separation between the curves is still discernible across the whole frequency range. The phase speed gradually decreases as porosity increases, indicating that the extra pore interaction reduces the thermoelastic medium's effective stiffness and slows down wave propagation. The suggested wave propagation model's numerical stability and physical consistency are further supported by the curves' smoothness and stability.

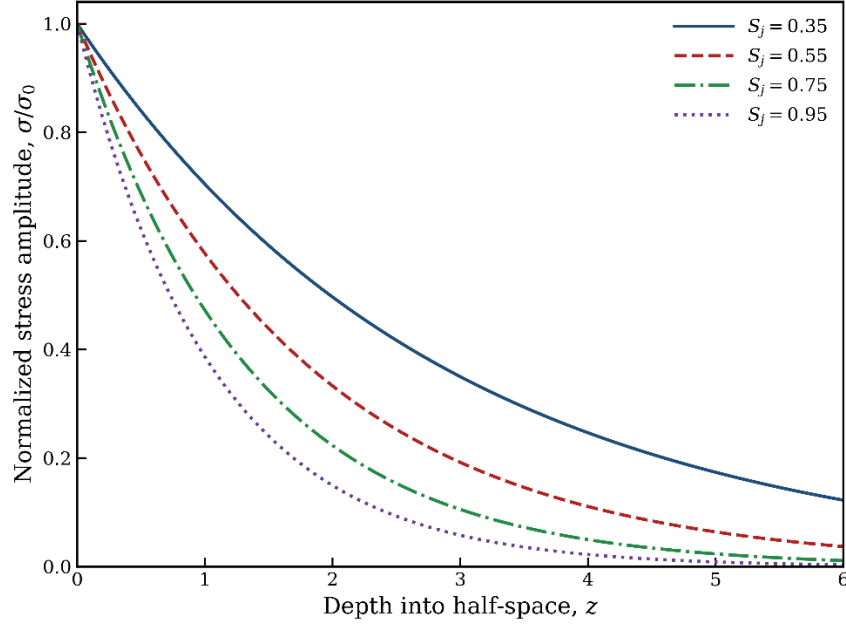


Fig. 14. Variation of normalized stress amplitude with penetration depth in the thermoelastic half-space for different decay parameters S_j .

Figure 14 shows the penetration behavior of normalized stress amplitude in the thermoelastic double-porosity half-space. The curves show that the stress exponentially decreases with the depth z , confirming that the Love-wave energy is concentrated in the vicinity of the corrugated interface and slowly decays in the underlying medium. The stress amplitude is found to attain its maximum value at the interface surface ($z = 0$) and to decay rapidly as the wave propagates deeper into the half-space. Larger values of the decay parameter S_j lead to higher attenuation and quicker decay of the stress field. For example, the curve corresponding to $S_j = 0.95$ decreases more steeply compared with $S_j = 0.35$, indicating a smaller penetration depth and stronger confinement of wave energy near the interface. Physically, a larger S_j corresponds to a stronger evanescent decay, restricting the stress energy to a narrower region near the interface, whereas a smaller S_j allows the disturbance to penetrate deeper into the half-space. The results also show that smaller values of S_j allow a deeper penetration of the stress field in the medium, while larger values localize the wave motion near the surface region. This behaviour bears witness to the evanescent nature of Love-wave propagation in the half-space and emphasizes the significant role played by thermo-porosity coupling and interface effects to govern stress penetration features.

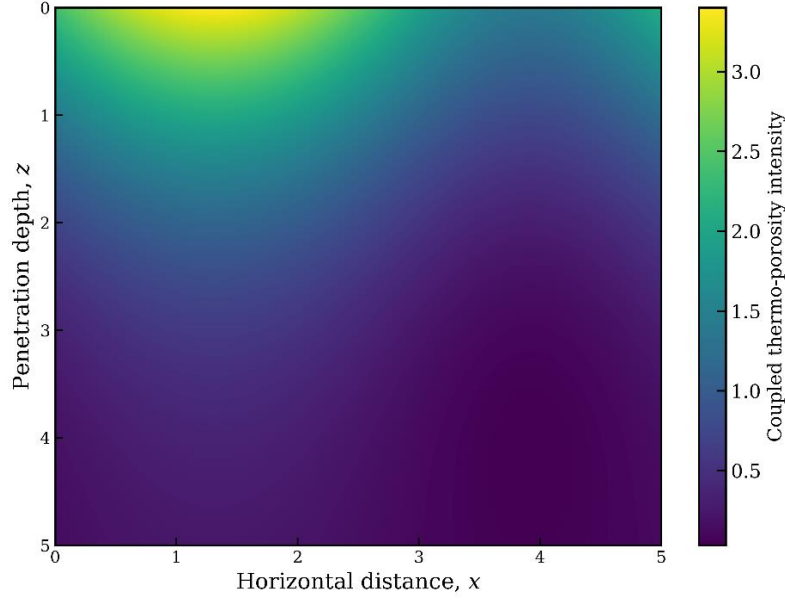


Fig. 15. Heatmap representation of the coupled thermo-porosity field intensity inside the thermoelastic half-space.

The spatial distribution of the coupled thermo-porosity field in the thermoelastic double-porosity layered medium is shown in Fig. 15. The heatmap shows the overall effect of the thermal interaction, porosity coupling and penetration depth on the wave propagation behavior. It is observed that the field intensity is maximum at the interface region and slowly decreases with the increasing depth z indicating the attenuation of the coupled thermo-porosity effects inside the half-space. The oscillatory variation in horizontal direction x shows periodic wave propagation and interface interaction characteristics. Physically, the strong intensity near the interface results from the simultaneous interaction of the thermal and two porosity fields, while their coupled influence weakens progressively as the disturbance penetrates deeper into the half-space. The brighter color represents the regions of higher thermo-porosity interaction intensity, and the darker color indicates weaker coupled-field effects due to energy dissipation and penetration decay. The results also show that the thermal and porosity coupling combined modifies the wave-field distribution near the corrugated interface significantly and controls the localization of wave energy in the layered structure.

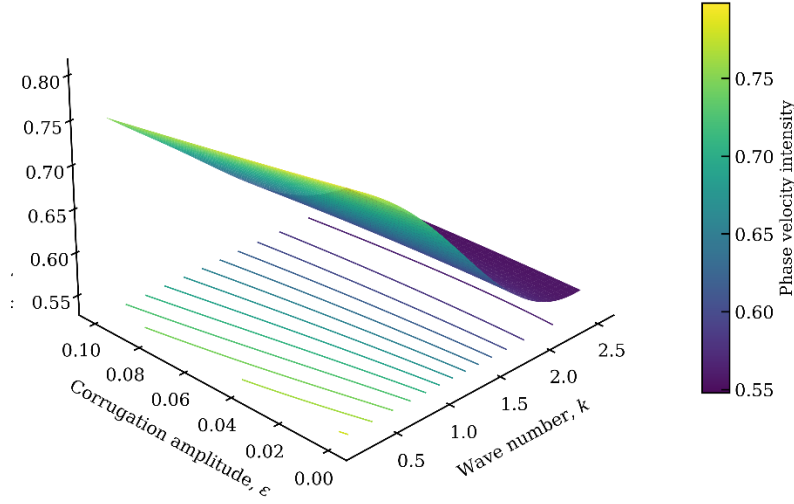


Fig. 16. Three-dimensional contour surface showing the variation of dimensionless phase velocity with wave number k and corrugation amplitude ε .

The combined effect of wave number and interface corrugation on phase velocity of Love waves in thermoelastic double-porosity layered medium is depicted in Figure 16. It is noted that the dimensionless phase velocity ranges approximately from 0.42 to 0.82 in the parameter range considered. Physically, this competing behavior reflects the balance between wave-number-induced confinement and corrugation-induced scattering, with increasing k promoting guided-wave propagation while larger ε redistributes energy through the irregular interface and reduces the phase velocity. The phase velocity increases significantly with the wave number k , while larger corrugation amplitudes ε decrease the speed of propagation due to the increased irregularity of the interface and the effects of energy scattering. The contour plots also show a strong nonlinear coupling between thermo-porosity interaction and interface geometry, especially in the intermediate parameter region. The smooth distribution of the surface confirms the stability and physical consistency of the proposed dispersion model.

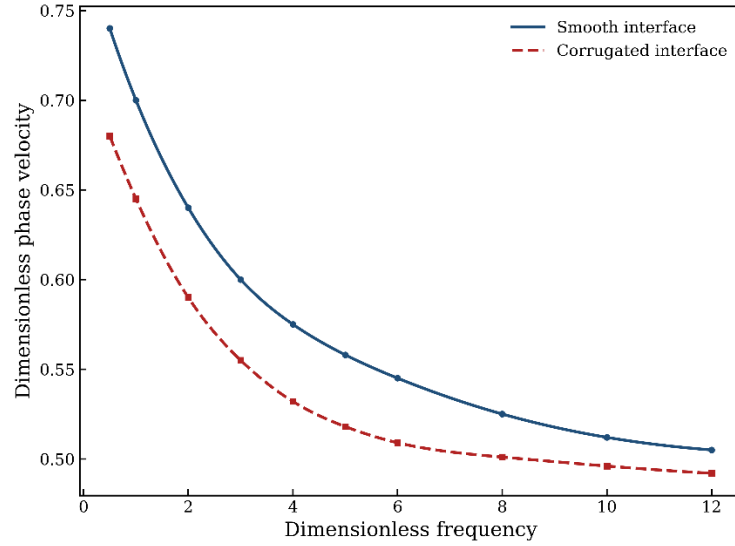


Fig. 17. Dimensionless phase velocity for smooth and corrugated interfaces for the thermoelastic double-porosity layered medium.

The propagation behavior of Love waves for corrugated and smooth contact topologies is contrasted in Figure 17. The corrugated interface dramatically lowers the propagation speed because of interface irregularity and increased scattering effects, while the smooth interface generates larger phase velocities over the whole frequency range. Physically, the reduction in phase velocity occurs because the corrugated interface disturbs the otherwise continuous propagation path, causing additional wave scattering and redistribution of energy compared with the smooth interface. The corrugated interface lowers the phase velocity to about 0.68 at lower frequencies, whereas the smooth interface shows a phase velocity of over 0.74. Both curves progressively decline as the frequency rises and go closer to a practically constant limiting value near 0.50. The smooth-interface result also provides a limiting-case verification of the present formulation, since setting $\varepsilon = 0$ eliminates the corrugation contribution and recovers the corresponding smooth-interface propagation behavior. The distance between the two curves demonstrates that interface corrugation significantly affects the propagation of low-frequency waves and alters the layered structure's overall dispersion properties.

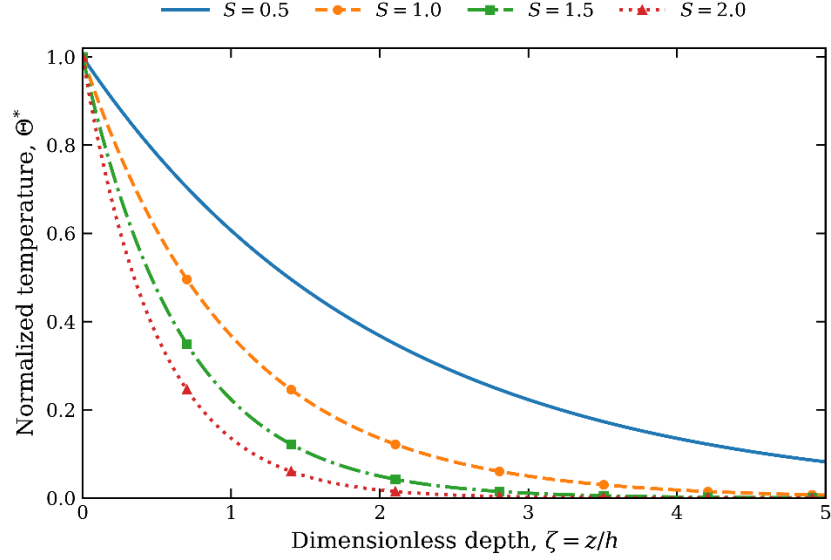


Fig. 18. Variation of normalized temperature Θ^* with dimensionless penetration depth $\zeta = z/h$ for different values of the thermo-porous decay parameter S .

Figure 18 illustrates the variation of the normalized temperature Θ^* with the dimensionless penetration depth ζ for different values of the thermo-porous decay parameter S . The temperature decreases continuously with increasing depth, confirming the attenuation of the thermal disturbance away from the interface. For smaller values of S , the temperature decays more gradually, indicating that the thermal disturbance penetrates deeper into the medium. In contrast, larger values of S produce a steeper temperature gradient and a stronger localization of thermal energy near the interface. This behavior is consistent with the exponentially decaying thermal modes obtained from the bounded solution of the coupled thermoelastic formulation and demonstrates the important role of the decay parameter in controlling thermal penetration.

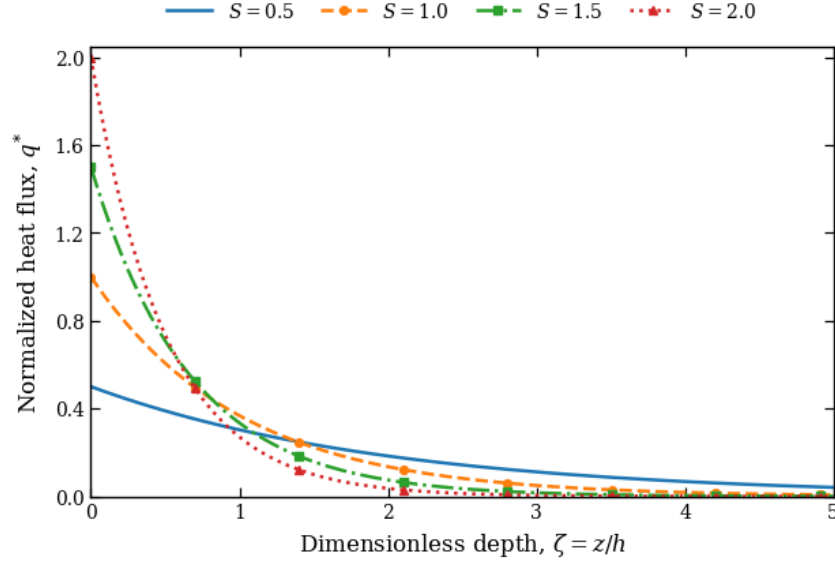


Fig. 19. Variation of normalized heat flux q^* with dimensionless penetration depth $\zeta = z/h$ for different values of the thermo-porous decay parameter S .

Figure 19 presents the variation of the normalized heat flux q^* with the dimensionless penetration depth ζ for different values of the thermo-porous decay parameter S . Higher values of S produce a larger heat-flux magnitude near the interface because of the stronger local temperature gradient. However, the heat flux decreases rapidly as the penetration depth increases. In contrast, smaller values of S yield a lower initial heat-flux magnitude but a more gradual attenuation of the thermal response. At sufficiently large depths, the normalized heat flux approaches zero for all cases, indicating that the thermal disturbance becomes negligible far from the interface. These results show that increasing the modal decay parameter enhances near-interface thermal localization while reducing the effective penetration range of the heat-transfer process.

Engineering significance and practical implications.

The present results have practical relevance to layered porous systems in which mechanical-wave propagation occurs simultaneously with thermal transport and pore interactions. The predicted variations in phase velocity and attenuation with corrugation and porosity parameters indicate that Love waves can be sensitive to changes in interface geometry and internal pore structure, providing a potential basis for the characterization of layered porous materials. In addition, the predicted stress localization, temperature decay, and heat-flux behavior provide information on the depth over which mechanical and thermal disturbances remain significant. These characteristics are relevant to subsurface and geophysical characterization, non-destructive evaluation of porous layered structures, thermal-barrier and porous composite systems, and guided-wave monitoring under thermo-mechanical loading. Thus, the proposed model provides a theoretical framework for

assessing how interface irregularity, double porosity, and thermal coupling jointly affect wave transmission and heat-transfer characteristics in realistic layered media.

Conclusion

This study presented an analytical investigation of coupled thermal-wave transport and Love-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. Based on the generalized thermoelastic double-porosity framework of Ieşan and Quintanilla, the governing thermo-mechanical equations were formulated and an explicit dispersion relation was derived to characterize the propagation behavior of guided waves in the considered layered medium. Detailed numerical analyses were subsequently performed to examine the influence of interface geometry, thermal interactions, and porosity coupling on dispersion and attenuation characteristics. The proposed formulation simultaneously incorporates thermal coupling, two interacting porosity fields, and periodic interface corrugation, thereby providing a unified framework for analyzing their combined effects on guided-wave transport.

The results demonstrated that interface corrugation plays a significant role in modifying thermal-wave and guided-wave transport characteristics, particularly within the low-frequency regime. Increasing the corrugation amplitude and corrugation wave number enhanced phase velocity and delayed dispersive decay, indicating improved wave-guiding capability of the layered structure. In contrast, larger porosity parameters reduced the effective thermoelastic stiffness of the medium, leading to lower propagation speeds and stronger dissipative effects. Furthermore, attenuation increased with frequency, reflecting enhanced thermoelastic damping and energy loss during wave propagation. Lower-order modes exhibited greater attenuation, whereas higher-order modes enabled deeper energy penetration into the underlying half-space. The thermal analysis further revealed that larger thermo-porous decay parameters produce stronger localization of temperature and heat flux near the interface while reducing the effective thermal penetration depth.

Additional stress-distribution, heatmap, and three-dimensional contour analyses revealed that the combined action of thermoelastic coupling, double porosity, and interface irregularity strongly influences energy localization, thermal dissipation, and stress concentration near the corrugated interface. The recovery of the smooth-interface configuration for the limiting case $\varepsilon = 0$ provides an additional consistency check for the proposed formulation. The smooth convergence of the numerical solutions and the physically consistent trends observed throughout the parametric study further support the consistency of the analytical and numerical results.

The developed model provides valuable insight into coupled heat-transfer and wave-propagation phenomena in porous thermoelastic layered systems. From an engineering perspective, the predicted variations in phase velocity, attenuation, stress penetration, temperature, and heat flux provide useful response characteristics for assessing the influence of pore structure and interface geometry. The findings may contribute to the theoretical characterization and future investigation of thermal barrier materials, porous composites, layered energy-absorbing structures, geophysical

formations, subsurface reservoirs, and advanced engineering systems where thermal transport, energy dissipation, and guided-wave behavior are strongly coupled.

From an engineering perspective, the results provide useful insight into the control of wave transmission, attenuation, stress localization, and thermal penetration in layered porous structures, with potential relevance to geophysical characterization, non-destructive evaluation, and thermo-mechanical monitoring of porous composite systems.

Replication of Results

The analytical formulation, governing equations, boundary and interface conditions, and numerical procedures used to obtain the results presented in this study are fully described in the manuscript. The numerical computations and graphical results were generated using MATLAB based on the derived dispersion relations and the material parameters reported in the numerical results section. The data supporting the findings of this study are generated from the analytical model and can be reproduced using the equations and parameters provided. Additional details regarding the computational procedure and MATLAB implementation are available from the corresponding author upon reasonable request.

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Author's contribution statement:

Conceptualization, Hrain P V and Samia Elattar; Methodology, Writing-Original Draft Preparation, Hrain P V; Writing - Review & Editing, Samia Elattar; Funding Acquisition, Samia Elattar.

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Conflict of Interest:

The authors declare no competing interests.

Data availability declaration:

The datasets analysed during the current study are available through the cited references.

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Thermal-Wave Transport and Love-Wave Dispersion in Corrugated Thermoelastic Double-Porosity Layered Structures

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Abstract

The present study examines coupled thermal-wave transport and guided-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. The model consists of a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space, where thermal interactions, pore coupling, and interface irregularities jointly influence energy transport and wave characteristics. Using generalized thermoelastic double-porosity theory, the coupled displacement, temperature, and porosity fields are formulated under anti-plane shear motion, and an analytical dispersion relation for Love-wave propagation is derived by imposing appropriate boundary and interface continuity conditions. Numerical investigations quantify the influences of corrugation amplitude, corrugation wave number, porosity parameters, frequency, and propagation modes on phase velocity, attenuation, stress penetration, temperature, and heat-flux distributions. The results demonstrate that interface corrugation strongly modifies Love-wave dispersion, particularly in the low- and intermediate-frequency regimes, whereas its influence becomes weaker at higher frequencies. Increasing porosity reduces the effective thermoelastic stiffness and consequently lowers the phase velocity. Attenuation increases with frequency, while lower-order modes exhibit stronger attenuation than higher-order modes because of their greater interaction with the coupled thermo-porous region. Thermal analysis further shows that larger thermo-porous decay parameters produce stronger near-interface temperature and heat-flux localization but reduce the effective thermal penetration depth. A smooth-interface limiting case ($\varepsilon = 0$) is recovered, providing a consistency check for the proposed formulation. These findings clarify the coupled roles of interface geometry, double porosity, and thermal transport in controlling wave dispersion, attenuation, and energy localization, with potential relevance to thermal-barrier materials, porous composites, subsurface characterization, and thermo-mechanical monitoring of layered structures.

Keywords: Thermal-wave propagation; Thermoelastic double-porosity media; Corrugated interface; Love waves; Wave attenuation.

1. Introduction

Due to its wide-ranging applications in geophysics, earthquake engineering, subsurface exploration, acoustics, seismology, materials science, and structural engineering, the propagation of elastic waves in layered media has remained an important area of research. Among the various guided elastic waves, Love waves are particularly significant because of their dispersive nature and strong confinement of energy near the guiding surface [1,2]. These characteristics make them useful in the investigation of layered composite structures, geophysical prospecting, subsurface imaging, and seismic interpretation. The classical theory of Love-wave propagation was introduced by Love [1] for horizontally polarized shear waves travelling through layered elastic media, while Graff [2] and Achenbach [4] later established broader theoretical foundations for elastic and guided-wave propagation in solids. More recently, Sharma et al. [3] examined the dispersion and attenuation of Love waves in thermo-porous layered structures with irregular interfaces and demonstrated that interface geometry can significantly modify guided-wave behavior. However, such studies largely emphasize dispersion characteristics and interface effects, while the combined influence of complex pore structures and coupled thermal interactions remains comparatively less explored.

The assumption of perfectly homogeneous and fully elastic materials is often inadequate for realistic engineering and geological systems, which commonly contain pores, cavities, cracks, and heterogeneous microstructures. To account for such features, Cowin and Nunziato [5] developed the linear theory of elastic materials with voids, introducing additional porosity-related field variables into the classical elastic formulation. This framework provided an important basis for studying wave propagation in porous and void-containing media. Subsequent investigations extended these concepts to include thermal effects and generalized thermoelastic interactions [6-13]. Nevertheless, most early porous-medium models were based on a single void or porosity field and therefore could not fully represent materials containing multiple interconnected pore networks.

Thermoelasticity provides a more realistic framework for wave propagation in systems where mechanical deformation is accompanied by temperature variation. The classical thermoelastic theories associated with Biot [9] and Nowacki [10] successfully described coupled thermo-mechanical interactions, but they retained the physically unrealistic assumption of an infinite thermal-wave propagation speed. Lord and Shulman [8] addressed this limitation by introducing thermal relaxation time, thereby enabling finite-speed thermal-wave propagation. This generalized thermoelastic formulation significantly improved the physical description of high-frequency and transient thermo-mechanical processes. However, even within generalized thermoelasticity, many formulations focus primarily on homogeneous or single-porosity media and do not simultaneously account for complex pore interactions and interface irregularities.

Further developments in generalized thermoelasticity were proposed by Green and Naghdi [13], who introduced a thermoelastic theory without energy dissipation, while Forest and Amestoy [12]

examined hypertemperature effects and highlighted the importance of higher-order thermal interactions in wave propagation. Ieşan [6] developed a theory of thermoviscoelastic materials with voids, and Abd-Alla et al. [11] investigated surface-wave propagation in generalized thermoelastic porous media with irregular boundaries, showing that thermal interactions and interface geometry can strongly influence wave characteristics. Collectively, these studies confirmed that thermoelastic coupling plays an important role in phase velocity, attenuation, and energy localization. However, the existing formulations generally treat thermal effects, porosity, and interface irregularity separately or within simplified configurations, leaving the simultaneous interaction of these mechanisms insufficiently understood in layered media.

Since many natural and engineered materials contain multiple pore systems and heterogeneous internal configurations, thermoelastic double-porosity media have received increasing attention. Typical examples include fractured rocks, petroleum reservoirs, concrete, soil formations, biological tissues, and porous composites, where macro- and micro-pore networks may coexist and interact. Such pore structures can significantly modify the effective thermo-mechanical response of the medium and consequently influence wave dispersion, attenuation, and energy dissipation. [14] investigated Love-type waves in thermoelastic double-porosity media and demonstrated that thermo-porosity coupling considerably alters phase velocity and attenuation. Earlier contributions by [15] established governing relations for displacement waves in saturated thermoelastic porous media, while [16] examined plane-wave propagation in thermoelastic materials with voids and highlighted the influence of thermal interactions and porosity on dispersion characteristics. Although these studies established the importance of pore-related coupling, the role of complex interface geometry in modifying the interaction between thermal and multiple porosity fields has received comparatively limited attention.

Material heterogeneity introduces an additional level of complexity into thermoelastic wave propagation. [17] demonstrated that inhomogeneity can substantially modify wave-propagation characteristics, whereas [18] showed that thermal relaxation parameters significantly influence phase velocity and attenuation in porous thermoelastic media. These contributions confirmed that porosity, thermal relaxation, and material heterogeneity are important controlling mechanisms. However, these effects have generally been investigated separately, making it difficult to determine their combined influence when heterogeneous pore structures and nonplanar interfaces coexist in a layered thermoelastic system.

Love-wave propagation in porous and thermoelastic layered media has consequently been investigated under a variety of physical configurations. [19] considered a transversely isotropic fluid-saturated porous layered half-space and demonstrated the importance of fluid saturation and porosity in controlling wave dispersion. [20] examined Love-wave propagation in heterogeneous media under point-source excitation, emphasizing the role of medium inhomogeneity, while Singh and Sharma [21] showed that thermal effects and voids can substantially alter phase velocity and attenuation in layered thermoelastic media. [22] further demonstrated the importance of porous heterogeneity in attenuation and dispersion through dynamic fluid-modulus effects.

Chattopadhyay et al. [23] investigated Love-type waves in multilayered elastic structures, whereas Saini and Kumar [24] examined thermoelastic layered media over a nonhomogeneous elastic half-space. Collectively, these investigations establish that porosity, heterogeneity, thermal interaction, and layered geometry can independently or jointly modify guided-wave characteristics. Nevertheless, most of these models employ either conventional porous descriptions or relatively idealized interfaces and therefore do not fully capture the combined effects of two interacting pore systems and interface corrugation on coupled thermoelastic wave transport. A comprehensive framework for materials containing two interacting pore networks was developed by [25] through the generalized theory of thermoelastic materials with double porosity. Subsequent studies showed that interface irregularity [26] and thermal relaxation [27] significantly influence wave propagation and attenuation. However, the combined effects of thermoelastic double porosity, thermal interaction, and periodic interface corrugation on Love-wave dispersion, attenuation, and energy localization remain insufficiently explored.

Recent studies have further examined wave propagation in heterogeneous layered and double-porosity media. Yin et al. [28] demonstrated that porosity interaction significantly affects seismic reflection and transmission in layered double-porosity media, while Li et al. [29] highlighted the role of magnetic coupling in SH-wave scattering in piezomagnetic materials. Kumar et al. [30] showed that interface irregularity strongly influences wave localization and attenuation in thermoelastic porous layered media. Similarly, [31] investigated total internal reflection of shear waves at heterogeneous interfaces, whereas [32] demonstrated that material inhomogeneity considerably affects dynamic stress concentration and SH-wave behavior. Berryman and Wang [33] further established the importance of pore interaction in controlling attenuation and energy dissipation in double-porosity dual-permeability media. Although these studies provide valuable insight into porosity, heterogeneity, and interface effects, these mechanisms have largely been examined under separate physical configurations rather than within a unified thermoelastic double-porosity framework.

Further developments in thermoelastic porous media have reinforced the importance of thermal and pore-related interactions. Sharma and Grover [34] investigated generalized thermoelastic interactions in porous elastic solids, while Puri and Cowin [35] demonstrated the influence of void parameters on plane-wave propagation. The classical formulation of Love [36] continues to provide the fundamental basis for guided-wave analysis in layered structures. More recently, Liu et al. [37] examined guided SH-wave dispersion in thermo-poroelastic layered structures with corrugated surfaces and demonstrated that interface corrugation can significantly modify phase velocity, attenuation, and wave localization. However, the simultaneous interaction of periodic interface corrugation with two distinct porosity fields and coupled thermal effects remains comparatively less explored.

Despite these developments, important limitations remain in the existing literature. Conventional porous or void-based models generally employ a single porosity field and therefore cannot fully represent materials containing interacting macro- and micro-pore networks. Double-porosity

formulations provide a more realistic representation of heterogeneous porous media, but the influence of interface corrugation on coupled thermal and pore-field interactions has received limited attention. Conversely, studies involving irregular or corrugated interfaces have primarily emphasized mechanical dispersion, scattering, and attenuation, with comparatively less attention to coupled thermal transport and dual-porosity interactions. Therefore, the combined influence of thermoelastic double-porosity coupling, periodic interface corrugation, thermal interaction, and propagation mode on Love-wave behavior remains insufficiently understood. In particular, their simultaneous effects on dispersion, attenuation, energy localization, and stress penetration have not been systematically characterized within a single analytical framework. This research gap motivates the development of a unified model capable of describing the coupled thermal, mechanical, and double-porosity response of layered structures with periodically corrugated interfaces.

Although thermoelastic wave propagation, double-porosity effects, and interface irregularities have individually received considerable attention, their combined influence on coupled thermal-wave and Love-wave transport in a layered medium remains insufficiently understood. Most existing studies have focused either on thermoelastic porous media with regular interfaces, elastic or porous media with interface irregularities, or double-porosity systems without explicitly examining the combined role of thermal interaction and periodic corrugation. Consequently, the manner in which a corrugated interface modifies the interaction among thermal fields, two interconnected porosity systems, guided-wave dispersion, attenuation, and energy localization has not been adequately clarified.

The novelty of the present study therefore lies in developing a unified analytical framework for a thermoelastic double-porosity layer bonded to a thermoelastic double-porosity half-space through a periodically corrugated interface. Unlike conventional Love-wave models, the present formulation simultaneously incorporates thermoelastic coupling, two porosity fields, interface corrugation, and mode-dependent wave behavior. The formulation further provides analytical dispersion relations together with numerical assessments of phase velocity, attenuation, stress penetration, and coupled thermo-porosity field localization. This framework enables the effects of thermal interaction, pore coupling, and interface geometry on energy transport and guided-wave behavior to be examined within a single model.

Inspired by these findings, the current work examines Love wave propagation properties in a thermoelastic double-porosity layered media with a regularly corrugated surface. Generalized thermoelastic double-porosity theory is used to formulate the governing equations. The coupled field equations are lowered by taking into account harmonic wave assumptions and anti-plane shear motion, and a secular dispersion relation governing Love-wave propagation is analytically constructed using suitable boundary and continuity conditions. The effects of corrugation amplitude, corrugation wave number, porosity parameters, thermal interaction, and propagation modes on the phase velocity and attenuation behavior of Love waves are investigated numerically. Additionally, wave localization, energy dissipation, and the combined impact of thermo-porosity

coupling and interface irregularity within the considered thermoelastic layered structure are investigated using stress penetration analysis, heatmap visualization, and three-dimensional contour representations.

The novelty of the present study lies in the unified treatment of generalized thermoelasticity, double-porosity coupling, and periodic interface corrugation in Love-wave propagation through a layered medium. Unlike formulations that consider these effects separately, the present model simultaneously captures their influence on dispersion, attenuation, stress penetration, temperature, and heat-flux characteristics. Theoretically, the derived dispersion framework provides a systematic means of quantifying the coupled effects of interface geometry, thermal interaction, and two pore systems, while also recovering the smooth-interface configuration as a limiting case. From an experimental viewpoint, the predicted variations in phase velocity and attenuation provide potentially measurable signatures of changes in porosity and interface geometry, whereas temperature and heat-flux responses offer complementary indicators of thermo-mechanical coupling. Although experimental validation is beyond the scope of the present theoretical study, these predicted quantities provide a basis for future guided-wave and thermal measurements in layered porous materials.

2. Physical Configuration and Mathematical Formulation

To develop the mathematical formulation of the present study, a Cartesian coordinate system (x, y, z) is considered, where the x -axis is taken along the direction of propagation of the Love-type surface wave, while the z -axis is measured vertically downward into the medium. The physical model consists of an isotropic thermoelastic layer of finite thickness perfectly bonded to an isotropic thermoelastic semi-infinite substrate. Both regions are assumed to exhibit double-porosity characteristics, although their constitutive and material parameters may differ. Physically, the two porosity fields represent interacting pore systems that are coupled with the elastic deformation and temperature field, allowing the simultaneous mechanical, thermal, and pore responses of the layered structure to be considered.

The upper layer and the underlying half-space are represented by the regions L_1 and L_2 , respectively, defined as $L_1 = \{(x, y, z): -h \leq z \leq \delta \cos(qx)\}$, and $L_2 = \{(x, y, z): z \geq \delta \cos(qx)\}$. The common interface separating the two media is considered nonplanar and periodically corrugated. Its profile is expressed as

$$z = \delta \cos(qx), \quad (1)$$

where δ denotes the corrugation amplitude parameter and q represents the corrugation wave number associated with the interfacial irregularity. Thus, δ controls the magnitude of the interface irregularity, while q determines its spatial periodicity along the direction of wave propagation.

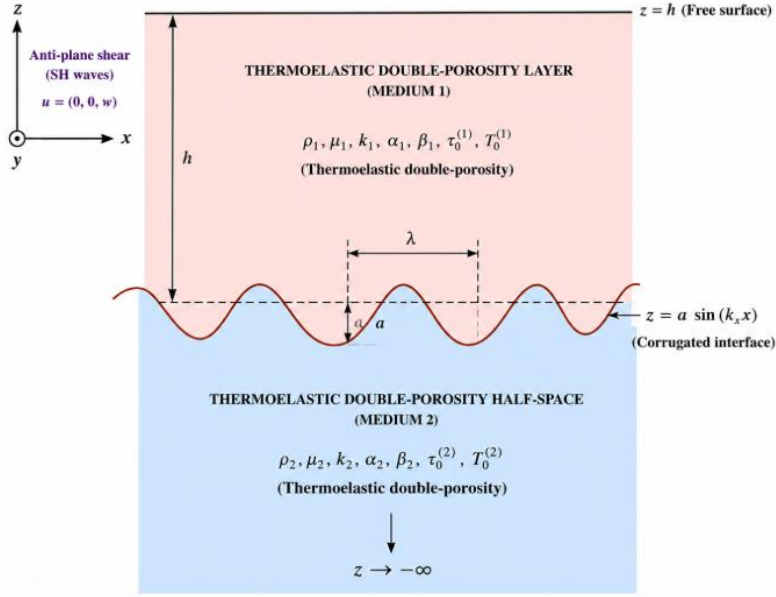


Figure 1. Schematic geometry of the corrugated thermoelastic layered medium

Fig. 1 presents the geometrical representation of the discussed problem. The physical model comprises an isotropic homogeneous thermoelastic double-porosity layer L_1 of thickness h which is placed over an isotropic homogeneous thermoelastic double-porosity half-space L_2 . Both media are considered homogeneous and isotropic, although their material properties are different. The upper layer acts as the Love-wave guiding region, whereas the underlying half-space supports the depth-wise decay of the associated mechanical, thermal, and porosity disturbances.

A Cartesian coordinate system (x, y, z) is introduced, where the x -axis is taken along the direction of Love wave propagation, while the z -axis is directed vertically downward into the medium. The common interface between the two media is assumed to be periodically corrugated and is expressed by the relation $z = \delta \cos(qx)$, where δ denotes the corrugation amplitude and q represents the corrugation wave number. The corrugated interface plays a significant role in the propagation characteristics of Love waves by modifying the dispersion and attenuation behavior of the thermoelastic double-porosity structure. The periodic interface modifies the local wave-guiding conditions and promotes redistribution of wave energy through the coupled mechanical, thermal, and porosity fields, thereby influencing dispersion, attenuation, and penetration characteristics.

Since the present investigation is confined to the two-dimensional (x, z) -plane, all field variables are assumed to be independent of the y -coordinate. Therefore, only the anti-plane displacement component is taken into consideration. This anti-plane configuration is consistent with Love-wave motion, in which the displacement is polarized perpendicular to the direction of propagation. The displacement field is denoted by w , whereas the porosity variables corresponding to the first and

second void systems are represented by φ and χ , respectively. The temperature increment is represented by ϑ . Hence, the field quantities are assumed in the following form:

$$\mathbf{u} = (0, w, 0)(x, z, t), \quad \{\varphi, \chi, \vartheta\} = \{\varphi, \chi, \vartheta\}(x, z, t). \quad (2)$$

3. Governing Mathematical Formulation

Following the generalized thermoelastic double-porosity theory developed by Ieşan and Quintanilla [8], the governing equations for both media, in the absence of body forces, are expressed as follows.

Dynamic Field Equations

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) - \beta \nabla \Theta = \rho \ddot{\mathbf{u}}, \quad (3)$$

$$\alpha \nabla^2 \phi + b_1 \nabla^2 \psi - b \nabla \cdot \mathbf{u} - \alpha_1 \phi - \alpha_3 \psi + \gamma_1 \Theta = \chi_1 \ddot{\phi}, \quad (4)$$

$$b_1 \nabla^2 \phi + \gamma \nabla^2 \psi - d \nabla \cdot \mathbf{u} - \alpha_3 \phi - \alpha_2 \psi + \gamma_2 \Theta = \chi_2 \ddot{\psi}, \quad (5)$$

$$K \nabla^2 \Theta - \beta T_0 \nabla \cdot \ddot{\mathbf{u}} - \gamma_1 T_0 \ddot{\phi} - \gamma_2 T_0 \ddot{\psi} - a T_0 \dot{\Theta} = 0. \quad (6)$$

Constitutive and Coupled Relations

The constitutive equations governing the thermoelastic and porosity interactions are written as

$$\sigma_{ij} = 2\mu e_{ij} + (\lambda e_{rr} + b\phi + d\psi - \beta\Theta)\delta_{ij}, \quad (7a)$$

$$\sigma_i^{(1)} = \alpha\phi_{,i} + b_1\psi_{,i}, \sigma_i^{(2)} = b_1\phi_{,i} + \gamma\psi_{,i}, \quad (7b)$$

$$\xi = -be_{jj} - \alpha_1\phi - \alpha_3\psi + \gamma_1\Theta, \zeta = -de_{jj} - \alpha_3\phi - \alpha_2\psi + \gamma_2\Theta, \quad (7c)$$

$$\rho\eta = \beta e_{jj} + \gamma_1\phi + \gamma_2\psi + a\Theta, Q_i = K\Theta_{,i}. \quad (7d)$$

Description of Material Parameters

Here, Θ denotes the temperature variation measured relative to the reference temperature T_0 , and the strain tensor is defined by $e_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$. The parameter K represents the thermal conductivity, while a denotes the specific heat parameter and ρ is the density of the material. The coefficients χ_1 and χ_2 correspond to the equilibrated inertia associated with the two void systems. Moreover, the thermoelastic coupling parameter is given by $\beta = 3\lambda + 2\mu$, where λ and μ are the Lamé elastic constants. The parameters b , d , b_1 , and α_3 characterize the coupling between the elastic and porous fields, whereas α_1 , α_2 , γ_1 , and γ_2 are constitutive coefficients associated with porosity and thermal interactions, respectively [8].

Substituting the assumed displacement representation given in equation (2) into the governing equations (3)-(6), the coupled system reduces to the following simplified form:

$$\mu \nabla^2 u = \rho \frac{\partial^2 u}{\partial t^2}, \quad (8)$$

$$\nabla^2(\alpha\phi + b_1\psi) - \alpha_1\phi - \alpha_3\psi + \gamma_1\Theta = \chi_1 \frac{\partial^2 \phi}{\partial t^2}, \quad (9)$$

$$\nabla^2(b_1\phi + \gamma\psi) - \alpha_3\phi - \alpha_2\psi + \gamma_2\Theta = \chi_2 \frac{\partial^2 \psi}{\partial t^2}, \quad (10)$$

$$K \nabla^2 \Theta - \gamma_1 T_0 \frac{\partial \phi}{\partial t} - \gamma_2 T_0 \frac{\partial \psi}{\partial t} - c_v \frac{\partial \Theta}{\partial t} = 0, \quad (11)$$

Where

$$c_v = aT_0.$$

These reduced equations describe the coupled propagation behavior of the displacement, porosity, and thermal fields within the thermoelastic double-porosity medium.

4. Harmonic Wave Solution

To analyze the propagation behavior of Love waves, the field variables are assumed to vary harmonically with space and time. Accordingly, the displacement, porosity, and temperature fields are expressed in the form

$$\{u, \phi, \psi, \Theta\}(x, z, t) = \{\bar{u}, \bar{\phi}, \bar{\psi}, \bar{\Theta}\}(z) e^{i(kx - \omega t)}, \quad (12)$$

where ω denotes the angular frequency and k represents the wave number.

Substituting the harmonic wave representation given in equation (12) into the reduced governing equations (8)-(11), the coupled partial differential equations transform into the following ordinary differential system:

$$\left(\frac{d^2}{dz^2} - k^2 + \frac{\omega^2}{c_t^2} \right) \bar{u} = 0, \quad (13)$$

$$\alpha \frac{d^2 \bar{\phi}}{dz^2} + b_1 \frac{d^2 \bar{\psi}}{dz^2} + (\omega^2 \chi_1 - \alpha_1 - \alpha k^2) \bar{\phi} + (-\alpha_3 - b_1 k^2) \bar{\psi} + \gamma_1 \bar{\Theta} = 0, \quad (14a)$$

$$b_1 \frac{d^2 \bar{\phi}}{dz^2} + \gamma \frac{d^2 \bar{\psi}}{dz^2} + (-\alpha_3 - b_1 k^2) \bar{\phi} + (\omega^2 \chi_2 - \alpha_2 - \gamma k^2) \bar{\psi} + \gamma_2 \bar{\Theta} = 0, \quad (14b)$$

$$K \left(\frac{d^2 \bar{\Theta}}{dz^2} - k^2 \bar{\Theta} \right) - \gamma_1 T_0 \dot{\bar{\phi}} - \gamma_2 T_0 \dot{\bar{\psi}} - c_v \dot{\bar{\Theta}} = 0. \quad (14c)$$

Introducing the shear-wave velocity $c_t = \sqrt{\frac{\mu}{\rho}}$, and using the relation $\omega = ck$, equation (13) can be rewritten as

$$\frac{d^2 \bar{u}^j}{dz^2} - k^2 \left(1 - \frac{c^2}{c_t^2} \right) \bar{u}^j = 0. \quad (15)$$

The above equations govern the propagation of coupled thermoelastic Love waves in the double-porosity layered structure.

The displacement field corresponding to the finite layer (L_1) and the underlying half-space (L_2) may be expressed in the following form:

$$\bar{u}^{(1)}(z) = G_1^{(1)} \cos(R_1 z) + G_2^{(1)} \sin(R_1 z), \quad (16)$$

$$\bar{u}^{(2)}(z) = G_1^{(2)} e^{-R_2 z}, \quad (17)$$

where $G_1^{(j)}$ and $G_2^{(1)}$ are arbitrary constants. The quantities R_1 and R_2 are given by $R_1 = k \sqrt{\frac{c^2}{c_{t1}^2} - 1}$, $R_2 = k \sqrt{1 - \frac{c^2}{c_{t2}^2}}$. To ensure bounded wave propagation in the half-space region, the exponentially increasing solution associated with equation (17) is omitted.

Further, equations (14a)-(14c) can be rewritten in a more compact form as

$$\alpha \bar{\phi}'' + b_1 \bar{\psi}'' - Q_{11} \bar{\phi} - Q_{12} \bar{\psi} + Q_{1\theta} \bar{\Theta} = 0, \quad (18a)$$

$$b_1 \bar{\phi}'' + \gamma \bar{\psi}'' - Q_{21} \bar{\phi} - Q_{22} \bar{\psi} + Q_{2\theta} \bar{\Theta} = 0, \quad (18b)$$

$$K \bar{\Theta}'' + Q_{3\phi} \bar{\phi} + Q_{3\psi} \bar{\psi} - Q_{33} \bar{\Theta} = 0, \quad (18c)$$

where

$$\begin{aligned} Q_{11} &= \alpha_1 + \alpha k^2 - \omega^2 \chi_1, Q_{12} = \alpha_3 + b_1 k^2, Q_{1\theta} = \gamma_1, \\ Q_{21} &= \alpha_3 + b_1 k^2, Q_{22} = \alpha_2 + \gamma k^2 - \omega^2 \chi_2, Q_{2\theta} = \gamma_2, \\ Q_{3\phi} &= i\omega \gamma_1 T_0, Q_{3\psi} = i\omega \gamma_2 T_0, Q_{33} = K k^2 - i\omega c_v. \end{aligned}$$

The above coupled system may alternatively be represented in matrix form as

$$L \begin{pmatrix} \bar{\phi} \\ \bar{\psi} \\ \bar{\Theta} \end{pmatrix} = 0, \quad (19a)$$

where

$$L = \begin{pmatrix} \alpha \frac{d^2}{dz^2} - Q_{11} & b_1 \frac{d^2}{dz^2} - Q_{12} & Q_{1\theta} \\ b_1 \frac{d^2}{dz^2} - Q_{21} & \gamma \frac{d^2}{dz^2} - Q_{22} & Q_{2\theta} \\ Q_{3\phi} & Q_{3\psi} & K \frac{d^2}{dz^2} - Q_{33} \end{pmatrix}. \quad (19b)$$

The matrix approach gives a concise representation of the coupled thermoelastic double-porosity wave system and assists in deriving the dispersion relation for Love-wave propagation in the considered layered structure.

For the existence of a nontrivial solution in $\bar{\phi}$, the variables $\bar{\psi}$ and $\bar{\Theta}$ are eliminated from equations (18a)-(18b). Consequently, the determinant of the coefficient matrix L must vanish, i.e.,

$$\det L = 0.$$

This condition yields the following governing differential equation:

$$\sum_{n=0}^3 (-1)^n E_n \nabla^{2(3-n)} \bar{\phi} = 0, \quad (20)$$

where the coefficients E_n are defined as

$$\begin{aligned} E_0 &= K(\alpha\gamma - b_1^2), \\ E_1 &= -[K(\alpha Q_{22} + \gamma Q_{11} - 2b_1 Q_{21}) + Q_{33}(\alpha\gamma - b_1^2)] + [Q_{2\Theta}(b_1 Q_{3\phi} - \alpha Q_{3\psi}) + Q_{1\Theta}(b_1 Q_{3\psi} \\ &\quad - \gamma Q_{3\phi})], \\ E_2 &= K(Q_{11} Q_{22} - Q_{12} Q_{21}) + Q_{33}(\alpha Q_{22} + \gamma Q_{11} - 2b_1 Q_{21}) \\ &\quad - Q_{1\Theta}(Q_{3\phi} Q_{22} - Q_{3\psi} Q_{21}) + Q_{2\Theta}(Q_{3\psi} Q_{11} - Q_{3\phi} Q_{12}), \\ E_3 &= -Q_{33}(Q_{11} Q_{22} - Q_{12} Q_{21}) + Q_{1\Theta}(Q_{3\phi} Q_{22} - Q_{3\psi} Q_{21}) \\ &\quad - Q_{2\Theta}(Q_{3\psi} Q_{11} - Q_{3\phi} Q_{12}). \end{aligned}$$

Since the field variables are mutually coupled through equations (18e)-(18g), the variables $\bar{\psi}$ and $\bar{\Theta}$ can be expressed in terms of $\bar{\phi}$ as

$$\bar{\psi} = \sum_{j=1}^3 \xi_{2j} \bar{\phi}_j, \quad \bar{\Theta} = \sum_{j=1}^3 \xi_{3j} \bar{\phi}_j, \quad (21)$$

where

$$\begin{aligned} \xi_{2j} &= \frac{Q_{1\Theta} p_j - m_j Q_{2\Theta}}{n_j Q_{2\Theta} - Q_{1\Theta} q_j}, \\ \xi_{3j} &= \frac{m_j q_j - n_j p_j}{n_j Q_{2\Theta} - Q_{1\Theta} q_j}, \end{aligned}$$

with

$$\begin{aligned} m_j &= \alpha\mu_j - Q_{11}, \quad n_j = b_1\mu_j - Q_{12}, \\ p_j &= b_1\mu_j - Q_{21}, \quad q_j = \gamma\mu_j - Q_{22}, \\ r_j &= K\mu_j - Q_{33}. \end{aligned}$$

Equation (20) gives rise to three distinct wave modes, which may be written as $\bar{\phi} = \sum_{j=1}^3 \phi_j$, where each mode satisfies

$$\left(\frac{d^2}{dz^2} - k^2 + \frac{\omega^2}{c_j^2}\right)\phi_j = 0. \quad (22)$$

Hence, the complete solution corresponding to equations (20)-(21) can be expressed as

$$\{\phi, \psi, \Theta\} = \sum_{j=1}^3 \{1, \xi_{2j}, \xi_{3j}\} (U_j e^{S_j z} + V_j e^{-S_j z}) e^{i(kx - \omega t)}, \quad (23)$$

where U_j and V_j are arbitrary constants, $\omega = ck$, $S_j = k \sqrt{1 - \frac{c^2}{c_j^2}}$. The above expressions describe the coupled thermo-porous wave modes propagating through the considered layered structure.

5. Interface and Boundary Constraints

Since the governing equations are formulated for a bounded layered medium, suitable boundary conditions must be imposed to characterize the coupled stress, displacement, porosity, and thermal behavior at the boundaries of the structure.

1. The upper surface of the layer L_1 , located at $z = -H$, is assumed to be traction-free. Therefore, the shear stress component satisfies

$$\sigma_{23} = 0 \text{ at } z = -H. \quad (24a)$$

Further, the upper boundary is considered thermally insulated, which leads to

$$\frac{\partial \Theta}{\partial z} = 0 \text{ at } z = -H. \quad (24b)$$

2. The regions L_1 and L_2 are assumed to remain in perfect mechanical and thermal contact. Consequently, the following continuity conditions are imposed at the irregular interface

$$z = \varepsilon \sin(kx).$$

Continuity of displacement field

$$u^{(1)} = u^{(2)},$$

$$u^{(1)}(0) + \varepsilon \sin(kx) \frac{\partial u^{(1)}}{\partial z} = u^{(2)}(0) + \varepsilon \sin(kx) \frac{\partial u^{(2)}}{\partial z}. \quad (25a)$$

Continuity of stress field

$$\begin{aligned} \sigma_{23}^{(1)} &= \sigma_{23}^{(2)}, \\ \sigma_{32}^{(1)} + \varepsilon \sin(kx) \frac{\partial \sigma_{32}^{(1)}}{\partial z} &= \sigma_{32}^{(2)} + \varepsilon \sin(kx) \frac{\partial \sigma_{32}^{(2)}}{\partial z}. \end{aligned} \quad (25b)$$

Continuity of porosity fields

$$\phi^{(1)} = \phi^{(2)}, \psi^{(1)} = \psi^{(2)}. \quad (25c)$$

Continuity of thermal field

$$\Theta^{(1)} = \Theta^{(2)}. \quad (25d)$$

Since the interface geometry is irregular and described by $z = \varepsilon \sin(kx)$, $\varepsilon \ll 1$, the boundary conditions are expanded about the mean interface position $z = 0$ by applying the Taylor series approximation. Higher-order terms associated with the irregularity parameter are neglected due to the smallness of ε .

Thus, for any field quantity A , the approximation becomes $A(z = \varepsilon \sin(kx)) \approx A(0) + \varepsilon \sin(kx) \frac{\partial A}{\partial z}$. These boundary conditions govern the interaction of the coupled thermoelastic and porosity fields across the corrugated interface separating the two media.

6. Dispersion Relation Formulation

Substituting the solutions obtained from equations (16)-(17) and (23) into the boundary conditions given by equations (24a)-(25d), a system of linear algebraic equations involving the unknown constants E_j , F_j , and G_j , where $j = 1, 2, 3$, is obtained. The resulting system may be expressed in compact matrix form as

$$(E + \varepsilon E_0)X = 0, \quad (26a)$$

where

$$X = \left[G_1^{(1)} \ G_2^{(1)} \ G_1^{(2)} \ E_1^{(1)} \ E_2^{(1)} \ E_3^{(1)} \ F_1^{(1)} \ F_2^{(1)} \ F_3^{(1)} \ F_1^{(2)} \ F_2^{(2)} \ F_3^{(2)} \right]^T,$$

and E is a 12×12 coefficient matrix defined by

$$E = \begin{bmatrix} F_{3 \times 3} & 0 \\ 0 & H_{9 \times 9} \end{bmatrix}, E_0 = \begin{bmatrix} F_{0 \ 3 \times 3} & 0 \\ 0 & H_{0 \ 9 \times 9} \end{bmatrix}.$$

The matrix F is given by

$$F = \begin{bmatrix} R_1 e^{-R_1 H} & -R_1 e^{R_1 H} & 0 \\ \mu_1 R_1 & -\mu_1 R_1 & \mu_2 R_2 \\ 1 & 1 & -1 \end{bmatrix}, \quad (26b)$$

while

$$F_0 = \begin{bmatrix} R_1 & -R_1 & R_2 \\ \mu_1 R_1^2 & \mu_1 R_1^2 & \mu_2 R_2^2 \\ 0 & 0 & 0 \end{bmatrix}. \quad (26c)$$

The above matrix representation characterizes the coupled thermoelastic and porosity interactions responsible for Love-wave propagation in the layered structure with a corrugated interface.

The entries of the coupled-field block H are generated using the thermo-porous wave numbers $S_j^{(i)}$ together with the coupling parameters $H_j^{(i)}$ and $B_j^{(i)}$, and are written as

$$\begin{aligned} J_{1,j} &= H_j^{(1)} S_j^{(1)} e^{-S_j^{(1)} H}, J_{1,j+3} = -H_j^{(1)} S_j^{(1)} e^{S_j^{(1)} H}, \\ J_{2,j} &= B_j^{(1)} S_j^{(1)} e^{-S_j^{(1)} H}, J_{2,j+3} = -B_j^{(1)} S_j^{(1)} e^{S_j^{(1)} H}, \\ J_{3,j} &= H_j^{(1)} S_j^{(1)}, J_{3,j+3} = -H_j^{(1)} S_j^{(1)}, J_{3,j+6} = H_j^{(2)} S_j^{(2)}, \\ J_{4,j} &= B_j^{(1)} S_j^{(1)}, J_{4,j+3} = -B_j^{(1)} S_j^{(1)}, J_{4,j+6} = B_j^{(2)} S_j^{(2)}, \\ J_{5,j} &= J_{5,j+3} = -J_{5,j+6} = 1, \\ J_{6,j} &= J_{6,j+3} = \xi_{2j}^{(1)}, J_{6,j+6} = -\xi_{2j}^{(2)}, \\ J_{7,j} &= J_{7,j+3} = \xi_{3j}^{(1)}, J_{7,j+6} = -\xi_{3j}^{(2)}, \\ J_{8,j} &= -J_{8,j+3} = K^{(1)} S_j^{(1)} \xi_{3j}^{(1)}, \\ J_{8,j+6} &= K^{(2)} S_j^{(2)} \xi_{3j}^{(2)}, \\ J_{9,j} &= \xi_{3j}^{(1)} S_j^{(1)} e^{-S_j^{(1)} H}, J_{9,j+3} = -\xi_{3j}^{(1)} S_j^{(1)} e^{S_j^{(1)} H}, \end{aligned}$$

where

$$H_j^{(i)} = \alpha^{(i)} + b_1^{(i)} \xi_{2j}^{(i)},$$

and

$$B_j^{(i)} = b_1^{(i)} + \gamma^{(i)} \xi_{2j}^{(i)}.$$

Similarly, the coupled-field block H_0 takes the form

$$\begin{aligned} J_{1,j}^{(1)} &= S_j^{(1)}, J_{1,j+3}^{(1)} = -S_j^{(1)}, J_{1,j+6}^{(1)} = S_j^{(2)}, \\ J_{2,j}^{(1)} &= \xi_{2j}^{(1)} S_j^{(1)}, J_{2,j+3}^{(1)} = -\xi_{2j}^{(1)} S_j^{(1)}, J_{2,j+6}^{(1)} = \xi_{2j}^{(2)} S_j^{(2)}, \\ J_{3,j}^{(1)} &= \xi_{3j}^{(1)} S_j^{(1)}, J_{3,j+3}^{(1)} = -\xi_{3j}^{(1)} S_j^{(1)}, J_{3,j+6}^{(1)} = \xi_{3j}^{(2)} S_j^{(2)}, \\ J_{4,j}^{(1)} &= K^{(1)} S_j^{(1)2}, J_{4,j+6}^{(1)} = K^{(2)} S_j^{(2)2}. \end{aligned}$$

For a nontrivial solution of equation (26a) to exist, the determinant of the matrix E must vanish. Therefore,

$$|E^{(0)} + \varepsilon E^{(1)}| = 0. \quad (27)$$

Expanding the determinant in equation (27) and neglecting the terms containing second and higher powers of ε ($\varepsilon \ll 1$), one obtains

$$|E^{(0)}| + \varepsilon \frac{\partial |E|}{\partial \varepsilon} = 0, \quad (28a)$$

which further reduces to

$$|E| = |F| \cdot |J| = 0. \quad (28b)$$

By setting $|F| = 0$, and using the elements of matrix F from equation (26b), the governing relation corresponding to the classical mechanical wave front is obtained as $\mu_1 R_1 (e^{-R_1 H} - e^{R_1 H}) = \mu_2 R_2 (e^{-R_1 H} + e^{R_1 H})$. Substituting the expressions for R_1 and R_2 , the dispersion relation for the first Love-wave front becomes

$$\tan \left(kH \sqrt{\frac{c^2}{c_{t1}^2} - 1} \right) = \frac{\mu_2 \sqrt{1 - \frac{c^2}{c_{t2}^2}}}{\mu_1 \sqrt{\frac{c^2}{c_{t1}^2} - 1}}. \quad (29)$$

Equation (29) coincides with the classical dispersion relation for Love waves [36]. This wave mode remains unaffected by the porosity parameters, thermal field, and interface corrugation effects.

Furthermore, from equation (26a), the dispersion relation associated with the second wave front is obtained as

$$|J| + \varepsilon J_0 = 0, \quad (30)$$

where the perturbation term εJ_0 arises due to the first-order correction associated with the irregular boundary conditions and depends explicitly on the vertical wave numbers S_j and the thermoporous coupling coefficients.

After simplification, the explicit dispersion equation governing the second Love-wave front can be written as

$$\det[J(c, k, \omega)] + \varepsilon k \mathcal{F}(\omega, k, S_j, R, \xi_{2j}, \xi_{3j}) = 0. \quad (31a)$$

where the irregularity correction function $\mathcal{F}$ is written as $\mathcal{F} = \sum_{j=1}^3 [\mathcal{F}_j^{(\phi)} + \mathcal{F}_j^{(\psi)} + \mathcal{F}_j^{(\Theta)} + \mathcal{F}_j^{(\text{mech})}]$, and the individual components are defined as follows.

Void coupling (first kind)

$$\mathcal{F}_j^{(\phi)} = \frac{S_j^{(1)} - S_j^{(2)}}{S_j^{(1)} + S_j^{(2)}} (H_j^{(1)} S_j^{(1)} + H_j^{(2)} S_j^{(2)}), \quad (31b)$$

with

$$H_j^{(l)} = \alpha^{(l)} + b_1^{(l)} \xi_{2j}^{(l)}.$$

Void coupling (second kind)

$$\mathcal{F}_j^{(\psi)} = \frac{S_j^{(1)} - S_j^{(2)}}{S_j^{(1)} + S_j^{(2)}} (B_j^{(1)} S_j^{(1)} + B_j^{(2)} S_j^{(2)}) \cdot \frac{\xi_{2j}^{(1)} + \xi_{2j}^{(2)}}{2}, \quad (31c)$$

where

$$B_j^{(l)} = b_1^{(l)} + \gamma^{(l)} \xi_{2j}^{(l)}, l = 1, 2.$$

Thermal coupling

$$\mathcal{F}_j^{(\Theta)} = \frac{S_j^{(1)} - S_j^{(2)}}{S_j^{(1)} + S_j^{(2)}} (K_1 S_j^{(1)} \xi_{3j}^{(1)} + K_2 S_j^{(2)} \xi_{3j}^{(2)}) \frac{\xi_{3j}^{(1)} + \xi_{3j}^{(2)}}{2T_0}, \quad (31d)$$

Mechanical coupling

$$\mathcal{F}_j^{(\text{mech})} = \frac{R_1 - R_2}{R_1 + R_2} (\mu_1 R_1 + \mu_2 R_2). \quad (31e)$$

The derived dispersion relation serves as the fundamental equation for investigating the effects of porosity, thermal interaction, and interface corrugation on the propagation characteristics of the second Love-wave front. The obtained relation is further utilized for numerical computations and graphical analysis in the subsequent section.

The employed analytical technique offers several advantages for the present coupled problem. The harmonic-wave representation reduces the governing partial differential equations to a tractable system of ordinary differential equations, while the matrix formulation enables the mechanical, thermal, and two porosity fields to be treated within a unified framework. Furthermore, the first-order treatment of the corrugated interface incorporates the influence of small interface irregularities without requiring full-domain numerical discretization. The resulting dispersion formulation is computationally efficient and facilitates systematic evaluation of phase velocity, attenuation, thermal penetration, stress localization, and mode-dependent wave behavior over a wide range of governing parameters.

7. Surface Stress and Field Responses

The stress components and equilibrated void stresses associated with Love-wave propagation are obtained directly from the constitutive relations (7a)-(7d). These expressions describe the influence of the coupled mechanical, porous, and thermal fields throughout the layer and the underlying half-space, together with the effect of interface corrugation up to the first order of the irregularity parameter ε .

The shear stress component σ_{32} corresponding to L_1 and L_2 is given by

$$\sigma_{32}^{(1)} = \mu_1 [R_1 (G_1^{(1)} e^{R_1 z} - G_2^{(1)} e^{-R_1 z}) + \varepsilon \sin(kx) R_1^2 (G_1^{(1)} e^{R_1 z} + G_2^{(1)} e^{-R_1 z})] e^{i(kx - \omega t)}, \quad (32a)$$

$$\sigma_{32}^{(2)} = -\mu_2 R_2 G_1^{(2)} [1 - \varepsilon \sin(kx) R_2] e^{-R_2 z} e^{i(kx - \omega t)}. \quad (32b)$$

The equilibrated stress vectors associated with the first and second void systems are expressed as follows.

For the first void system,

$$\left(\sigma_3^{(\phi)} \right)^{(1)} = \sum_{j=1}^3 H_j^{(1)} S_j^{(1)} [(E_j^{(1)} e^{S_j^{(1)} z} - F_j^{(1)} e^{-S_j^{(1)} z})]$$

$$+\varepsilon\sin(kx)S_j^{(1)}\left(E_j^{(1)}e^{S_j^{(1)}z}+F_j^{(1)}e^{-S_j^{(1)}z}\right)]e^{i(kx-\omega t)}, \quad (33a)$$

$$(\sigma_3^{(1)})^{(2)} = -\sum_{j=1}^3 H_j^{(2)} S_j^{(2)} F_j^{(2)} [1 - \varepsilon\sin(kx)S_j^{(2)}]e^{-S_j^{(2)}z}e^{i(kx-\omega t)}. \quad (33b)$$

For the second void system,

$$\begin{aligned} \left(\sigma_3^{(2)}\right)^{(1)} &= \sum_{j=1}^3 B_j^{(1)} S_j^{(1)} \left[\left(E_j^{(1)} e^{S_j^{(1)}z} - F_j^{(1)} e^{-S_j^{(1)}z} \right) \right. \\ &\quad \left. + \varepsilon\sin(kx)S_j^{(1)}\left(E_j^{(1)}e^{S_j^{(1)}z}+F_j^{(1)}e^{-S_j^{(1)}z}\right) \right] e^{i(kx-\omega t)}, \end{aligned} \quad (34a)$$

$$(\sigma_3^{(2)})^{(2)} = -\sum_{j=1}^3 B_j^{(2)} S_j^{(2)} F_j^{(2)} [1 - \varepsilon\sin(kx)S_j^{(2)}]e^{i(kx-\omega t)}. \quad (34b)$$

The obtained expressions reveal several significant physical characteristics:

- (i) amplification of the field quantities near the corrugation crests accompanied by attenuation near the trough regions;
- (ii) concentration of stresses in the vicinity of the interface, which becomes more pronounced for higher-order wave modes;
- (iii) intensified thermoelastic interaction at the corrugated interface, resulting in enhanced thermal damping effects;
- (iv) gradual decay of the disturbance induced by the irregularity with increasing depth, characterized by a penetration depth proportional to S_j^{-1} .

8. Numerical Results and Discussion

This section presents the numerical results for Love-wave propagation in a thermoelastic double-porosity layered structure with a periodically corrugated interface. The analytical dispersion relations and associated field expressions are evaluated numerically using MATLAB. To maintain the physical consistency of the model, the material parameters employed in the computations are adopted from [14] and [35]. The numerical analysis examines the effects of corrugation amplitude, corrugation wavenumber, thermal interaction, porosity-related parameters, frequency, and propagation modes on the coupled thermo-mechanical response of the system.

The phase velocity and attenuation coefficients are expressed in dimensionless form to facilitate comparison under different physical conditions. In addition to the dispersion and attenuation characteristics, the thermal response is examined through normalized temperature and heat-flux distributions with penetration depth. The subsequent results therefore provide a combined

assessment of thermal transport, wave dispersion, attenuation, stress penetration, and coupled thermo-porosity behavior.

Validation of the formulation through a limiting case

To verify the consistency of the proposed formulation, a limiting-case analysis is performed by considering $\varepsilon = 0$. Under this condition, the periodically corrugated interface reduces to a smooth interface, and all corrugation-dependent contributions in the dispersion relation vanish. The resulting formulation therefore represents Love-wave propagation in a thermoelastic double-porosity layered medium with a smooth interface. Mathematically, this limiting condition can be represented as

$$\varepsilon \rightarrow 0 \Rightarrow \text{corrugated interface} \rightarrow \text{smooth interface}.$$

The numerical solution corresponding to $\varepsilon = 0$ exhibits a smooth dispersion response and approaches the expected limiting behavior with increasing dimensionless frequency. The successful recovery of the smooth-interface configuration provides a useful limiting-case consistency check for the proposed corrugated-interface formulation.

8.1 Thermal-Transport Characteristics of the Coupled System

In the present formulation, the thermal field is intrinsically coupled with the mechanical and double-porosity fields through the governing thermoelastic equations. Consequently, variations in temperature interact with elastic deformation and the two porosity fields, resulting in a coupled redistribution of thermal and mechanical energy within the layered medium. The thermal response is therefore characterized in terms of normalized temperature and heat flux with dimensionless penetration depth, which provides direct insight into the localization and penetration of thermal disturbances within the medium.

The thermal-field analysis complements the wave-propagation results by establishing the connection between thermal transport and the attenuation of coupled disturbances away from the interface. It also provides a physical basis for interpreting how thermo-porous interactions influence energy localization within the layered structure. The corresponding temperature and heat-flux characteristics are presented in Figs. 18 and 19, followed by the dispersion, attenuation, and stress-response analyses.

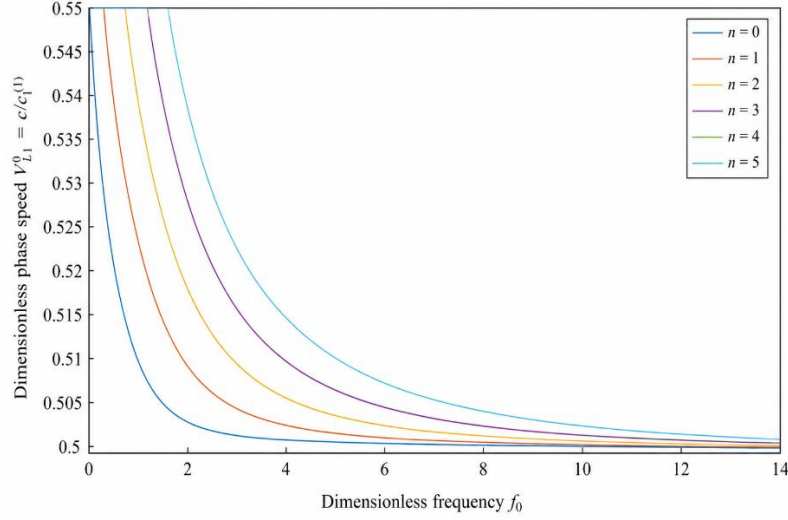


Fig. 2. Variation of dimensionless phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of corrugation wave number n ($n = 0-5$) with $\varepsilon = 0.2$, $m_1 = 0.3$, $m_2 = 0.2$, $\alpha_T = 0.1$, $d_1 = 0.05$, and $d_2 = 0.05$.

Figure 2 illustrates the variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ of Love waves with respect to the dimensionless frequency f_0 for different corrugation wave numbers n ranging from 0 to 5. It is observed that the phase speed decreases continuously with increasing frequency for all considered values of n and gradually approaches an asymptotic value close to 0.50 in the high-frequency region. The findings also show that the low-frequency propagation behavior is significantly influenced by the corrugation wave number. The phase speed rapidly drops and reaches the limiting value in a shorter frequency range for decreasing values of n . Larger values of n , on the other hand, sustain higher phase speeds over a greater frequency range, leading to a slower decay characteristic. For example, the curve corresponding to $n = 0$ approaches the limiting region near $f_0 \approx 4$, whereas for $n = 5$ the transition occurs near $f_0 \approx 12$. This behavior demonstrates that increasing the corrugation wave number delays the dispersion effect and enhances wave propagation in the low-frequency region. Physically, increasing the corrugation wave number strengthens the interaction between the guided wave and the periodically irregular interface, thereby modifying interfacial energy confinement and delaying the dispersive reduction in phase velocity.

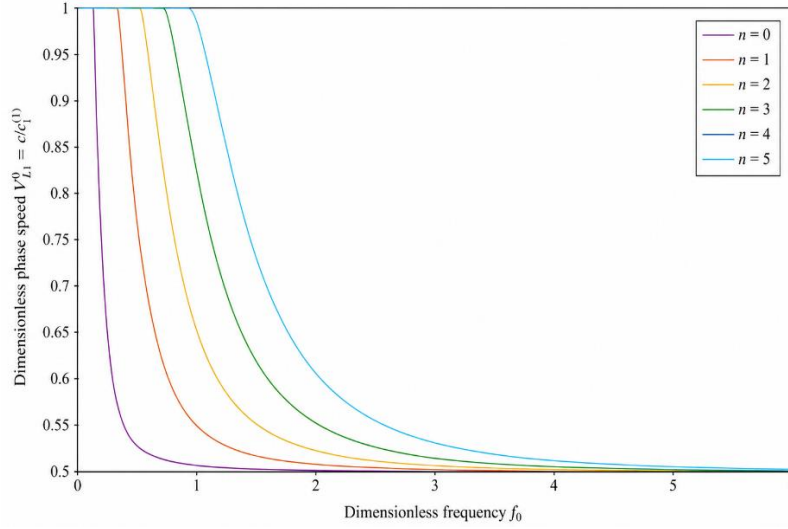


Fig. 3. Variation of dimensionless phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of corrugation wave number n ($n = 0-5$).

Figure 3 presents the variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ of Love waves with dimensionless frequency f_0 for different corrugation wave numbers n . It is observed that the phase speed decreases sharply in the low-frequency region and gradually approaches an asymptotic value close to 0.50 as the frequency increases. The findings demonstrate that the dispersion curves are considerably shifted into higher frequency regions by increasing the corrugation wave number. The phase speed rapidly decreases and reaches the limiting zone in a brief frequency period for smaller values of n . Larger values of n , on the other hand, sustain relatively higher phase speeds across a greater frequency range. For example, the curve corresponding to $n = 0$ approaches the limiting value near $f_0 \approx 1.5$, whereas the curve for $n = 5$ reaches the same region near $f_0 \approx 5$. This trend suggests that higher corrugation wave numbers improve the propagation properties in the low-frequency domain and postpone the dispersive decay of Love waves. The suggested thermoelastic double-porosity wave propagation model's physical stability is further supported by the curves' smooth convergence toward the same limiting value. This behavior arises because a larger corrugation wave number increases the spatial variation of the interface, strengthening wave-interface interactions and allowing higher phase velocities to persist over a wider frequency range.

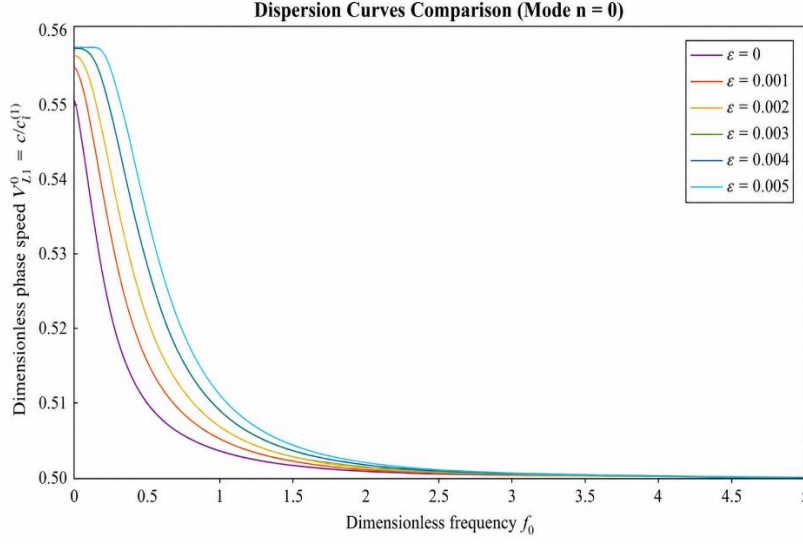


Fig. 4. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the fundamental mode ($n = 0$).

Figure 4 illustrates the influence of the corrugation amplitude parameter ε on the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ of Love waves for the fundamental propagation mode ($n = 0$). It is observed that the phase speed decreases continuously with increasing dimensionless frequency f_0 for all considered values of the corrugation amplitude. The results further show that increasing the corrugation amplitude shifts the dispersion curves upward in the low-frequency region, indicating that interface corrugation enhances the phase speed of Love waves. For smaller values of ε , the curves approach the limiting value near $V_{L_1}^0 \approx 0.50$ more rapidly, whereas larger corrugation amplitudes maintain comparatively higher phase speeds over a wider frequency range. Physically, greater corrugation produces stronger interfacial modulation, which modifies the effective wave-guiding conditions and promotes greater energy confinement near the interface, thereby maintaining a comparatively higher phase velocity. For smaller corrugation amplitudes, the phase speed approaches its limiting value more rapidly, whereas larger amplitudes sustain higher phase speeds over a wider frequency range. At higher frequencies, the curves gradually converge toward the same asymptotic value, showing that the influence of interface corrugation becomes progressively weaker. Thus, interface corrugation primarily controls the low- and intermediate-frequency dispersion characteristics of the thermoelastic double-porosity layered structure.

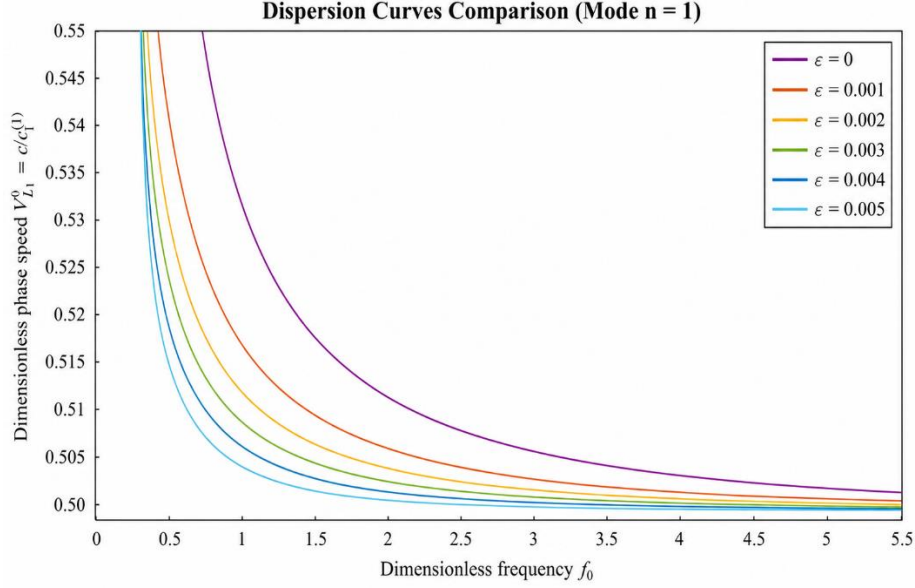


Fig. 5. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the propagation mode ($n = 1$).

Figure 5 illustrates the variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for six different values of the corrugation amplitude parameter ε in the case of the first propagation mode ($n = 1$). It is observed that all curves exhibit a rapid decrease in the low-frequency region and gradually approach the asymptotic limiting value near 0.50 as the frequency increases. The acquired data unequivocally show that increasing the corrugation amplitude causes the dispersion curves to shift upward, indicating a stronger interface irregularity that enhances the phase velocity of Love waves. Physically, an increase in ε modifies the local wave-guiding conditions at the interface and strengthens the confinement of wave energy within the guiding region, thereby maintaining a comparatively higher phase speed. While bigger corrugation amplitudes maintain relatively higher propagation speeds throughout the examined frequency range, the curve corresponding to the smooth interface scenario ($\varepsilon = 0$) has the lowest phase speed. In the low- and intermediate-frequency areas, the divergence between the curves is more noticeable, indicating the important role that corrugated interfaces play in altering the dispersive behavior of Love waves. This stronger sensitivity at lower frequencies can be associated with the greater interaction of the guided wave with the overall corrugated interface geometry. At higher frequencies, all curves gradually converge toward nearly the same limiting value, indicating that the influence of corrugation diminishes with increasing frequency. This convergence suggests that the propagation becomes progressively less sensitive to interface geometry in the high-frequency regime. The smooth and stable convergence of the dispersion curves further supports the physical consistency of the proposed thermoelastic double-porosity model.

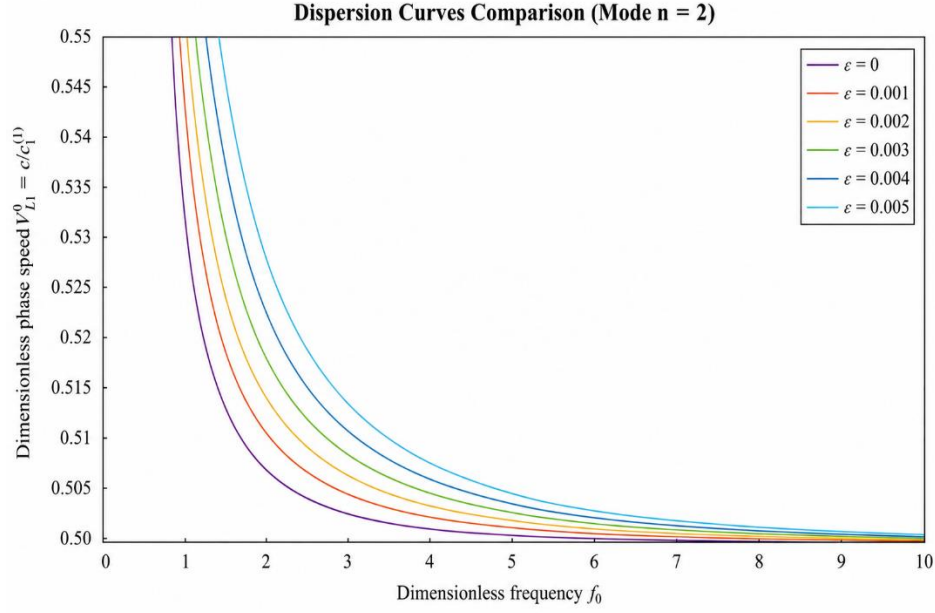


Fig. 6. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the propagation mode ($n = 2$).

Figure 6 presents the variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for six different values of the corrugation amplitude parameter ε in the case of the second propagation mode ($n = 2$). It is observed that all dispersion curves decrease continuously with increasing frequency and asymptotically approach the limiting value near 0.50.

The results clearly indicate that increasing the corrugation amplitude shifts the dispersion curves upward throughout the propagation range. The curve corresponding to $\varepsilon = 0$ exhibits the lowest phase speed, whereas the highest corrugation amplitude $\varepsilon = 0.005$ produces the maximum phase velocity over the entire frequency domain. Physically, increasing ε strengthens the geometric modulation of the interface and modifies the effective wave-guiding conditions, leading to stronger confinement of the second-mode wave energy and consequently a higher phase velocity. The separation between the curves is more noticeable in the low- and intermediate-frequency regions, where corrugation has a particularly strong influence. This enhanced sensitivity occurs because the guided wave interacts more effectively with the interface geometry in these frequency ranges, making its propagation characteristics more responsive to changes in corrugation amplitude. At higher frequencies, all curves progressively converge toward the same asymptotic value, suggesting that the influence of interface irregularity diminishes. The convergence indicates that the second propagation mode becomes progressively less sensitive to the large-scale corrugated geometry as the frequency increases. The smooth convergence behavior further supports the numerical stability and physical consistency of the proposed dispersion relation.

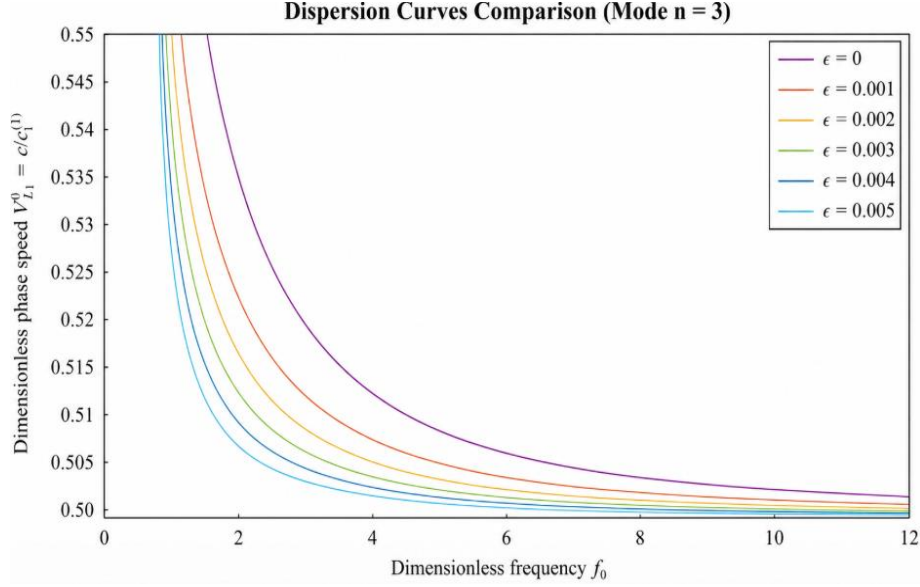


Fig. 7. Comparison of dispersion curves for different corrugation amplitudes ε ($0 \leq \varepsilon \leq 0.005$) corresponding to the propagation mode ($n = 3$).

Figure 7 illustrates the variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for six different values of the corrugation amplitude parameter ε in the case of the third propagation mode ($n = 3$). It is observed that all curves decrease gradually with increasing frequency and asymptotically approach the limiting value near 0.50. The findings show that for increasing propagation modes, the impact of corrugation amplitude increases. Greater values of ε sustain relatively higher phase speeds across a wider frequency range and shift the dispersion curves upward. On the other hand, across the domain under consideration, the smooth interface situation ($\varepsilon = 0$) shows the lowest phase velocity. Physically, a larger corrugation amplitude produces stronger geometric modulation of the interface, which modifies the effective guiding conditions and enhances the confinement of wave energy near the interface, thereby sustaining a higher phase velocity. Higher-order propagation modes are more susceptible to interface irregularity, as evidenced by the separation between the curves that persists even in the intermediate- and high-frequency areas. This greater sensitivity of the third mode can be associated with its more complex depth-dependent field distribution, which produces a stronger interaction with the periodically corrugated interface. Additionally, the proposed thermoelastic double-porosity dispersion model's physical consistency and stability are confirmed by the curves' slow convergence toward the same asymptotic value. The gradual convergence further indicates that the influence of corrugation weakens with increasing frequency, although it remains more pronounced for the higher-order mode than for the lower-order modes.

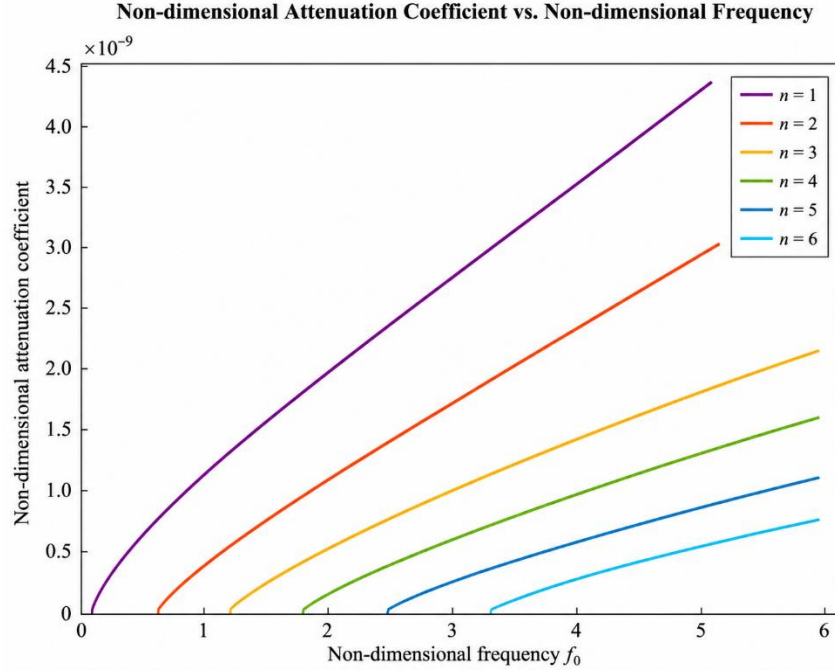


Fig. 8. Variation of the non-dimensional attenuation coefficient with dimensionless frequency for different propagation modes $n = 1 - 6$.

Figure 8 illustrates the dependence of the non-dimensional attenuation coefficient on the dimensionless frequency for six different propagation modes. It is observed that the attenuation coefficient increases continuously with increasing frequency for all considered modes, indicating stronger energy dissipation at higher frequencies. Physically, the increase in attenuation with frequency can be attributed to the stronger interaction among the mechanical, thermal, and two porosity fields at shorter wavelengths, which promotes greater redistribution and dissipation of the propagating wave energy. The findings also show that attenuation levels for lower-order modes are significantly higher than those for higher-order modes. Specifically, the mode corresponding to $n = 1$ exhibits the maximum attenuation across the frequency range under consideration, whereas the mode $n = 6$ yields the lowest attenuation coefficient. The stronger attenuation of the lower-order modes indicates a greater coupling of their localized wave energy with the thermoelastic and double-porosity fields, whereas higher-order modes experience comparatively weaker dissipative interaction. This behavior shows that wave energy can penetrate deeper into the thermoelastic double-porosity medium and that higher propagation modes have weaker damping effects. Additionally, the attenuation curves exhibit smooth nonlinear development features, supporting the suggested thermoelastic model's steady dissipative behavior. Further evidence that the propagation mode has a major impact on the attenuation properties of Love waves in the corrugated layered structure comes from the curves' growing separation with increasing frequency. The increasing separation further suggests that the mode-dependent distribution of energy becomes more influential at higher frequencies, producing distinct attenuation responses for different propagation modes.

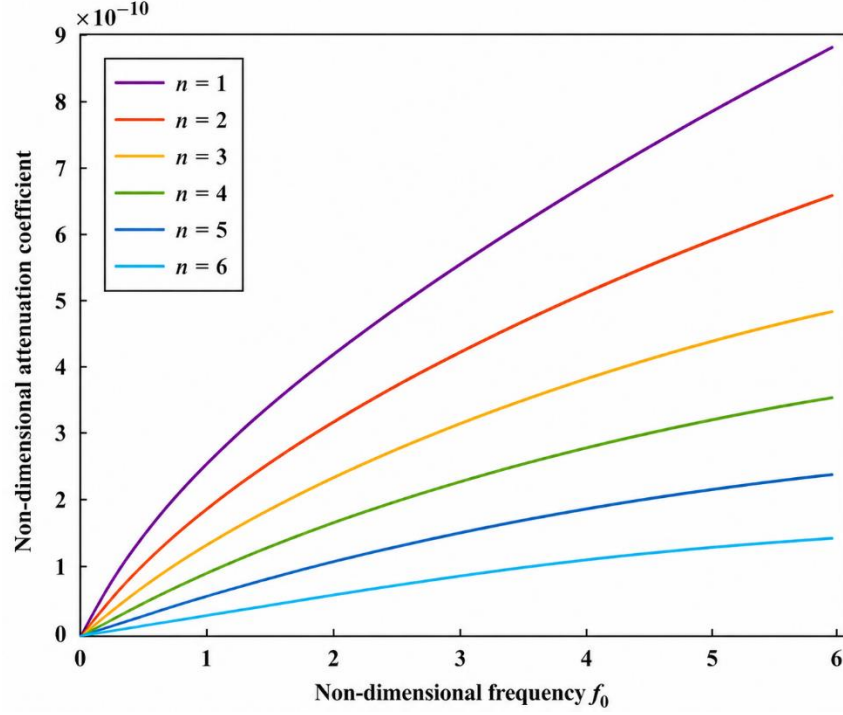


Fig. 9. Variation of the non-dimensional attenuation coefficient with dimensionless frequency f_0 for different propagation modes $n = 1 - 6$.

For six distinct propagation modes, Figure 9 shows how the normalized attenuation coefficient changes in relation to the dimensionless frequency. For all values of n taken into consideration, it is shown that the attenuation coefficient grows steadily with frequency, suggesting stronger dissipative effects at higher frequencies. Physically, higher frequencies strengthen thermo-porosity interactions, resulting in greater energy dissipation. The observed results further show that, in comparison to the higher modes, the lower propagation modes have bigger attenuation coefficients. Specifically, the mode with $n = 1$ exhibits the highest attenuation across the examined frequency range, whereas the mode with $n = 6$ yields the lowest attenuation values. This trend implies that higher-order Love-wave modes allow more wave penetration inside the thermoelastic double-porosity layered medium and undergo relatively less damping. Lower-order modes interact more strongly with the coupled thermo-porous region and therefore experience greater attenuation than higher-order modes. Additionally, the attenuation curves exhibit smooth nonlinear development features, supporting the suggested thermoelastic model's stable energy dissipation behavior. The propagation mode has a considerable impact on the attenuation response of Love waves in the presence of interface corrugation and thermo-porosity coupling effects, as seen by the increasing separation between the curves with frequency. The increasing curve separation reflects the stronger influence of mode-dependent energy distribution at higher frequencies.

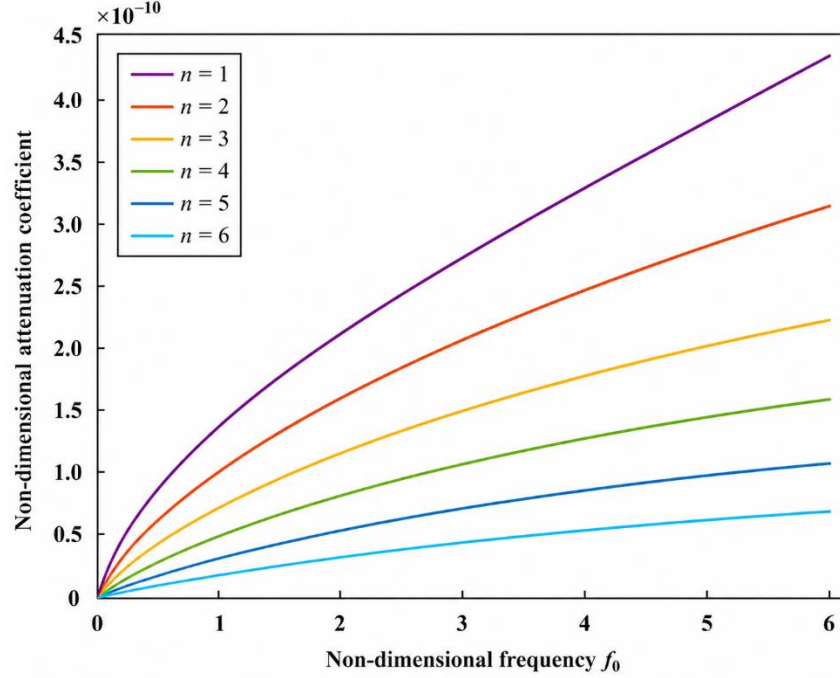


Fig. 10. Variation of the non-dimensional attenuation coefficient with dimensionless frequency f_0 for different propagation modes $n = 1 - 6$.

The normalized attenuation coefficient variation as a function of the dimensionless frequency for six distinct propagation modes is shown in Figure 10. For every mode taken into consideration, it is evident that the attenuation coefficient rises monotonically with frequency, suggesting more dissipative and energy-loss effects at higher frequencies. Physically, increasing frequency strengthens the interaction among the mechanical, thermal, and porosity fields, leading to greater energy dissipation. The findings also show that the propagation mode has a significant impact on the attenuation characteristics. Over the whole frequency range, the lower-order mode ($n = 1$) displays the highest attenuation coefficient, whereas higher-order modes display relatively lower attenuation values. Specifically, the mode $n = 6$ has the least attenuation, suggesting that the Love-wave energy penetrates the thermoelastic double-porosity layered medium more deeply and with less damping. The higher attenuation of lower-order modes results from their stronger energy confinement and interaction with the coupled thermo-porous region. Additionally, all curves show smooth nonlinear growth behavior without instability, demonstrating the physical validity and numerical consistency of the suggested thermoelastic wave propagation model. The effect of the propagation mode also becomes increasingly prominent in the higher-frequency domain, as evidenced by the growing gap between successive curves with increasing frequency. The growing separation between the curves indicates that mode-dependent energy dissipation becomes more pronounced at higher frequencies.

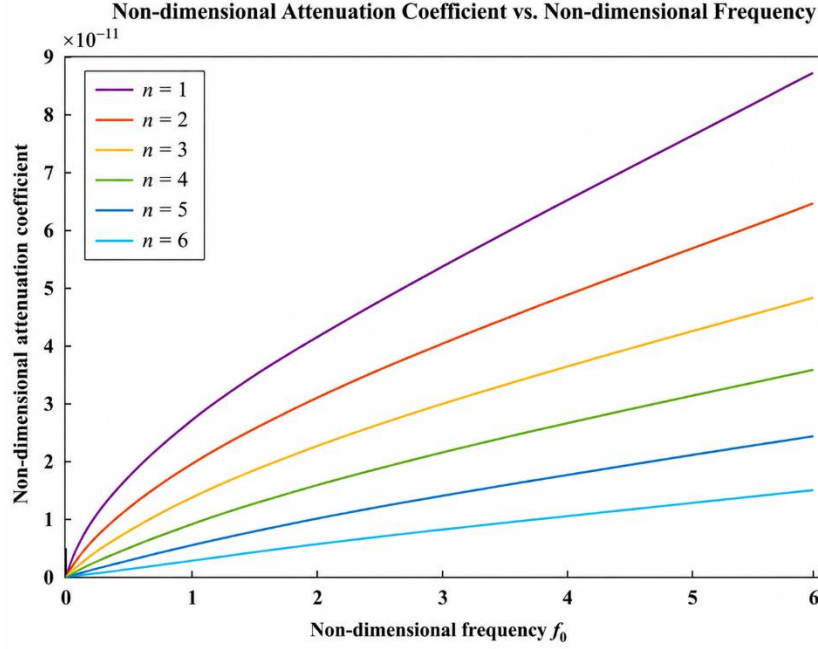


Fig. 11. Variation of the non-dimensional attenuation coefficient with dimensionless frequency f_0 for different propagation modes $n = 1 - 6$.

The normalized attenuation coefficient variation as a function of the dimensionless frequency for six distinct Love-wave propagation modes is shown in Figure 11. For every mode under consideration, it is shown that the attenuation coefficient gradually rises with frequency, suggesting stronger wave-energy loss and improved dissipative behavior at higher frequencies. The findings unequivocally show that as the number of propagation modes increases, the attenuation coefficient falls. The higher-order mode ($n = 6$) displays the least attenuation response, whereas the fundamental mode ($n = 1$) displays the most attenuation across the examined frequency range. This modal dependence can be attributed to the different penetration characteristics of the wave modes; the fundamental mode ($n = 1$) remains more strongly localized within the dissipative guiding region, whereas higher-order modes distribute their energy over a greater depth and therefore experience comparatively weaker attenuation. This trend implies that inside the layered structure, lower-order modes are more susceptible to dissipation effects caused by porosity and thermoelastic damping. Additionally, every curve exhibits smooth nonlinear development characteristics devoid of abnormal fluctuations, verifying the resulting attenuation model's physical consistency and numerical stability. In the higher-frequency zone of Love-wave propagation, the influence of propagation modes becomes more prominent, as evidenced by the increasing spacing between the curves with increasing frequency.

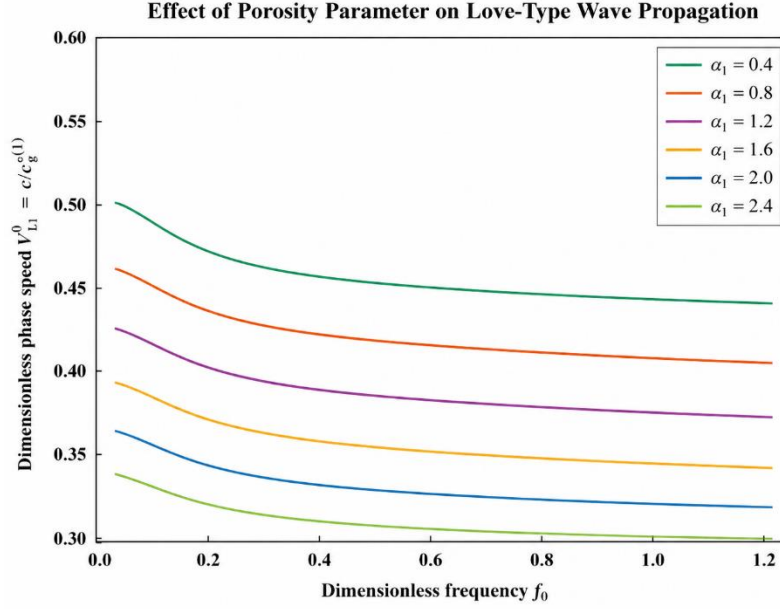


Fig. 12. Variation of the normalized phase speed $V_{L1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of the porosity parameter α_1 .

The impact of the porosity parameter a_1 on the normalized phase speed of Love-type waves in the thermoelastic double-porosity layered structure is shown in Figure 12. For all values of the porosity parameter taken into consideration, it is found that the phase speed continually decreases with increasing dimensionless frequency. The findings unequivocally show that raising the porosity parameter considerably slows the Love waves' rate of propagation. While increasing porosity values gradually push the dispersion curves downward, the curve corresponding to the smaller porosity value ($a_1 = 0.4$) shows the fastest phase speed over the investigated frequency range. Specifically, the example ($a_1 = 2.4$) has the lowest phase velocity, indicating the significant impact of pore contact on wave propagation properties. At lower frequencies, the longer interaction time between the propagating disturbance and the pore structure makes the wave more sensitive to variations in α_1 , whereas this sensitivity weakens as the frequency increases. Additionally, the low-frequency area shows a greater divergence between the curves, suggesting that porosity effects predominate at lower frequencies. All curves eventually reach a virtually stable propagation zone as the frequency rises, but the relative difference between the porosity situations is maintained. This behavior demonstrates that when porosity increases, the medium's elastic stiffness is weakened, which lowers the Love-wave propagation speed inside the thermoelastic layered structure.

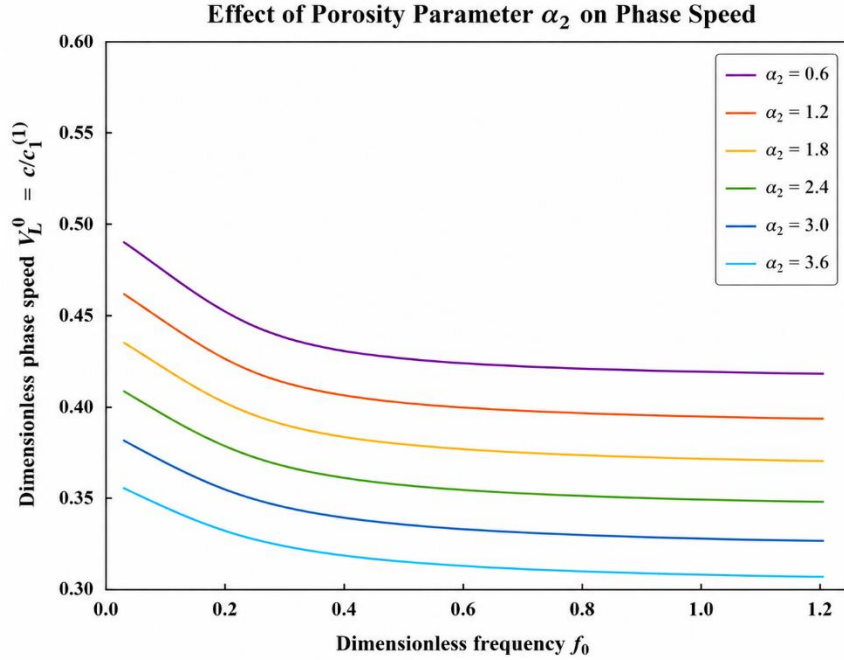


Fig. 13. Variation of the normalized phase speed $V_{L_1}^0 = c/c_1^{(1)}$ with dimensionless frequency f_0 for different values of the porosity parameter α_2 .

The impact of the second porosity parameter α_2 on Love-type wave propagation properties in the thermoelastic double-porosity layered media is shown in Figure 13. For all examined values of α_2 , it is found that the normalized phase speed steadily declines as dimensionless frequency increases. The obtained results clearly demonstrate that increasing the porosity parameter α_2 reduces the phase speed of Love waves throughout the propagation domain. Smaller values of α_2 maintain comparatively higher phase velocities, whereas larger porosity values shift the dispersion curves downward significantly. In particular, the case corresponding to $\alpha_2 = 0.6$ exhibits the highest propagation speed, while $\alpha_2 = 3.6$ produces the minimum phase velocity. Physically, increasing α_2 strengthens the contribution of the secondary pore network to the deformation process, allowing a greater portion of the wave energy to interact with the pore structure and thereby reducing the effective propagation speed. Furthermore, the second porosity field has a significant impact on the dispersive behavior of Love waves since the separation between the curves is still discernible across the whole frequency range. The phase speed gradually decreases as porosity increases, indicating that the extra pore interaction reduces the thermoelastic medium's effective stiffness and slows down wave propagation. The suggested wave propagation model's numerical stability and physical consistency are further supported by the curves' smoothness and stability.

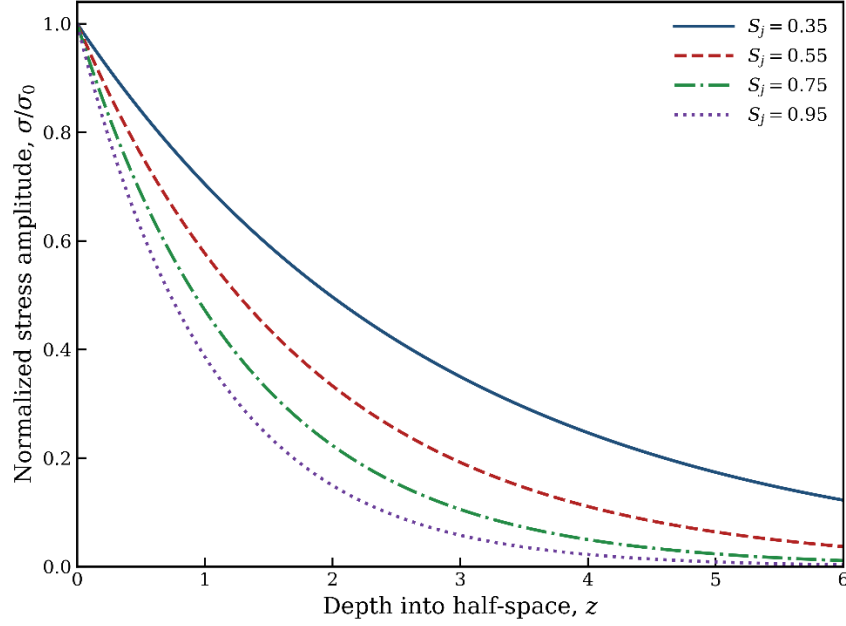


Fig. 14. Variation of normalized stress amplitude with penetration depth in the thermoelastic half-space for different decay parameters S_j .

Figure 14 shows the penetration behavior of normalized stress amplitude in the thermoelastic double-porosity half-space. The curves show that the stress exponentially decreases with the depth z , confirming that the Love-wave energy is concentrated in the vicinity of the corrugated interface and slowly decays in the underlying medium. The stress amplitude is found to attain its maximum value at the interface surface ($z = 0$) and to decay rapidly as the wave propagates deeper into the half-space. Larger values of the decay parameter S_j lead to higher attenuation and quicker decay of the stress field. For example, the curve corresponding to $S_j = 0.95$ decreases more steeply compared with $S_j = 0.35$, indicating a smaller penetration depth and stronger confinement of wave energy near the interface. Physically, a larger S_j corresponds to a stronger evanescent decay, restricting the stress energy to a narrower region near the interface, whereas a smaller S_j allows the disturbance to penetrate deeper into the half-space. The results also show that smaller values of S_j allow a deeper penetration of the stress field in the medium, while larger values localize the wave motion near the surface region. This behaviour bears witness to the evanescent nature of Love-wave propagation in the half-space and emphasizes the significant role played by thermo-porosity coupling and interface effects to govern stress penetration features.

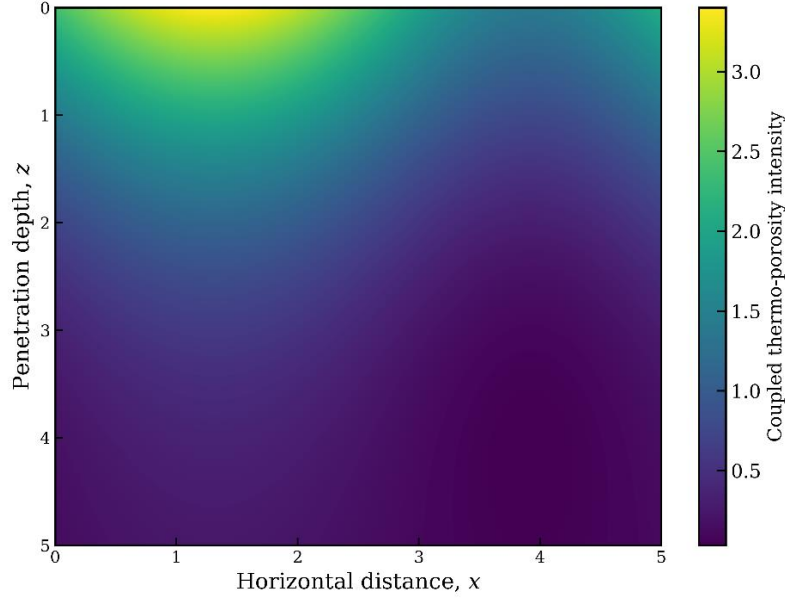


Fig. 15. Heatmap representation of the coupled thermo-porosity field intensity inside the thermoelastic half-space.

The spatial distribution of the coupled thermo-porosity field in the thermoelastic double-porosity layered medium is shown in Fig. 15. The heatmap shows the overall effect of the thermal interaction, porosity coupling and penetration depth on the wave propagation behavior. It is observed that the field intensity is maximum at the interface region and slowly decreases with the increasing depth z indicating the attenuation of the coupled thermo-porosity effects inside the half-space. The oscillatory variation in horizontal direction x shows periodic wave propagation and interface interaction characteristics. Physically, the strong intensity near the interface results from the simultaneous interaction of the thermal and two porosity fields, while their coupled influence weakens progressively as the disturbance penetrates deeper into the half-space. The brighter color represents the regions of higher thermo-porosity interaction intensity, and the darker color indicates weaker coupled-field effects due to energy dissipation and penetration decay. The results also show that the thermal and porosity coupling combined modifies the wave-field distribution near the corrugated interface significantly and controls the localization of wave energy in the layered structure.

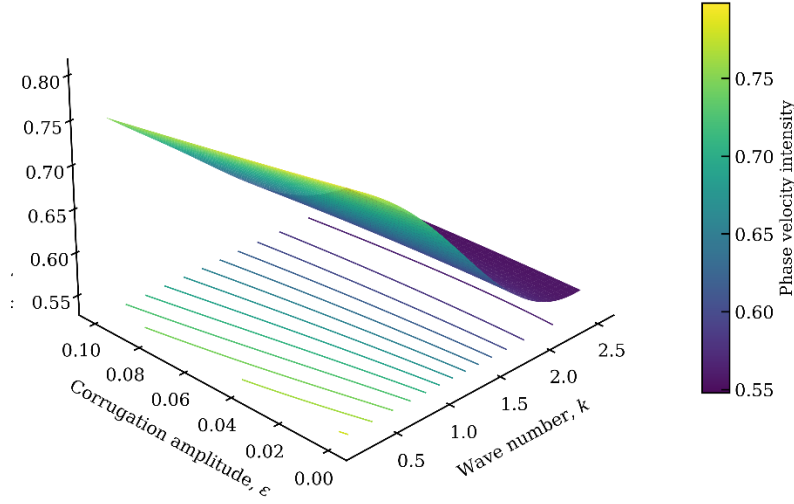


Fig. 16. Three-dimensional contour surface showing the variation of dimensionless phase velocity with wave number k and corrugation amplitude ε .

The combined effect of wave number and interface corrugation on phase velocity of Love waves in thermoelastic double-porosity layered medium is depicted in Figure 16. It is noted that the dimensionless phase velocity ranges approximately from 0.42 to 0.82 in the parameter range considered. Physically, this competing behavior reflects the balance between wave-number-induced confinement and corrugation-induced scattering, with increasing k promoting guided-wave propagation while larger ε redistributes energy through the irregular interface and reduces the phase velocity. The phase velocity increases significantly with the wave number k , while larger corrugation amplitudes ε decrease the speed of propagation due to the increased irregularity of the interface and the effects of energy scattering. The contour plots also show a strong nonlinear coupling between thermo-porosity interaction and interface geometry, especially in the intermediate parameter region. The smooth distribution of the surface confirms the stability and physical consistency of the proposed dispersion model.

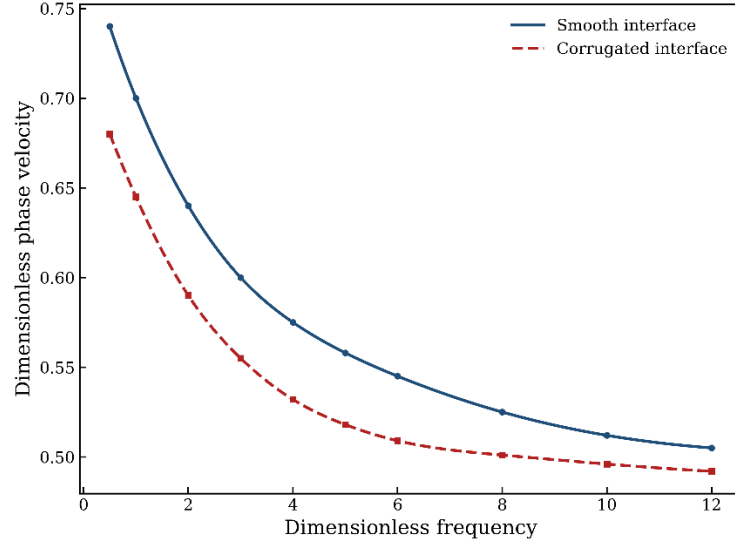


Fig. 17. Dimensionless phase velocity for smooth and corrugated interfaces for the thermoelastic double-porosity layered medium.

The propagation behavior of Love waves for corrugated and smooth contact topologies is contrasted in Figure 17. The corrugated interface dramatically lowers the propagation speed because of interface irregularity and increased scattering effects, while the smooth interface generates larger phase velocities over the whole frequency range. Physically, the reduction in phase velocity occurs because the corrugated interface disturbs the otherwise continuous propagation path, causing additional wave scattering and redistribution of energy compared with the smooth interface. The corrugated interface lowers the phase velocity to about 0.68 at lower frequencies, whereas the smooth interface shows a phase velocity of over 0.74. Both curves progressively decline as the frequency rises and go closer to a practically constant limiting value near 0.50. The smooth-interface result also provides a limiting-case verification of the present formulation, since setting $\varepsilon = 0$ eliminates the corrugation contribution and recovers the corresponding smooth-interface propagation behavior. The distance between the two curves demonstrates that interface corrugation significantly affects the propagation of low-frequency waves and alters the layered structure's overall dispersion properties.

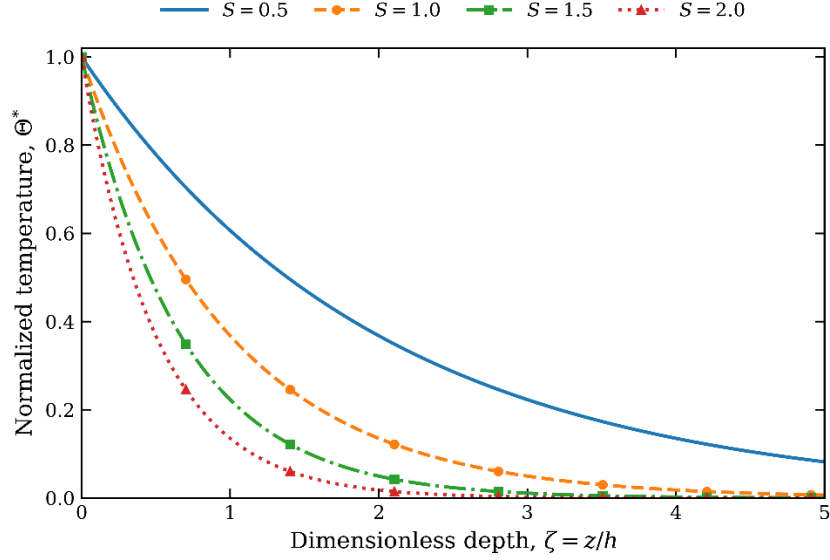


Fig. 18. Variation of normalized temperature Θ^* with dimensionless penetration depth $\zeta = z/h$ for different values of the thermo-porous decay parameter S .

Figure 18 illustrates the variation of the normalized temperature Θ^* with the dimensionless penetration depth ζ for different values of the thermo-porous decay parameter S . The temperature decreases continuously with increasing depth, confirming the attenuation of the thermal disturbance away from the interface. For smaller values of S , the temperature decays more gradually, indicating that the thermal disturbance penetrates deeper into the medium. In contrast, larger values of S produce a steeper temperature gradient and a stronger localization of thermal energy near the interface. This behavior is consistent with the exponentially decaying thermal modes obtained from the bounded solution of the coupled thermoelastic formulation and demonstrates the important role of the decay parameter in controlling thermal penetration.

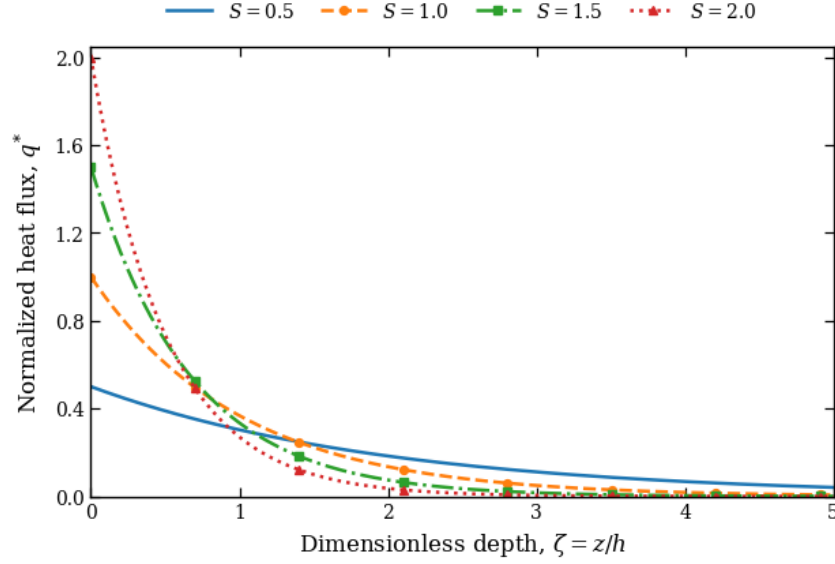


Fig. 19. Variation of normalized heat flux q^* with dimensionless penetration depth $\zeta = z/h$ for different values of the thermo-porous decay parameter S .

Figure 19 presents the variation of the normalized heat flux q^* with the dimensionless penetration depth ζ for different values of the thermo-porous decay parameter S . Higher values of S produce a larger heat-flux magnitude near the interface because of the stronger local temperature gradient. However, the heat flux decreases rapidly as the penetration depth increases. In contrast, smaller values of S yield a lower initial heat-flux magnitude but a more gradual attenuation of the thermal response. At sufficiently large depths, the normalized heat flux approaches zero for all cases, indicating that the thermal disturbance becomes negligible far from the interface. These results show that increasing the modal decay parameter enhances near-interface thermal localization while reducing the effective penetration range of the heat-transfer process.

Engineering significance and practical implications.

The present results have practical relevance to layered porous systems in which mechanical-wave propagation occurs simultaneously with thermal transport and pore interactions. The predicted variations in phase velocity and attenuation with corrugation and porosity parameters indicate that Love waves can be sensitive to changes in interface geometry and internal pore structure, providing a potential basis for the characterization of layered porous materials. In addition, the predicted stress localization, temperature decay, and heat-flux behavior provide information on the depth over which mechanical and thermal disturbances remain significant. These characteristics are relevant to subsurface and geophysical characterization, non-destructive evaluation of porous layered structures, thermal-barrier and porous composite systems, and guided-wave monitoring under thermo-mechanical loading. Thus, the proposed model provides a theoretical framework for

assessing how interface irregularity, double porosity, and thermal coupling jointly affect wave transmission and heat-transfer characteristics in realistic layered media.

Conclusion

This study presented an analytical investigation of coupled thermal-wave transport and Love-wave propagation in thermoelastic double-porosity layered structures containing a periodically corrugated interface. Based on the generalized thermoelastic double-porosity framework of Ieşan and Quintanilla, the governing thermo-mechanical equations were formulated and an explicit dispersion relation was derived to characterize the propagation behavior of guided waves in the considered layered medium. Detailed numerical analyses were subsequently performed to examine the influence of interface geometry, thermal interactions, and porosity coupling on dispersion and attenuation characteristics. The proposed formulation simultaneously incorporates thermal coupling, two interacting porosity fields, and periodic interface corrugation, thereby providing a unified framework for analyzing their combined effects on guided-wave transport.

The results demonstrated that interface corrugation plays a significant role in modifying thermal-wave and guided-wave transport characteristics, particularly within the low-frequency regime. Increasing the corrugation amplitude and corrugation wave number enhanced phase velocity and delayed dispersive decay, indicating improved wave-guiding capability of the layered structure. In contrast, larger porosity parameters reduced the effective thermoelastic stiffness of the medium, leading to lower propagation speeds and stronger dissipative effects. Furthermore, attenuation increased with frequency, reflecting enhanced thermoelastic damping and energy loss during wave propagation. Lower-order modes exhibited greater attenuation, whereas higher-order modes enabled deeper energy penetration into the underlying half-space. The thermal analysis further revealed that larger thermo-porous decay parameters produce stronger localization of temperature and heat flux near the interface while reducing the effective thermal penetration depth.

Additional stress-distribution, heatmap, and three-dimensional contour analyses revealed that the combined action of thermoelastic coupling, double porosity, and interface irregularity strongly influences energy localization, thermal dissipation, and stress concentration near the corrugated interface. The recovery of the smooth-interface configuration for the limiting case $\varepsilon = 0$ provides an additional consistency check for the proposed formulation. The smooth convergence of the numerical solutions and the physically consistent trends observed throughout the parametric study further support the consistency of the analytical and numerical results.

The developed model provides valuable insight into coupled heat-transfer and wave-propagation phenomena in porous thermoelastic layered systems. From an engineering perspective, the predicted variations in phase velocity, attenuation, stress penetration, temperature, and heat flux provide useful response characteristics for assessing the influence of pore structure and interface geometry. The findings may contribute to the theoretical characterization and future investigation of thermal barrier materials, porous composites, layered energy-absorbing structures, geophysical

formations, subsurface reservoirs, and advanced engineering systems where thermal transport, energy dissipation, and guided-wave behavior are strongly coupled.

From an engineering perspective, the results provide useful insight into the control of wave transmission, attenuation, stress localization, and thermal penetration in layered porous structures, with potential relevance to geophysical characterization, non-destructive evaluation, and thermo-mechanical monitoring of porous composite systems.

Replication of Results

The analytical formulation, governing equations, boundary and interface conditions, and numerical procedures used to obtain the results presented in this study are fully described in the manuscript. The numerical computations and graphical results were generated using MATLAB based on the derived dispersion relations and the material parameters reported in the numerical results section. The data supporting the findings of this study are generated from the analytical model and can be reproduced using the equations and parameters provided. Additional details regarding the computational procedure and MATLAB implementation are available from the corresponding author upon reasonable request.

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Author's contribution statement:

Conceptualization, Hrain P V and Samia Elattar; Methodology, Writing-Original Draft Preparation, Hrain P V; Writing - Review & Editing, Samia Elattar; Funding Acquisition, Samia Elattar.

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Conflict of Interest:

The authors declare no competing interests.

Data availability declaration:

The datasets analysed during the current study are available through the cited references.

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